

Intra-Category Multi-Choice Preferences Learning and Assortment Recommendation in E-Commerce

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Abstract: This paper proposes a framework for learning customer preferences and optimizing e-commerce assortments, focusing on intra-category multi-choice behavior, where customers buy multiple items within the same category. Unlike traditional discrete choice models (DCMs) that assume single-product purchases, e-commerce data show frequent intra-category multi-product purchases, especially during promotions. To capture this, we introduce the multi-choice rank list model (MC-RLM), which accounts for both multi-choice and substitution effects. Each customer type is defined by a preference ranking and an intended purchase quantity (IPQ), allowing selection of up to IPQ products. The MC-RLM adheres to the regularity axiom and aligns with random utility theory.

We present the multi-choice market discovery (MCMD) algorithm to estimate the MC-RLM, extending single-purchase methods to multi-purchase settings. We also introduce behavior-reveal-preference (BRP) rules, using customer behavior data (e.g., clicks, cart additions) to enhance preference estimation. Given the NP-hardness of the assortment optimization problem, we analyze the performance of the revenue-ordered assortments heuristic and provide guarantees. The problem is formulated as a mixed-integer linear program (MILP) that can generate personalized recommendations based on real-time customer data. Extensive numerical experiments, including a case study using Tmall data, demonstrate that the MC-RLM outperforms models such as the Independent Choice Model (ICM) and the Multi-Purchase Multinomial Logit (MP-MNL) in predictive accuracy, with BRP rules further enhancing performance. Synthetic experiments confirm that accurately modeling multi-purchase behavior significantly boosts expected revenue.

Key words: e-commerce, assortment optimization, multi-purchase behavior, non-parametric choice model, click-stream data

1 Introduction

Understanding customer preferences and leveraging this knowledge to strategically plan product assortments are crucial for maximizing revenue in retail. Discrete choice models (DCMs) are widely used to model customer demand for substitutable products and assist decision-makers in assortment planning (see Gallego and Topaloglu 2019 for a comprehensive review). Most of these models assume that each customer purchases at most one product at a time, which is a reasonable premise for high-priced items like cars or

smartphones. However, intra-category multi-purchase¹ behavior, where customers buy multiple products within the same category, is common in practice. For example, consumers may purchase several lipsticks in different shades or various clothing styles for different occasions. Real-world data support this phenomenon. According to Bai et al. (2023), data from a leading flash-sales e-retailer show that in categories like underwear and dresses, over 40% of customers buy multiple distinct items within the same category. Our analysis of a publicly available dataset from Tmall, one of China’s largest e-commerce platforms, further corroborates this behavior (details provided in Section 6.1). Such multi-purchases occur more frequently during promotional campaigns, e.g., Figure 1, when well-known brands offer steep discounts for a limited time.

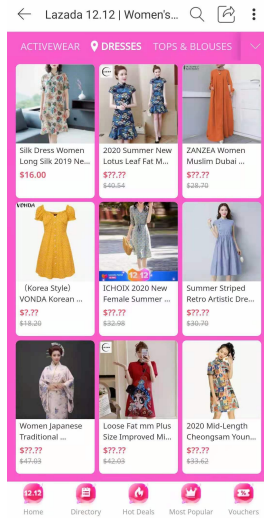


Figure 1 A dress recommendation during a campaign on Lazada

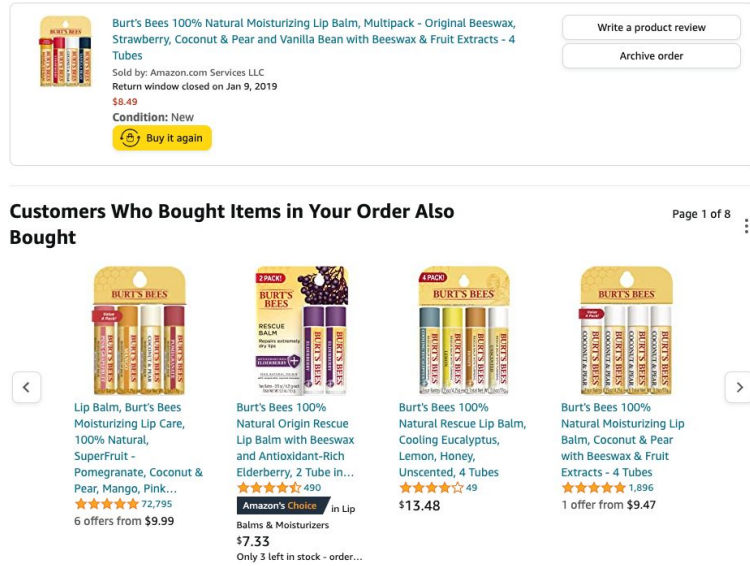


Figure 2 A lip balm recommendation following a checkout on Amazon

Despite its prevalence, intra-category multi-purchase behavior, and the associated assortment planning, have received limited attention in the existing literature, particularly in the context of e-commerce retail. While substantial research has addressed assortment planning under traditional single-purchase choice models in brick-and-mortar settings (e.g., Ryzin and Mahajan 1999, Mahajan and Van Ryzin 2001b), studies focused on learning customer preferences and planning assortments for e-commerce are gaining increasing attention (e.g., Li et al. 2018, Aouad et al. 2019). In physical stores, tracking customer-product interactions beyond sales transactions is challenging. In contrast, e-retailers can easily collect extensive customer behavior data, including search queries, browsing patterns, click histories, and shopping cart activity. These data, often referred to as *side information* (Aouad et al. 2019, Farias and Li 2019), provide valuable insights for predicting future purchases and brand loyalty (Laudon and Traver 2013), enabling retailers to better understand customer preferences and make more informed decisions. Moreover, e-commerce platforms

¹ In this paper, we use *multi-purchase* and *multi-choice* interchangeably; both refer to choosing multiple distinct alternatives. Conversely, we use *single-purchase*, *single-choice*, and *single-choice* interchangeably to denote choosing a single alternative.

offer various methods of recommending products beyond merely displaying items on shelves, such as front-page banner advertisements and personalized product suggestions based on browsing history. Consequently, retailers must adjust assortment recommendations at different stages of the customer journey. In this work, we focus on modeling intra-category multi-purchase behavior and developing assortment planning strategies for various e-commerce scenarios, collectively referred to as assortment recommendations.

The effectiveness of an assortment decision critically depends on accurately capturing customers' preferences and understanding their choice behaviors, particularly whether they purchase single or multiple products. Under traditional models, where customers purchase at most one product, offering a wide range of alternatives could lead to cannibalization, as lower-profit products may detract from the sales of higher-profit items. Retailers often limit their assortments to maximize revenue. However, when customers exhibit intra-category multi-purchase behavior, this conventional insight may not apply. Providing a broader assortment could increase revenue, as customers may intend to purchase multiple items. To illustrate this point, consider the following example:

EXAMPLE 1 (ASSORTMENT OPTIMIZATION WITH INTRA-CATEGORY MULTI-PURCHASES). Suppose a retailer offers two products, i_1 and i_2 , with profits of \$10 and \$9, respectively. The market consists of two types of customers in equal proportions. Type A customers prefer i_1 over i_2 and both over the no-purchase option, while Type B customers prefer i_2 over i_1 and both over the no-purchase option. If each customer intends to purchase at most one product, the optimal assortment is to offer only i_1 . Both customer types will purchase i_1 , yielding an expected revenue of \$10. Offering both products leads to cannibalization: Type A customers buy i_1 , but Type B customers switch to i_2 , reducing the expected revenue to \$9.50. However, if all customers intend to purchase multiple products, offering both i_1 and i_2 becomes optimal. Both customer types purchase both products, yielding an expected revenue of \$19, a 90% increase over offering only i_1 . ■

This example illustrates that ignoring such multi-purchase behavior can lead to suboptimal assortment decisions and significant revenue loss. Beyond assortment planning, considering intra-category multi-purchase behavior offers a novel understanding of e-commerce recommendations. Traditional DCMs assume customers purchase at most one product per category, suggesting that recommending additional similar products after a purchase would be ineffective. However, many e-commerce platforms routinely recommend products within the same category even after a customer has checked out (e.g., Figure 2), recognizing that customers may be willing to purchase additional items from the same category. By incorporating the multi-purchase behavior into consideration, we reconcile this apparent inconsistency between traditional theory and observed practices. Customers may indeed be open to purchasing additional products within the same category, making such post-purchase recommendations logical and effective. This insight highlights the importance of accounting for multi-choice behavior in assortment recommendation strategies.

In summary, this paper presents a comprehensive framework for e-commerce assortment planning that leverages customer behavioral data to learn preferences and optimizes assortment recommendations for intra-category multi-choice scenarios. The key contributions of this work are as follows:

First, we introduce the multi-choice rank list model (MC-RLM) to capture customers' intra-category multi-choice and substitution behaviors. In the MC-RLM, each customer type is represented by a preference ranking over all products (including the no-purchase option) and an intended purchase quantity q . If the number of available products that the customer prefers exceeds q , they select the q highest-ranked items; otherwise, they purchase all preferred products that rank higher than the outside option. The MC-RLM adheres to the regularity axiom and provides a flexible framework for modeling rational multi-purchase behavior within a single category. Moreover, we demonstrate its consistency with random utility theory by proving its equivalence to the proposed multi-choice random utility model (MC-RUM). We also demonstrate that both the independent choice model (ICM), which assumes independent product choices, and the multi-purchase multinomial logit model (Bai et al. 2023) are special cases of both MC-RLM and MC-RUM.

Second, we propose a comprehensive estimation framework for e-commerce settings that enables efficient learning of customer preferences under the MC-RLM. We introduce the multi-choice market discovery (MCMD) algorithm, an extension of the approach by van Ryzin and Vulcano (2015) for single-purchase rank list models, adapted to account for multi-choice behavior. The MCMD algorithm typically yields sparse models, capturing the population with only a few customer types assigned positive arrival probabilities while still accurately representing the population. Additionally, we introduce behavior-reveal-preference (BRP) rules, a novel concept that integrates customer behavior data, such as viewing, clicking, and adding to cart, into the estimation process of rank list models. BRP rules are flexible and adaptable to various data types, allowing for a more precise representation of customer preferences.

Third, given the NP-hardness of solving the assortment recommendation problem under the MC-RLM, we first analyze the widely used revenue-ordered assortments heuristic and demonstrate its performance guarantee when the underlying model is an MC-RLM. Additionally, we formulate the assortment recommendation problem as a mixed-integer linear program (MILP), which can be solved efficiently using state-of-the-art solvers to obtain optimal solutions. Typically, when there are fewer than 80 products and fewer than 20 customer types, the MILP can be solved in under one second. Furthermore, we demonstrate that personalized recommendations can be generated by re-solving the MILP with updated customer types based on real-time customer behavior data.

Lastly, we conduct extensive numerical experiments to validate the effectiveness of the proposed MC-RLM. Using a publicly available dataset from Tmall, we demonstrate that the MC-RLM effectively captures intra-category multi-purchase behavior and achieves a lower root mean square error (RMSE) compared to the ICM and MP-MNL models in most cases. Accounting for multi-purchase behavior by the MC-RLM significantly improves predictive accuracy over the single-choice rank list model (SC-RLM) and the

multinomial logit (MNL) model. Additionally, incorporating the BRP rules in the estimation of both the MC-RLM and SC-RLM results in substantial improvements in predictive accuracy. Furthermore, a series of synthetic experiments show that accurately modeling multi-purchase behavior leads to higher expected revenue in assortment optimization.

1.1 Organization

The remainder of the paper is organized as follows. Section 2 provides a comprehensive overview of related literature. Section 3 introduces the MC-RLM model and its key properties. In Section 4, we present an algorithm for estimating MC-RLM parameters using historical behavior data. Section 5 discusses the assortment recommendation problem, including a performance guarantee for the revenue-ordered assortment heuristic, an MILP formulation for exact optimization, and methods for personalized recommendation based on real-time behavior. Section 6 presents a case study using Tmall data to demonstrate the value of modeling multi-purchase behavior and incorporating behavioral data into estimation, along with synthetic experiments that further highlight the importance of multi-purchase-aware assortment decisions. Finally, Section 7 concludes the paper, and proofs are provided in Section EC.1.

2 Literature Review

This section briefly reviews the literature on customer choice models, assortment optimization, and the consider-then-choose framework relevant to our work.

2.1 Customer Choice Models

Single-purchase discrete choice models, which assume that each customer selects at most one item, are widely used to describe decision-making among a finite set of alternatives. These models are broadly divided into parametric and non-parametric streams. A well-studied parametric class is the Random Utility Model (RUM), introduced by Thurstone (1927) and extensively examined in both economics and operations research (see Anderson et al. 1992, Gallego and Topaloglu 2019 for reviews). The most popular version, the Multinomial Logit (MNL) model, assumes that each alternative is associated with a utility comprising both deterministic and random components, with the latter assumed to be i.i.d. Gumbel distributed (McFadden 1974). Among non-parametric models, the Rank List Model (RLM) is particularly noteworthy. It assumes that consumers have individual preference rankings for all products and select the highest-ranked available option. This model originates from psychology and economics literature (e.g., Block and Marschak 1960, Falmagne 1978) and was introduced to operations research by Mahajan and Van Ryzin (2001a,b). Block and Marschak (1960) demonstrated that RLM is equivalent to the RUM when deterministic utilities are fixed. While RLM offers a more flexible representation of preferences than many parametric choice models, estimating the distribution of rankings is computationally challenging due to the factorial growth in

possible rankings. To address this, Farias et al. (2013) proposed a robust optimization approach, and van Ryzin and Vulcano (2015) developed a market discovery algorithm using maximum likelihood estimation. Both methods demonstrate that, despite the large number of possible preference rankings, a limited number of preference types with positive probability can effectively capture the preferences of the entire customer population. In this paper, we extend the RLM and RUM to capture intra-category multi-purchase behavior, and adapt the market discovery algorithm to estimate the proposed choice model.

While single-purchase models have been extensively studied, they fail to explain the prevalent multi-purchase behavior observed in practice. In response, the marketing literature has explored market basket analysis, which seeks to understand the interdependence between products purchased together during a single shopping trip (Russell et al. 1999). This research typically focuses on examining cross-category dependence (including complementarity and substitutability) and independence across different product categories. Various multi-purchase choice models have been proposed for basket choice analysis, such as the Multivariate Probit (MVP) model, the Multivariate Logit (MVL) model, the Multivariate Multinomial Logit (MVMNL) model (a generalization of the MVL model), etc. (see Seetharaman et al. 2005, Aurier and Mejia 2014 for reference). These models assume that products are grouped into different categories, and a customer can simultaneously select products from multiple categories while restricting their choices to at most one item per category. Although originally developed to uncover cross-category relationships for marketing strategies, these models are increasingly being applied to inform operations decisions. Tulabandhula et al. (2023) introduce a variant of the MVL model, called the Bundle Multivariate Logit (BundleMVL-K) model², which assumes customers can purchase a bundle of up to $K \geq 1$ products (constant K is predetermined). Jasin et al. (2023) study both uncapacitated and capacitated assortment optimization, under the MVMNL model and develop several FPTAS for these problems under specific conditions. Our work differs from market basket analysis by focusing on intra-category substitution and multi-purchase behavior, rather than cross-category relationships. While our model is not suited for cross-category complementarity, extending it with basket choice models to capture general *pick-any* behavior is a promising future direction.

Aside from the application of basket choice models in operations research, various multi-purchase choice models have also been introduced for different contexts. Huh and Li (2021) and Zhang et al. (2021) study pricing and assortment optimization under the Multiple-Discrete Choice (MDC) model, a deterministic variant of the Multiple Discrete-Continuous Extreme Value (MDCEV) model introduced by Bhat (2005). While the MDC model accommodates multiple purchases, it assumes divisible products, making it unsuitable for our setting. Luan et al. (2024) extend the MNL and mixed-MNL models to examine purchasing multiple units of a single alternative. More closely related to our work, Gallego and Wang (2019) propose the Threshold Utility Model (TUM) to capture multi-choice behavior within a category. However, they

² The BundleMVL-K model is also a special case of the MVMNL model where each category contains only one product.

assume a uniform purchasing capacity for all customers, which may not align with real-world scenarios. The most closely related works are Bai et al. (2023) and Abdallah et al. (2024). Bai et al. (2023) introduce the Multi-Purchase Multinomial Logit (MP-MNL) model, which extends the classic MNL model to the same intra-category multi-purchase setting. As shown in Section 3, the MP-MNL model is a special case of our proposed model. Abdallah et al. (2024) extend the RUM to account for intra-category multi-purchase behavior by predetermining an intended purchase quantity and introducing multiple no-purchase options, but do not address model estimation using historical data. Unlike most prior work, which relies on parametric models that manually specify the structure of product utility, we adopt a more flexible non-parametric framework to capture intra-category multi-purchase behavior. Importantly, we propose a corresponding practical method for estimating the model using abundant customer-product interaction data, an approach that may be challenging for existing parametric models.

2.2 Assortment Optimization

The assortment optimization problem under various random utility models has been extensively studied. For example, Talluri and Van Ryzin (2004), Davis et al. (2013) assume multinomial logit demand; Gallego and Topaloglu (2014), Feldman and Topaloglu (2015) use nested logit; and Désir et al. (2014) apply mixed-MNL. Bai et al. (2023) show that assortment optimization under MP-MNL is NP-hard, even when the IPQ support has only two values, and provide two PTAS for approximating the optimal revenue. Abdallah et al. (2024) develop a heuristic solution using asymptotic analysis.

Assortment optimization under the classic RLM has also been explored. Honhon et al. (2012) were the first to solve the problem for specific cases, where preference rankings form paths in a binary tree, allowing the problem to be solved by finding the shortest path. Paul et al. (2018) extended this to a more general tree structure. Belloni et al. (2008) reformulated the single-purchase RLM as MIP, providing an approximate solution. Bertsimas and Mišić (2019) revisited the problem with a new MIP formulation and used Benders Decomposition to find the exact optimum. Building on their work, we extend their MIP formulation to solve the intra-category multi-purchase assortment optimization under our proposed MC-RLM. Additionally, inspired by Aouad et al. (2018), who studied the best-possible approximability bounds for assortment planning under the RLM, we analyze the performance of the widely used revenue-ordered assortments heuristic under the MC-RLM.

Our work also relates to the literature on recommender systems, which began with collaborative filtering in the 1990s (Resnick et al. 1994, Hill et al. 1995) and has been widely adopted in e-commerce and digital content platforms (see Ricci et al. 2010 for a comprehensive review). Although modern recommender systems incorporate customer and product features, they typically focus on ranking individual products to maximize predicted relevance or user engagement. In contrast, our work complements this literature by

explicitly modeling intra-category multi-purchase and substitution behavior, enabling assortment-level recommendations that better reflect customer behavior in categories (e.g., fashion or grocery) where multiple, mutually substitutable items are often purchased within a single shopping occasion.

2.3 Consider-then-Choose Framework

Our estimation process incorporates BRP rules by leveraging customer browsing data, closely relating to the consider-then-choose framework in the literature. In this framework, customers first form a consideration set from the offer set before making a final choice. Since Howard and Sheth (1969) introduced the concept, numerous studies have demonstrated its effectiveness in explaining purchasing behavior. The approach is particularly relevant to e-commerce, where evaluating thousands of products is impractical (Aouad et al. 2019). While Li et al. (2018) employed a two-stage MNL-based model to optimize assortments under the consider-then-choose framework, our work adopts a more explicit approach by directly leveraging BRP rules. Specifically, we assume that customer-product interactions reveal ordinal preferences: purchased items are preferred to carted items, which in turn are preferred to merely viewed items. This structure provides a practical method for inferring preferences from clickstream data, and the flexibility of BRP rules allows for adjustment based on empirical patterns and expert knowledge.

3 Modeling Intra-Category Multi-Choice Behavior

In this section, we formally introduce the *Multi-Choice Rank List Model* (MC-RLM) to capture intra-category multi-purchase behavior and its connection to random utility theory. We assume that customers may buy multiple different products from the same category, but they will only purchase one unit of each product. This assumption is applicable to various categories, such as books, clothing, cosmetics, and digital content, and is commonly adopted in the literature (e.g., Bai et al. 2023, Abdallah et al. 2024).

We define $N = \{1, 2, \dots, N\}$ as the set of alternatives with $N \geq 2$, and let 0 denote the outside (no-purchase) option. We denote $V \subseteq N$ as the offer set and $P \subseteq V$ as the purchase set. For any set S , we let $S^+ = S \cup \{0\}$, and for any integer n , we let $[n] = \{1, \dots, n\}$. We assume V is non-empty and the outside option is always available, meaning choices are made from V^+ . A preference ranking over N^+ is a bijection σ from N^+ to $[N + 1]$, where $\sigma(n)$ denotes the rank of product n , and $\sigma^{-1}(i)$ denotes the product ranked i -th. We assume no ties: product n is preferred to n' under σ if and only if $\sigma(n) < \sigma(n')$.

Recall that the classic *Single-Choice Rank List Model* (SC-RLM) assumes each customer chooses the offered product that ranks highest according to their specific preference. To account for multi-purchases, we introduce a non-negative integer q , the customer's *intended purchase quantity* (IPQ), representing the number of distinct alternatives (excluding the outside option) they intend to buy if all products are available. When $q = 0$, the customer intends to buy nothing. A customer type is defined by the tuple (q, σ) , with T representing all possible types. A type (q, σ) customer purchases up to q products based on the preference

ranking σ . In MC-RLMs, the IPQ, q , can be regarded as the budget for a customer. If enough preferred products are offered, a customer selects the top q according to her σ . If fewer than q preferred products are available above the outside option, the customer purchases only those offered. For example, a customer of the type $(2, [2, 3, 1, 0, 4])$ will buy products 2 and 3 if the assortment is $\{1, 2, 3, 4\}$, but will choose only product 3 if the offer set is $\{3, 4\}$. Moreover, we define $(0, [0])$ as the sole customer type with $q = 0$, representing customers who do not intend to make any purchase, such as those who are merely browsing. For these customers, the preference list is irrelevant. Additionally, customers of other types will leave without purchasing if none of the available products are preferred over the outside option.

For the MC-RLM, we define $\lambda : T \rightarrow [0, 1]$ as the probability mass function of customer types with $\lambda_{q,\sigma} \geq 0$ representing the probability that an arriving customer belongs to type (q, σ) . Here, λ satisfies $\sum_{(q,\sigma) \in T} \lambda_{q,\sigma} = 1$, ensuring that the total probability across all types sums to one. Additionally, we define an *interest list*, $I_{(q,\sigma)} = \{n \in N : \sigma(n) < \sigma(0)\}$ as the set of products ranked higher than the outside option in σ , representing the products that type (q, σ) customers consider for purchase. To express the choice probability for each possible order, we define $T_{P,V}$ as the collection of customer types corresponding to the purchase decision (P, V) , i.e., choosing set P when V is offered, as follows: $T_{P,V} = \{(q, \sigma) \in T : q = |P|, \max_{i \in P} \sigma(i) < \min_{j \in V \setminus P} \sigma(j)\} \cup \{(q, \sigma) \in T : q > |P|, \max_{i \in P} \sigma(i) < \sigma(0), \sigma(0) < \min_{k \in V \setminus P} \sigma(k)\}$. In the expression, the first set includes customer types who find sufficient products of interest in V , choosing the q highest-ranked products offered, and these q products happen to be P . The second set comprises customer types who find fewer products of interest than their IPQ, and P is all the available products in their interest lists. Then, in the MC-RLM, the probability that a randomly arriving customer purchases set P when set $V \supseteq P$ is offered is given by:

$$\pi_{P,V} = \sum_{(q,\sigma) \in T_{P,V}} \lambda_{q,\sigma}. \quad (1)$$

Indeed, the multi-choice rank list model is a generalization of the classic single-choice rank list model. That is, if all customers intend to buy at most one product, i.e., $q \leq 1$ for all $(q, \sigma) \in T$, then the MC-RLM reduces to the SC-RLM. Besides, we can show that the MC-RLM satisfies the regularity axiom, a property that holds for many single-purchase discrete choice models, particularly for the random utility models (Block and Marschak 1960). This axiom states that the introduction of new products will not increase the choice probability of any existing product. Next we show that even in the presence of multi-purchase, MC-RLM still satisfies regularity.

PROPOSITION 1. *The regularity axiom is satisfied by the multi-choice rank list model. Specifically, for any nested assortments $V \subseteq V' \subseteq N$ and for any product $n \in V \cup \{0\}$, it holds that $\mathbb{P}(n, V) \geq \mathbb{P}(n, V')$. Here, $\mathbb{P}(n, V) := \sum_{P \subseteq (V \setminus \{n\})} \pi_{P \cup \{n\}, V}$ for $n \neq 0$, and $\mathbb{P}(0, V) := \pi_{\emptyset, V}$.*

Proposition 1 shows that in the MC-RLM, the probability of selecting a product does not increase as the offer set expands. Not all choice models satisfy this regularity axiom; for example, certain BundleMVL models can violate it due to positive correlations among alternatives, as illustrated in Example EC.1. While MC-RLM may be less suitable for cross-category settings, its compliance with the regularity property aligns with the economic intuition that adding options should not make existing products more attractive, thereby enhancing its interpretability and reliability in modeling intra-category consumer behavior.

Leveraging these findings, in subsequent subsections, we will explore how this newly proposed model relates to other existing discrete choice models. Particularly, we will demonstrate that the MC-RLM aligns with random utility theory and is capable of representing independent choices among products.

3.1 Random Utility Justification of the MC-RLM

According to Block and Marschak (1960), the classic single-choice rank list model is equivalent to the classic single-choice random utility model. From our previous discussion, the MC-RLM extends the SC-RLM to accommodate multi-purchase scenarios and maintains the regularity axiom commonly observed in single-choice random utility models. Inspired by these insights, we introduce the *Multi-Choice Random Utility Model* (MC-RUM) as a generalization of the classic random utility model.

For the MC-RUM, we assume that there are $N + 1$ random variables $U_0, U_1, \dots, U_N$, representing the utilities of the outside option and N alternatives, respectively. The random utility U_n models the internal value that a random customer attaches to alternative $n \in \mathbb{N}^+$. For ease of representation, we denote $\mathbf{U} = (U_1, \dots, U_N)^\top$ and $\mathbf{U}^+ = (U_0, U_1, \dots, U_N)^\top$ as random utility vectors. Similar to the MC-RLM, we introduce an integer random variable $Q \in \{0, 1, \dots, N\}$ to represent a customer's intended purchase quantity. Note that in MC-RUMs, Q represents a random variable, whereas in MC-RLMs, a fixed integer q corresponds to a specific customer type. The MC-RUM assumes that Q and $\mathbf{U}^+$ are jointly distributed.

To ensure choice probabilities are well-defined, we adopt the *Noncoincidence Assumption*. Specifically, we assume that the probability of any two distinct alternatives having identical utility values is zero, i.e., $\mathbb{P}(U_i = U_j) = 0$ for all $i, j \in \mathbb{N}^+$, $i \neq j$. This guarantees that each customer perceives different utilities for all alternatives, thereby eliminating the possibility of ties in preferences and ensuring no two options are equally likely to be chosen.

Under the MC-RUM, each arriving customer has a realization $(q, \mathbf{u}^+)$ of $(Q, \mathbf{U}^+)$, which determines their purchase decision. Given the Noncoincidence Assumption, the probability that a randomly arriving customer chooses set P when assortment V (with $V \supseteq P$) is offered is given by

$$\pi_{P,V} = \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k). \quad (2)$$

This probability consists of two parts: the first reflects customers who intend to purchase exactly $|P|$ products, choosing the offered ones with the highest utilities, which happen to be P . The second describes

customers planning to buy more than $|P|$ products but who ultimately choose only the products in P , as these are the only offered ones with utilities exceeding that of the outside option. Additionally, it can be verified that Equation (2) is a valid expression of choice probability. That is, for all non-empty $V \subseteq N$, $\pi_{P,V} \geq 0 \forall P \subseteq V$ and $\sum_{P \subseteq V} \pi_{P,V} = \sum_{q=0}^N \sum_{P \subseteq V, |P|=q} \mathbb{P}(Q = q, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > q, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k) = 1$. Similarly to the MC-RLM, the MC-RUM also satisfies the regularity axiom (see Section EC.1 for details). The proposed MC-RUM is thus an intuitive extension of the classic random utility model, maintaining the key property of regularity. Furthermore, building on previous discussions and established results in single-purchase settings (Block and Marschak 1960), we demonstrate the equivalence between the MC-RLM and MC-RUM.

THEOREM 1. *Under the Noncoincidence Assumption, a choice probability distribution π is derived from a multi-choice rank list model if and only if there exists a joint distribution over $(Q, \mathbf{U}^+)$ such that for all non-empty subsets $V \subseteq N$ and for all $P \subseteq V$,*

$$\pi_{P,V} = \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k).$$

Theorem 1 establishes that the proposed MC-RLM adheres to random utility theory. More importantly, it implies that for any MC-RUM, we can construct an equivalent MC-RLM, and vice versa. In what follows, we discuss two important special cases of both the MC-RUM and the MC-RLM.

3.2 Special Cases of the MC-RUM and MC-RLM

In single-purchase settings, the multinomial logit (MNL) model, a special case of the RUM, is widely used for its closed-form choice probabilities, and the independent demand model (IDM), a special case of the RC-RLM, is valued for its simplicity in practice. In this subsection, we discuss extensions of both models in multi-purchase settings and explore their relationship with the proposed MC-RUM and MC-RLM.

Extension of the Multinomial Logit Model to Multi-Purchase Settings

Recall from the classic single-purchase RUM, when the utility is defined as the sum of a deterministic component and an i.i.d. Gumbel-distributed random perturbation, this model becomes the MNL model (McFadden 1974). To extend the MNL model to accommodate multi-purchase scenarios, Bai et al. (2023) propose the *Multi-Purchase Multinomial Logit* (MP-MNL) model. The MP-MNL model makes a simplifying assumption that the intended purchase quantity, Q , and the utilities of the products, $\mathbf{U}^+$, are independent. This implies that the perceived utility of individual products does not change based on the number of products a customer intends to buy. However, unlike the classic MNL model, which has a straightforward expression for choice probabilities, the choice probability of each product under the MP-MNL model might need to be calculated recursively, and the number of recursions can be as large as $O(2^N \cdot N^2)$ if Q is arbitrarily distributed (Bai et al. 2023).

A more general extension of the MNL model to multi-purchase settings can be derived directly from the MC-RUM by allowing Q and $\mathbf{U}^+$ to be jointly distributed. This generalization permits interactions between the intended purchase quantity and the product utilities, thus offering a more flexible approach. Both the MP-MNL model and this generalized extension are special cases of the MC-RUM. By Theorem 1, it follows directly that these extensions are also special cases of the multi-choice rank list model. In fact, other single-purchase choice models within the random utility model class, such as the multinomial probit model, can be similarly extended and are thus encompassed by the MC-RUM and MC-RLM.

Modeling Independent Product Choices in Multi-Purchase Settings

In single-purchase settings, the IDM is widely used because of its simplicity and effectiveness in operational decision-making (van Ryzin 2005), despite not considering product substitution. The IDM assumes product utilities follow a discrete distribution with $\mathbb{P}(\mathbf{U} = \mathbf{e}_i) = p_i^{IDM}$, where $\mathbf{e}_i$ is the unit vector with the i -th entry set to 1 (Mahajan and Van Ryzin 2001b). This model can also be treated as a special case of the classic SC-RLM by assuming that there are N distinct preference rankings, each exclusively ranking one product over the outside option (van Ryzin and Vulcano 2017). Both explanations indicate that each customer is interested in only one product.

Notably, the IDM requires $\sum_{i=1}^N p_i^{IDM} \leq 1$, allowing for the possibility of selecting the outside option. However, this constraint is unsuitable for multi-purchase scenarios. Moreover, if customers independently decide whether to purchase each product, some will inevitably purchase multiple items. Motivated by this observation, we propose the following *Independent Choice Model* (ICM), which extends the IDM to account for independent choices in multi-purchase settings.

DEFINITION 1 (INDEPENDENT CHOICE MODEL). A model is called the independent choice model if probabilities $p_i \in [0, 1]$ exist for each $i \in N$ such that, for any non-empty offered set $V \subseteq N$ and for any subset $P \subseteq V$, the choice probability of P is given by: $\pi_{P,V}^{ICM} = \prod_{i \in P} p_i \prod_{j \in V \setminus P} (1 - p_j)$, and 0 otherwise.

It can be verified that the ICM is well-defined, as $\sum_{P \subseteq V} \pi_{P,V}^{ICM} = 1$ for any non-empty $V \subseteq N$ (see Proposition EC.2 for more details). In addition, we can demonstrate that p_n is indeed the marginal choice probability of product n . Specifically, for any non-empty assortment V that includes n , we have $\mathbb{P}(n, V) = \sum_{P \subseteq (V \setminus \{n\})} \pi_{P \cup \{n\}, V}^{ICM} = p_n \sum_{P \subseteq (V \setminus \{n\})} \left(\prod_{i \in P} p_i \prod_{j \in (V \setminus \{n\}) \setminus P} (1 - p_j) \right) = p_n \cdot 1 = p_n$. This re-illustrates that the choice of a product is independent of all other products and is not affected by the assortment offered. Furthermore, we can show that the ICM is a special case of the MC-RUM.

PROPOSITION 2. A multi-choice random utility model reduces to an independent choice model when $\mathbb{P}(Q = N) = 1$ and all utilities U_n are independent of each other, with $U_0 = u_0$ being deterministic.

The intuition behind this finding is clear: since customers have no purchasing limits, they will choose product n if and only if $U_n \geq U_0$. Additionally, the choices for each product are made independently because the differences $U_n - U_0$ are independent among different products. By Theorem 1, we conclude that the ICM is also a special case of the MC-RLM. Bai et al. (2023) show that when the intended purchase quantity is sufficiently large, product choices under the MP-MNL model become independent of the assortment. However, this condition alone does not reduce the MP-MNL model to an ICM (see Claim EC.1 for reference); an additional assumption of a fixed outside-option utility is required.

So far, we have introduced the MC-RLM and MC-RUM, both grounded in random utility theory, satisfying the regularity property and accounting for product substitution effects. While the MC-RUM naturally models multiple purchases, its estimation is challenging, and even for the MP-MNL model, closed-form choice probabilities of the MNL model do not extend easily to the multi-purchase setting. In contrast, the independent choice model, though straightforward, ignores interrelationships among products. In the following sections, we show how the MC-RLM can be estimated with adjustments from single-purchase algorithms, without enumerating 2^N subsets or $N!$ rankings. We also explain how the preference ranking structure uses customer data to reflect relationships between alternatives, which enhances the model’s predictive power and is difficult to achieve with other models.

4 Model Estimation Using E-Commerce Behavior Data

In this section, we present an algorithm for estimating the MC-RLM that incorporates customer browsing behavior in e-commerce, beyond just transaction data. The algorithm extends the *Market Discovery* (MD) algorithm by van Ryzin and Vulcano (2015) for single-choice rank list models and introduces the *Behavior-Reveal-Preference* (BRP) rule, which can be applied to any ranking-based choice model.

4.1 Behavior-Reveal-Preference Rules

For e-commerce retailers, the vast amount of browsing and clickstream data offers valuable insights into consumer preferences, enabling more accurate demand predictions. When shopping online, customers typically engage in four key behaviors: *viewing*³, *clicking*⁴, *adding to the shopping cart*, and *purchasing*. To model the customer’s decision-making process during online shopping, we make the following assumption:

ASSUMPTION 1. *Customers generally follow this sequence when making a purchase:*

Viewing $\Rightarrow$ Clicking $\Rightarrow$ Adding to the Shopping Cart $\Rightarrow$ Purchasing.

This assumption captures the core online shopping process. While some platforms allow direct purchases without adding items to the shopping cart, we consider the *purchasing* step to include this action. Similarly,

³ *Viewing* refers to products being displayed to customers, recorded once a product appears on screen.

⁴ *Clicking* refers to customers interacting with a product by clicking on it for more detailed information.

if a platform permits adding items to the cart without viewing detailed information, we assume the *adding to the shopping cart* step subsumes the *clicking* action.

Consider an e-commerce platform offering N substitutable products within the same category, denoted by $N = \{1, 2, \dots, N\}$, all of which are accessible. Customers typically review only a subset of N , as they are unlikely to browse through many pages, especially on mobile devices. Products that appear on-screen are more likely to capture customer attention, and customers cannot purchase a product they have not viewed, consistent with Assumption 1. Therefore, rather than considering the entire product set, the products viewed by a customer, denoted by V , act as a proxy for the offer set. Additionally, we define K , C , and P as the sets of products clicked on, added to the shopping cart, and purchased, respectively. These sets can be empty, indicating that a customer may leave without making a purchase, i.e., choosing the outside option 0.

While only the purchasing action directly reflects customer choices, side information such as clicking and adding items to the shopping cart, combined with sales data, can provide valuable insights into customer preferences for products they have viewed (see, e.g., Farias and Li 2019). Before making a purchase, customers typically consider which viewed products to add to the cart, then finalize their decision among these selected items. Building on this, we introduce the concept of *Behavior-Reveal-Preference* (BRP) rules, which describe how customer preferences are expressed through different product interactions.

DEFINITION 2. [Behavior-Reveal-Preference Rules] For a customer, interactions with products reflect preferences according to the following rules:

- (R1) Products that are clicked are preferred over those that are viewed but not clicked.
- (R2) Products that are added to the shopping cart are preferred over those that are clicked but not added.
- (R3) Products that are purchased are preferred over those that are added to the shopping cart but not purchased, as well as to the outside option.

The rationale behind R2 and R3 is clear: customers typically do not engage with products they find unappealing. As for R1, clicking on a product indicates the customer's intent to explore further, while products that are not clicked may either lack appeal or already be familiar to the customer. These rules are valid when analyzing historical customer data for model estimation, as the purchase process is typically initiated only after a product is clicked, according to Assumption 1. Moreover, it is important to note that other customer behaviors, such as *adding to a collection* and *removing from the shopping cart*, also provide insights into preferences and could be integrated into this framework through additional rules, advised by expert insights or business context if needed. However, in this study, we focus on the four primary behaviors and the three established BRP rules for demand estimation.

Intuitively, integrating customer browsing behavior can enhance demand estimation. However, incorporating BRP rules poses challenges for most choice models. For instance, utility-based models must determine deterministic product utilities from browsing data while complying with BRP rules. To the best of

our knowledge, no existing methods have achieved this. In contrast, non-parametric models, such as rank list models, can more easily accommodate these rules. Specifically, for a customer of type (q, σ) who has viewed $V \neq \emptyset$, clicked on $K \subseteq V$, added $C \subseteq K$ to the shopping cart, and purchased $P \subseteq C$, the following conditions must be satisfied according to the BRP rules: i) $q \geq |P|$; ii) $\sigma(p) < \sigma(c)$, $\forall p \in P$, $\forall c \in C^+ \setminus P$; iii) $\sigma(c) < \sigma(k)$, $\forall c \in C$, $\forall k \in K \setminus C$; iv) $\sigma(k) < \sigma(v)$, $\forall k \in K$, $\forall v \in V \setminus K$; v) if $q > |P|$, $\sigma(v) > \sigma(0) > q$, $\forall v \in V \setminus P$. We would like to remark that condition (v) indicates that for customer types with $q > |P|$, the products viewed but not purchased must rank after the outside option; otherwise, additional products would have been chosen. Furthermore, for these customer types, the outside option must explicitly rank after the q -th position to avoid a contradiction with the intended purchase quantity q . This requirement is naturally satisfied by condition (ii) for customer types with $q = |P|$.

Unlike the consider-then-choose framework by Aouad et al. (2019), which assumes the consideration set includes all click-based behaviors, we utilize clickstream data in a more detailed way through BRP rules. These rules reflect a progressively increasing preference pattern from viewing to clicking, adding to the cart, and ultimately purchasing. As customers interact with the platform, their preferences are incrementally revealed. In the following section, we discuss how to incorporate the BRP rules into MC-RLM estimation.

4.2 Estimation of the Multi-Choice Rank List Model in the E-Commerce Setting

Suppose the true MC-RLM for the entire population includes all possible distinct customer types, which can total up to $O(q_{\max} \cdot N \cdot N!)$, where $q_{\max} \leq N$ denotes the maximum intended purchase quantity. For invalid customer types, the corresponding $\lambda = 0$. To simplify our analysis, we establish a bidirectional mapping that assigns each pair (q, σ) to a non-negative integer m (any such mapping is valid), with $(0, [0])$ designated as type $m = 0$. We denote type m as (q_m, σ_m) , with its appearance probability given by $\lambda_{q_m, \sigma_m} = \lambda_m$.

Suppose there are T recorded customers, indexed by $1, 2, \dots, T$, each of whom has viewed at least one of the N products—that is, no view set is empty. Without additional information, the choice probabilities are given by Equation (1). In e-commerce settings, where customer behavior histories are available, we can integrate the BRP rules from Definition 2 into the model estimation. Let $T_{P,C,K,V}$ represent the collection of customer types that can explain the tuple (P, C, K, V) . For any valid tuple (P, C, K, V) , we define:

$$T_{P,C,K,V} = \left\{ (q, \sigma) \in T : \begin{array}{ll} q \geq |P|, & \\ \sigma(p) < \sigma(c) & \forall p \in P, \forall c \in C^+ \setminus P, \\ \sigma(c) < \sigma(k) & \forall c \in C, \forall k \in K \setminus C, \\ \sigma(k) < \sigma(v) & \forall k \in K, \forall v \in V \setminus K, \\ \sigma(v) > \sigma(0) > q & \forall v \in V \setminus P, \text{ if } q > |P| \end{array} \right\}.$$

Then, the probability that customer t clicks on K_t , adds C_t to the shopping cart, and purchases P_t given that they have viewed V_t , denoted as $\mathbb{P}(P_t, C_t, K_t | V_t, \boldsymbol{\lambda})$, is given by: $\mathbb{P}(P_t, C_t, K_t | V_t, \boldsymbol{\lambda}) = \sum_{(q, \sigma) \in T_{P_t, C_t, K_t, V_t}} \lambda_{q, \sigma}$.

The estimation of the MC-RLM from behavioral data can be formulated as a maximum likelihood problem, with the log-likelihood function: $LL(\boldsymbol{\lambda}) = \sum_{t=1}^T \ln(\mathbb{P}(P_t, C_t, K_t | V_t, \boldsymbol{\lambda})) = \sum_{t=1}^T \ln\left(\sum_{(q, \sigma) \in T_{P_t, C_t, K_t, V_t}} \lambda_{(q, \sigma)}\right)$. The corresponding maximum log-likelihood estimation (MLLE) problem is given by:

$$(\text{MLLE}) : \max_{\lambda \geq 0} LL(\lambda) \\ \text{s.t. } \sum_{(q,\sigma) \in T} \lambda_{q,\sigma} = 1.$$

The goal is to find an optimal solution λ^* to Problem (MLLE). As discussed, the number of possible customer types becomes too large to solve it directly when N is large. This challenge is similar to that faced by the SC-RLM. A state-of-the-art method for estimating SC-RLMs is the *Market Discovery* (MD) algorithm proposed by van Ryzin and Vulcano (2015). For the MC-RLM, we propose the *Multi-Choice Market Discovery* (MCMD) algorithm, which extends the MD algorithm to handle the multi-choice rank list model and incorporate BRP rules. Indeed, the essence of the MCMD algorithm is similar to the MD algorithm, as both iteratively estimate customer types and arrival rates from historical data based on a maximum likelihood approach, rather than directly learning λ all at once. The estimated model remains sparse in customer types while still accurately capturing customer choice behaviors.

We now introduce the MCMD algorithm. Detailed descriptions are provided in Section EC.2, and the corresponding pseudo-code provided in Algorithm 1⁵. The algorithm follows these key parts: (a) Start with an initial set of customer types, assuming excluded types have zero probability of arrival. (b) In decreasing order of intended purchase quantity, identify new customer types that improve the model's likelihood based on historical data. (c) Output the estimated multi-choice rank list model, including only types with positive estimated arrival probabilities, until no new types are added or a stopping criterion is met.

As discussed, directly solving Problem (MLLE) is computationally infeasible, so we assume that only a subset of customer types has positive arrival probabilities and solve a restricted version of (MLLE) over this subset. The initial subset is constructed in part (a) and is iteratively expanded in step (b) (corresponding to Steps 6–20 in Algorithm 1) to improve model fit. The criterion for adding a new type is derived via duality analysis: if a binary vector $\omega \in \{0, 1\}^T$ can be found by solving Problem (EC.10) such that $\sum_{t \in [T]} \tilde{\mu}_t \omega_t > T$, then introducing the corresponding type improves the log-likelihood. Here, $\omega_t = 1$ indicates that the new type explains K_t, C_t, P_t given V_t . BRP rules play a crucial role in this process. Rather than identifying types that merely explain the transaction (P, V) , BRP rules guide the discovery of types that align with the full behavioral pattern (P, C, K, V) . In other words, they prevent the discovery of preference rankings that are consistent with the view–purchase pair but contradict the broader behavioral history. Another key advantage of BRP rules is that they render the entire preference ranking informative, not just the part prior to the outside option. Without BRP rules, transaction data reveal only the top-ranked available product prior to the outside option. By incorporating BRP rules, each stage of customer interaction—viewing, clicking, and adding to the cart—provides valuable signals, thereby enhancing prediction accuracy and personalized assortment recommendations. Similarly, the construction of the matrix $\tilde{A}$ also incorporates BRP rules.

⁵ In Algorithm 1, $\tilde{\cdot}$ denotes a temporary parameter, and $\hat{\cdot}$ indicates an estimated output.

$$\begin{aligned}
& \max_{\omega} \sum_{t \in [T]} \tilde{\mu}_t \omega_t \\
& \text{s.t. } \omega_t = 0 \quad \text{for all } t \in [T] \text{ with } |P_t| > q, \\
& \omega_t \leq \mathbb{I} \left(\begin{array}{l} \sigma(0) < \min_{v \in V_t} \sigma(v) \\ \max_{c \in C_t} \sigma(c) < \min_{k \in \{K_t \setminus C_t\}} \sigma(k) \\ \max_{k \in K_t} \sigma(k) < \min_{v \in \{V_t \setminus K_t\}} \sigma(v) \\ \min_{v \in V_t^+} \sigma(v) > q \end{array} \right) \quad \text{for all } t \in [T] \text{ with } P_t = \emptyset, \\
& \omega_t \leq \mathbb{I} \left(\begin{array}{l} \max_{p \in P_t} \sigma(p) < \min_{c \in \{C_t^+ \setminus P_t\}} \sigma(c) \\ \max_{c \in C_t} \sigma(c) < \min_{k \in \{K_t \setminus C_t\}} \sigma(k) \\ \max_{k \in K_t} \sigma(k) < \min_{v \in \{V_t \setminus K_t\}} \sigma(v) \\ \min_{v \in \{V_t^+ \setminus P_t\}} \sigma(v) > q \end{array} \right) \quad \text{for all } t \in [T] \text{ with } P_t \neq \emptyset, |P_t| = q, \\
& \omega_t \leq \mathbb{I} \left(\begin{array}{l} \sigma(0) < \min_{v \in \{V_t \setminus P_t\}} \sigma(v) \\ \max_{p \in P_t} \sigma(p) < \min_{c \in \{C_t^+ \setminus P_t\}} \sigma(c) \\ \max_{c \in C_t} \sigma(c) < \min_{k \in \{K_t \setminus C_t\}} \sigma(k) \\ \max_{k \in K_t} \sigma(k) < \min_{v \in \{V_t \setminus K_t\}} \sigma(v) \\ \min_{v \in \{V_t^+ \setminus P_t\}} \sigma(v) > q \end{array} \right) \quad \text{for all } t \in [T] \text{ with } P_t \neq \emptyset, |P_t| < q.
\end{aligned} \tag{EC.10}$$

However, identifying an improving type directly from Problem (EC.10) is nontrivial. To address this, we introduce binary variables $\mathbf{x} \in \{0, 1\}^{(N+1) \times (N+1)}$ to encode pairwise preference relationships, where $x_{ij} = 1$ if product i is preferred over product j , and $x_{ij} = 0$ otherwise. These variables enable the nonlinear indicator constraints to be reformulated as linear constraints (see Problem (EC.11), the MILP reformulation of Problem (EC.10)). Any feasible solution yields an improving customer type, with the corresponding preference ranking recovered via $\sigma(i) = \sum_{j \in \mathbb{N}^+, j \neq i} x_{ji} + 1$ for all $i \in \mathbb{N}^+$ (Step 17).

As will be shown in the case study using a real-world e-commerce dataset, the proposed algorithm takes an average of 18.94 seconds to estimate the MC-RLM using full behavioral data, yielding models with an average of around 6 customer types for categories with fewer than 60 products (see Table EC.16 for details). This highlights the tractability of the proposed algorithm in model estimation.

The MCMD algorithm is a natural extension of the MD algorithm. Unlike other multi-purchase choice models, such as the MP-MNL model, which significantly complicate estimation compared to the classic MNL model, our approach demonstrates that existing methods for single-purchase settings can be effectively extended to multi-purchase settings for rank list models with minimal modifications. The key modification involves designing a new MILP reformulation for Problem (EC.10) to handle multi-purchase cases and incorporate BRP rules. Additionally, a trick in Algorithm 1 is discovering new types in descending order of IPQ. Customer types with larger q can explain transactions with smaller quantities (see Example EC.2), leading to a sparser estimated model and faster downstream solving for assortment optimization. It is worth noting that non-identifiability is a fundamental issue in ranking-based models, where different ranking structures can explain the same choice data, making it impossible to uniquely determine the true preference rankings. This occurs because various sets of preference rankings can yield identical predictions

for customer choices, even with abundant data for estimation. Our study offers a practical method for estimating the MC-RLM from historical data, although it does not directly resolve the non-identifiability issue. One promising direction to address this challenge is using directed acyclic graph (DAG) representations to model dependencies between preference rankings, potentially narrowing down the set of plausible ranking structures. While DAG methods have been explored in single-purchase contexts (e.g., Susan et al. 2022), extending them to multi-purchase scenarios remains an open area for future research. Incorporating DAG frameworks in these settings could significantly enhance the identifiability of the MC-RLM.

5 Product Assortment Recommendation in E-Commerce

Thus far, we have discussed the estimation of the multi-choice rank list model using historical data. In this section, we address the challenge of product assortment recommendation and personalization in e-commerce, where demand follows a multi-choice rank list model.

In online retail, product assortment recommendations are typically displayed across various sections of the platform, including search result pages, homepage advertisements, and promotional highlights. Personalized recommendations are also integrated as customers browse products or proceed to checkout. Along with Figures 1 and 2 from the introduction, additional examples of such recommendations are shown in Figures 3 to 5. Figure 3 shows the search page for sticky notes on the Cainiao app, where products from this category can be displayed to customers. Figure 4 illustrates a cosmetics store on the Taobao app, recommending six related products when a lipstick is viewed. Figure 5 shows a recommendation on the Amazon shopping cart page, suggesting additional products from the same category as the item in the cart. The problems discussed in this section are motivated by these real-world examples.



Figure 3 Search Page of Sticky Notes on Cainiao

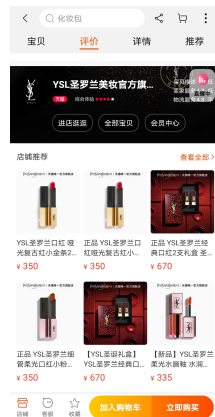


Figure 4 Lipsticks Recommendation on Taobao

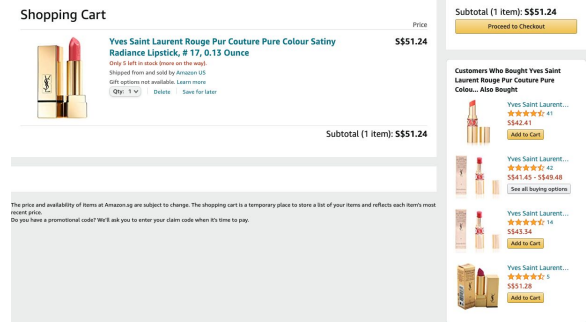


Figure 5 Lipstick Recommendation on the Shopping Cart Page on Amazon

We begin by discussing the complexity of the assortment recommendation problem under the multi-choice rank list model and evaluating the performance of the widely used revenue-ordered assortment

Algorithm 1: Multi-Choice Market Discovery Algorithm Incorporating BRP Rules**Input :** $\mathbb{N} = \{1, 2, \dots, N\}$, $(P_t, C_t, K_t, V_t)_{t \in [T]}$ // $V_t \neq \emptyset$ for all $t \in [T]$ **Output:** $\hat{T} = \{(0, [0]), (q_1, \sigma_1), (q_2, \sigma_2), \dots, (q_{\hat{M}}, \sigma_{\hat{M}})\}$, $\hat{\lambda} = (\lambda_0, \lambda_1, \lambda_2, \dots, \lambda_{\hat{M}})^\top$

```

1  $q_{\max} \leftarrow \max \{|P_1|, |P_2|, \dots, |P_T|\}$  ;
2 Construct the initial set of customer types,  $T_0$ , with  $(q_0, \sigma_0) = (0, [0])$ , using Algorithm 2 provided in
   Section EC.2.1 ;
3  $\tilde{T} \leftarrow T_0$ ,  $\tilde{M} \leftarrow |\tilde{T}| - 1$ , let  $\tilde{\lambda} \in \mathbb{R}^{|\tilde{T}|}$  represent the probability mass function regarding  $\tilde{T}$  ;
4 Construct  $\tilde{A} \in \{0, 1\}^{T \times (\tilde{M}+1)}$  with  $a_{t,1+m} = 1$  if  $(q_m, \sigma_m) \in \tilde{T}_{P_t, C_t, K_t, V_t}$  and 0 otherwise, introduce  $\tilde{\gamma} \in \mathbb{R}^T$ 
   with  $\tilde{A}\tilde{\lambda} = \tilde{\gamma}$  ;
5 Given  $\tilde{T}$ , calculate the optimal  $\tilde{\lambda}$  and  $\tilde{\mu}$  (the dual variable of  $\tilde{\gamma}$ ) by solving Problem (EC.8) and its
   dual problem ;
6 for  $q = q_{\max}$  down to 1 do
7   while true do
8     Given  $\tilde{\mu}$ , solve Problem (EC.11) ;
9     if Problem (EC.11) has a feasible solution then
10      | Get Problem (EC.11)'s optimal value, denoted  $\Gamma^*$ , and the optimal solution  $x_{ij} \forall i, j \in \mathbb{N}^+$ ;
11    else
12      | break ;
13    end
14    if  $\Gamma^* \leq T$  then
15      | break ;
16    end
17    Updated  $\tilde{T}$  by adding a new type  $(q_{\tilde{M}+1}, \sigma_{\tilde{M}+1})$  with  $q_{\tilde{M}+1} = q$  and  $\sigma_{\tilde{M}+1}(n) = \sum_{j \in \mathbb{N}^+, j \neq n} x_{jn} + 1$ 
      for all  $n \in \mathbb{N}^+$  to  $\tilde{T}$  ;
18     $\tilde{M} \leftarrow \tilde{M} + 1$  ;
19    Update  $\tilde{A} \in \{0, 1\}^{T \times (\tilde{M}+1)}$  with  $a_{t,1+m} = 1$  if  $(q_m, \sigma_m) \in \tilde{T}_{P_t, C_t, K_t, V_t}$  and 0 otherwise ;
20    Given  $\tilde{T}$ , calculate the optimal  $\tilde{\lambda}$  and  $\tilde{\mu}$  by solving Problem (EC.8) and its dual problem,
      respectively ;
21   end
22 end
23  $\hat{M} \leftarrow \tilde{M}$ ,  $\hat{T} \leftarrow \tilde{T}$ ,  $\hat{\lambda} \leftarrow \tilde{\lambda}$ .

```

heuristic. We then introduce a mixed-integer programming formulation to solve the problem optimally. Finally, we explore how online retailers can leverage customers' browsing history to recommend personalized assortments.

5.1 Assortment Recommendation under the Multi-Choice Rank List Model

In this subsection, we focus on assortment optimization for the entire customer population. This involves decisions such as determining which products to display for new customers, as shown in Figure 3, or selecting products to feature in a promotional campaign, as illustrated in Figure 1. We adopt the same setting as Section 4, where an e-commerce retailer offers N substitutable products with profit rates $r_1, \dots, r_N$. The goal of assortment recommendation is to choose a set of products for all customers to view. Customers interact with these items based on their preferences, but only the purchased products contribute to revenue.

Assume demand follows an MC-RLM with $T = \{(0, [0]), (q_1, \sigma_1), (q_2, \sigma_2), \dots, (q_M, \sigma_M)\}$ and λ_m representing the arrival probability for customer type m . Here, we consider only types with positive support. Let $S \subseteq N$ denote the product assortment the retailer intends to offer. Our objective is to identify the optimal assortment that maximizes expected profit, subject to business constraints defined by the feasible assortment set Ψ . The expected revenue of an assortment $S \subseteq N$ is given by $R(S) = \sum_{n \in S} r_n \mathbb{P}(n, S)$. Thus, the assortment recommendation problem, aimed at maximizing expected revenue and referred to as (GAR), can be formulated as follows:

$$\begin{aligned} \text{(GAR)} : \max_{\substack{S \subseteq N \\ \text{s.t. } S \in \Psi}} \sum_{n \in S} r_n \mathbb{P}(n, S) \end{aligned}$$

Note from Kohli and Krishnamurti (1989) that the unconstrained assortment optimization problem under the classic single-choice rank list model is already NP-hard. Hence, Problem (GAR) is NP-hard to solve.

PROPOSITION 3 (Theorem 1 in Kohli and Krishnamurti 1989). *The unconstrained assortment recommendation problem under the rank list model is NP-hard to solve, even if all customers intend to purchase at most one product and $r_n = 1$ for all $n \in N$.*

Furthermore, Aouad et al. (2018) shows that Problem (GAR) is NP-hard to approximate within $O(N^{1-\epsilon})$ for any fixed $\epsilon > 0$, even if $q_m = 1$ for all $m \in [M]$.

5.1.1 Revenue-Ordered Assortments in Multi-Purchase Settings

Given the NP-hardness of Problem (GAR), we examine the commonly used revenue-ordered assortments strategy in unconstrained assortment optimization. For simplicity, we assume that products are indexed in decreasing order of profit, i.e., $r_1 \geq r_2 \geq \dots \geq r_N$. Revenue-ordered assortments are thus defined as progressively inclusive sets: $\{r_1\}, \{r_1, r_2\}, \dots, \{r_1, r_2, \dots, r_N\}$, and the strategy selects the set that maximizes expected revenue. This simple heuristic has been effective across various choice models and is known to be optimal when demand follows the MNL model (Talluri and Van Ryzin 2004).

For ease of our analysis, we let $r_{\max} = \max\{r_1, r_2, \dots, r_N\}$, $r_{\min} = \min\{r_1, r_2, \dots, r_N\}$, $q_{\max} = \max\{q_1, q_2, \dots, q_M\}$. In the literature, Bai et al. (2023) shows that the best revenue-ordered assortment approximates the optimal expected revenue under the multi-purchase multinomial choice model within a

factor of $O(\ln(N))$ and Aouad et al. (2018) demonstrates that the best revenue-ordered assortment approximates the optimal expected revenue under the classic single-choice rank list model within a factor of $O\left(\lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right)$. Next, we analyze the performance of the revenue-ordered assortment strategy when the underlying demand follows the MC-RLM. We first introduce a lemma establishing a lower bound for a specific revenue-ordered assortment.

LEMMA 1. *For every customer of type $m \in [M]$, let h_m denote the highest profit within their interest list $\mathcal{I}_{(q_m, \sigma_m)}$. Let $\mathcal{S}_{h_m}$ be the revenue-ordered assortment with products whose profits are higher than or equal to h_m , and let $R(\mathcal{S}_{h_m})$ be the expected revenue of offering assortment $\mathcal{S}_{h_m}$ under an MC-RLM. Then, we have $\sum_{m=1}^M \lambda_m h_m \leq \left(\frac{5}{2} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right) \cdot R(\mathcal{S}_{h_{m^*}})$, where $m^* := \arg \max_{m \in [M]} h_m \sum_{i=1}^m \lambda_i$ (selecting an arbitrary one in cases where the maximum value is achieved by multiple customer types $m \in [M]$).*

Indeed, Lemma 1 establishes a lower bound for the expected revenue of a specific revenue-ordered assortment under the multi-choice rank list model. It is worth noting that the proof of this lemma resembles the approach used to prove Theorem 3 in Aouad et al. (2018). This similarity stems from the fact that the multi-choice rank list model is an extension of the classic single-choice rank list model. Consequently, we can always lower bound the expected revenue by assuming that the customer will purchase no more than one product, even if they intend to buy multiple items.

Leveraging Lemma 1, we can establish the following performance guarantee for the optimal revenue-ordered assortment to any unconstrained assortment optimization problem under the MC-RLM.

THEOREM 2. *Consider an unconstrained assortment optimization problem under the multi-purchase rank list model. The expected revenue achieved by the best revenue-ordered assortment approximates the optimal expected revenue within a factor of $O\left(q_{\max} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right)$.*

This result implies that the ratio between the optimal expected revenue R^* and the expected revenue of the best revenue-ordered assortment R_{RO}^* satisfies $\frac{R^*}{R_{\text{RO}}^*} \leq O\left(q_{\max} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right)$. Moreover, we denote $\lambda_{r_{\max}}$ as the aggregate appearance probability of all customer types whose interest list includes the product yielding the highest profit, $r_{\max}$. Then, we have the following corollary of Theorem 2.

COROLLARY 1. *The optimal expected revenue in the unconstrained assortment optimization problem under the multi-choice rank list model can be approximated within a factor of $O\left(q_{\max} \cdot \lceil \ln\left(\frac{1}{\lambda_{r_{\max}}}\right) \rceil\right)$.*

In summary, the widely used revenue-ordered assortment strategy provides an efficient and reliable solution with an acceptable performance guarantee for the unconstrained problem. While capacity constraints may apply in certain scenarios, such as in Figure 4, the flexibility of e-commerce platforms allows for display adjustments, such as resizing or adding a scroll bar, to accommodate more products. Therefore, this heuristic remains applicable when needed. However, it does not directly solve Problem (GAR). In the following section, we propose a tractable method to solve Problem (GAR) optimally.

5.1.2 Mixed-Integer Linear Programming Formulation for Assortment Recommendation

To solve Problem (GAR) exactly, we can reformulate it as a mixed-integer linear program. We define binary variables ξ_n , where $\xi_n = 1$ if product n is offered and 0 otherwise, and variables $z_{k,n}^{(m)}$, where $z_{k,n}^{(m)} = 1$ if product n is the k -th choice of a customer of type m and 0 otherwise. Suppose the feasible region has a linear structure represented as $\Psi = \{\xi \mid \mathbf{B}\xi \leq \mathbf{c}\}$ for some $\mathbf{B} \in \mathbb{R}^{b \times N}$ and $\mathbf{c} \in \mathbb{R}^b$, with b being a positive integer. If there are no specific requirements, $\mathbf{B}$ is set to a zero matrix and $\mathbf{c}$ to a zero vector. Then, (GAR) can be reformulated as follows:

$$\max_{\xi, z} \sum_{m=1}^M \sum_{n=1}^N \sum_{k=1}^{q_m} r_n \cdot \lambda_m \cdot z_{k,n}^{(m)}$$

$$\text{s.t.} \sum_{n=0}^N z_{k,n}^{(m)} = 1 \quad \forall m \in [M], k \in [q_m], \quad (3a)$$

$$\sum_{k=1}^{q_m} z_{k,n}^{(m)} \leq \xi_n \quad \forall m \in [M], n \in [N], \quad (3b)$$

$$\sum_{e: \sigma_m(e) > \sigma_m(n)} z_{1,e}^{(m)} \leq 1 - \xi_n \quad \forall m \in [M], n \in [N], \quad (3c)$$

$$\sum_{e: \sigma_m(e) > \sigma_m(n)} z_{k,e}^{(m)} \leq 1 - \xi_n + \sum_{l=1}^{k-1} z_{l,n}^{(m)} \quad \forall m \in [M], n \in [N], k \in [q_m] \setminus \{1\}, \quad (3d)$$

$$\sum_{e: \sigma_m(e) > \sigma_m(0)} z_{k,e}^{(m)} = 0 \quad \forall m \in [M], k \in [q_m], \quad (3e)$$

$$\mathbf{B}\xi \leq \mathbf{c}, \quad (3f)$$

$$\xi_n \in \{0, 1\} \quad \forall n \in [N], \quad (3g)$$

$$z_{k,n}^{(m)} \geq 0 \quad \forall m \in [M], i \in \{0\} \cup [N], k \in [q_m]. \quad (3h)$$

To help understand, we offer the following remarks on the MILP formulation (3): (a) Constraint (3a) ensures that customers of type (q_m, σ_m) select exactly one product for each of their first q_m choices; (b) Constraint (3b) ensures that a customer can pick product n only if it is offered and that each product can be picked at most once; (c) Constraint (3c) ensures that customers of type (q_m, σ_m) cannot select products less preferred than product n (according to σ_m) as their first choice if product n is offered; (d) Constraint (3d) ensures that a customer's k -th choice ($k \geq 2$) is less preferred than their $(k-1)$ -th choice. There are three cases: (i) if product n is not offered, less preferred products can be chosen; (ii) if product n is offered and previously picked, less preferred products can still be chosen; (iii) if product n is offered but not previously picked, less preferred products will not be chosen; (e) Constraint (3e) ensures that customers of type (q_m, σ_m) cannot pick products less preferred than the outside option according to σ_m ; (f) this MILP formulation does not require $z_{k,n}^{(m)}$ to be binary because the first five constraints ensure it for a given ξ . Note that although Constraints (3c) and (3d) appear similar, they cannot be combined. This is because, for a customer's first choice,

they select the highest-ranked product, whereas for subsequent choices, previously selected products must be excluded, requiring separate constraint formulations. Indeed, formulation (3) extends the one proposed by Bertsimas and Mišić (2019) for the product line design problem under the classic single-choice rank list model. Hence, if $q_m = 1$ for all $m \in [M]$, our formulation reduces to that of Bertsimas and Mišić (2019).

Note that Problem (3) involves $O(q_{\max}MN)$ constraints and $O(q_{\max}MN)$ variables, of which only N are binary. In practice, the models estimated by the MCMD algorithm are typically sparse. For example, in a real-world case study from an e-commerce company (presented later in Section 6.1), the estimated number of customer types does not exceed 10 across all tested categories, each involving fewer than 60 alternatives. This sparsity enables direct solution to optimality within acceptable computation time. Modern MILP solvers, such as Gurobi, are capable of solving moderately large instances efficiently. To assess the computational time of the MILP more rigorously, we conduct extensive numerical experiments, as detailed in Section EC.4. Specifically, we observe that solving an instance with 60 products, 10 customer types, and $q_{\max} = 3$ takes, on average, 0.32 seconds.

5.2 Personalized Assortment Recommendation

Online platforms nowadays have more and more opportunities to recommend personalized products to customers. We focus on four key scenarios where personalized recommendations can be made. First, when a customer searches for a product or category, a personalized assortment can be recommended (e.g., Figure 3). Second, when a customer clicks on products for more details, the platform can suggest products the customer has not viewed but may find interesting (e.g., Figure 4). Third, before checkout, the platform can recommend products the customer did not view or click (e.g., Figure 5). Fourth, after completing a purchase, the platform can suggest related products (e.g., Figure 2).

In the literature, personalized assortment recommendations have primarily relied on customers' demographic features (e.g., Cheung and Simchi-Levi 2017). While demographic and geographic data can provide insights into preferences, such information is often unavailable for new users or those not logged in, and growing privacy concerns limit its use. In response to these challenges, we propose a personalized recommendation framework based solely on customers' real-time browsing behavior.

When a customer views V , clicks K , adds C to the cart, and purchases P , their preferences are revealed, helping to infer their customer type. By observing these behaviors, we can narrow down the possible customer types to those consistent with the observed V , K , C , and P in accordance with BRP rules. Specifically,

- if $V \neq \emptyset, K \neq \emptyset$ are revealed, the customer must belong to:

$$T_{V,K}'' = \{(q, \sigma) \in T : q > 0, \sigma(k) < \sigma(v) \forall k \in K, v \in V \setminus K\};$$

- if $V \neq \emptyset, K \neq \emptyset, C \neq \emptyset$ are revealed, the customer must belong to:

$$T''_{V,K,C} = \left\{ (q, \sigma) \in T : \begin{array}{l} q > 0, \\ \sigma(c) < \sigma(k) \quad \forall c \in C, k \in K \setminus C, \\ \sigma(k) < \sigma(v) \quad \forall k \in K, v \in V \setminus K \end{array} \right\};$$

- if $V \neq \emptyset, K \neq \emptyset, C \neq \emptyset, P \neq \emptyset$ are revealed, the customer would belong to:

$$T''_{V,K,C,P} = \left\{ (q, \sigma) \in T : \begin{array}{l} q > |P|, \\ \sigma(p) < \sigma(c) \quad \forall p \in P, c \in C^+ \setminus P, \\ \sigma(c) < \sigma(k) \quad \forall c \in C, k \in K \setminus C, \\ \sigma(k) < \sigma(v) \quad \forall k \in K, v \in V \setminus K, \\ \sigma(v) > \sigma(0) > q \quad \forall v \in V \setminus P \end{array} \right\}.$$

Note that in the last scenario, customer types with $q = |P|$ are excluded, as such customers are not likely to consider purchasing additional products. Rather than addressing the preferences of the entire population, we can update the set of customer types T and their probabilities λ with T'' (such as $T''_{V,K}$, $T''_{V,K,C}$, or $T''_{V,K,C,P}$) and the corresponding conditional probabilities λ'' . This allows us to derive the optimal personalized assortment for specific customers by solving Problem (3) with the updated model. If a new customer does not match any type in the estimated model, the general assortment derived from the (GAR) problem can be recommended due to insufficient historical data. The optimal assortment can be recommended at each stage of the customer's shopping journey. To exclude previously viewed or clicked products, we can set $\xi_n = 0$ for those products, as incorporated into Constraint (3f).

As mentioned at the end of Section 5.1.2, MILP (3) can be solved within a second in many practical cases, owing to the sparsity of the estimated model. When browsing history is updated, the set of valid customer types can often be further reduced through BRP rules, making personalized recommendations more tractable in online settings. In instances where customer types cannot be sufficiently narrowed or the number of products is particularly large, the best revenue-ordered assortment serves as a computationally efficient and reliable substitute, with a provable performance guarantee. Note that the heuristic may select different assortments across instances, as it depends on the specific customer types and arrival probabilities.

Our assortment personalization strategy differs from existing approaches. Prior work in static settings typically relies on pre-specified choice models or structural assumptions. For example, El Housni and Topaloglu (2023) propose a mixture-of-MNL framework, and Xie and Wang (2020) incorporate network effects into the classic MNL model. Other studies focus on dynamic personalization via online model updates, including dynamic clustering (Bernstein et al. 2019), contextual bandits with low-rank utility structures (Kallus and Udell 2020), and high-dimensional online learning using random projection and convex optimization (Miao and Chao 2022). Chen et al. (2024) further addresses online recommendation under inventory constraints. Many of these methods also leverage demographic or contextual features to estimate segment-specific preferences. In contrast, our approach adopts a behavior-driven perspective: rather than estimating model parameters, we apply BRP rules that directly leverage browsing history to filter out

inconsistent customer types. This yields a simple yet effective personalization strategy that complements approaches based on parametric models. Moreover, when demographic data are available, our framework naturally allows for estimating separate multi-choice rank list models across segments, further capturing preference heterogeneity.

6 Numerical Experiments

We introduced the multi-choice rank list model to capture multi-purchase behavior within the same category while accounting for product substitution effects, and discussed its estimation using historical data and BRP rules, as well as its application in optimizing assortment recommendations to maximize revenue. This section presents case studies using real-world datasets to evaluate the model’s predictive accuracy, followed by synthetic experiments designed to assess the effectiveness of assortment optimization.

6.1 Predictive Performance: A Case Study with the Tmall Dataset

To evaluate the predictive performance of the proposed MC-RLM and to assess the impact of incorporating side information in estimation via BRP rules, we present a case study based on the publicly available REC-TMALL dataset provided by Tmall, one of China’s largest B2C platforms under Alibaba Group (Zhang et al. 2014, Zhong et al. 2015). The dataset contains 1.33 billion user-item interactions from April 1 to September 30, 2013, across over 8.1 million distinct items. It captures four types of user behavior: *click* (93.6%), *purchase* (0.95%), *add to cart* (3.01%), and *add to favorite* (2.43%), spanning 8,116 leaf categories under 90 parent categories. Each customer’s interactions within a leaf category are aggregated into a single data point, capturing the sequence of actions including clicking, adding to cart, and purchasing. For our experiments, we focus on some leaf categories that exhibit intra-category multi-purchase behavior, as detailed in Table 1. These categories cover diverse areas such as food, personal care, and home improvement⁶. Multi-purchase behavior constitutes a notable portion of purchases, ranging from 14.89% to 41.18%, emphasizing the importance of modeling intra-category multi-purchase behavior.

To assess the impact of accounting for customer multi-purchase behavior and product substitution effects, as well as the benefit of incorporating BRP rules in estimation and prediction, we benchmark the proposed multi-choice rank list model (MC-RLM) against the classic single-choice rank list model (SC-RLM), the independent choice model (ICM), the multinomial logit (MNL) model, and the multi-purchase multinomial logit (MP-MNL) model proposed by Bai et al. (2023). We randomly split the data for each leaf category into training and test sets in a 4:1 ratio to compare the predictive performance of different models. In most categories, non-purchase instances dominate, potentially biasing model estimation. To address this imbalance, we apply a down-sampling technique commonly used in the literature (e.g., Feldman et al.

⁶ Category IDs are numerical, with item keywords in simplified Chinese. We used *sklearn* feature extraction package and GPT-4 to generate concise English descriptions.

2022), randomly selecting non-purchase cases in proportion to purchase cases for model estimation, while retaining the original data for out-of-sample testing.

Since the SC-RLM and MNL models are not designed to handle multi-purchase behavior, we make necessary adjustments when estimating these models. For customers who purchase multiple products (i.e., $P = \{p_1, \dots, p_n\}$ with $n \geq 2$), we decompose the multi-purchase event into n separate single-purchase events and remove the original multi-purchase case. For example, if a customer's data is recorded as $(P = \{i_1, i_2, i_3\}, C, K, V)$, we replace it with three entries: $(P_1 = \{i_1\}, C, K, V)$, $(P_2 = \{i_2\}, C, K, V)$, and $(P_3 = \{i_3\}, C, K, V)$. Each entry retains the original cart (C), clicking (K), and viewing (V) data but is associated with only one purchased product. No adjustments are needed for single-purchase or non-purchase cases. This modification enables the SC-RLM and MNL models to be estimated while preserving the information from the data. In the experiments, we estimate the MC-RLM and SC-RLM using Algorithm 1, and the MNL and ICM models via maximum likelihood estimation, as described in Section EC.3. Since no existing method is available for estimating the MP-MNL model, we propose a maximum likelihood estimation approach that enumerates all possible permutations of the purchase sets (details are also provided in Section EC.3).

We use the root mean square error (RMSE) to evaluate the predictive performance of the models across the test set, indicating the average prediction error. For T_{test} customers with corresponding offer sets $V_1, \dots, V_{T_{test}}$, the RMSE is defined as: $RMSE = \sqrt{\frac{1}{\sum_{t \in T_{test}} |V_t|} \sum_{t \in T_{test}} \sum_{i \in V_t} (v_{t,i} - \hat{v}_{t,i})^2}$, where $v_{t,i} = 1$ if product i is purchased by customer t and 0 otherwise, and $\hat{v}_{t,i}$ is the predicted probability that product i is purchased by customer t . For all models, including those trained on single-purchase data, predictions are made at the offer-set level without decomposing multi-purchase observations, and each offered item is assigned a predicted purchase probability. This metric is referred to as the *hard RMSE* (Berbeglia et al. 2021), appropriate when ground truth probabilities are unavailable, and commonly used in discrete choice model evaluations. In addition to prediction accuracy, we calculate the corrected Akaike information criterion (AIC_c) as a model performance metric, defined as: $AIC_c = 2 \frac{\#p - LL + \#p(\#p+1)}{T - \#p - 1}$, where $\#p$ is the number of model parameters and LL is the model's optimal log-likelihood on the training data. Lower AIC_c values indicate better model fit. For rank list models, $\#p = |T|$, representing the number of customer types with positive arrival probabilities.

6.1.1 The Impact of Considering Intra-Category Multi-Purchase Behavior

We begin by examining the impact of ignoring multi-purchase behavior. While the dataset does not include customer view sets, the click set serves as a suitable proxy for the offer set, as it reflects the products customers considered (Aouad et al. 2019). Similarly, items added to the shopping cart provide a stronger indication of purchase intent, as customers typically make their final choice from these items. In this subsection, we conduct experiments using both the click set and the cart set as proxies for the offer set, estimating all five models and evaluating their performance. To isolate the effect of multi-purchase behavior, we focus

solely on offer set and purchase information, excluding the BRP rules from the estimation process and prediction. Due to space constraints and the similarity in results and insights, we present results based on the cart set here, while results based on the click set are provided in Section EC.5 for reference.

The estimation results are presented in Table 1, where the proportions of single- and multi-purchase behaviors are normalized across all purchase instances. Notably, both estimated RLMS exhibit a high degree of sparsity, with only a few customer types showing positive arrival probabilities. Moreover, the MC-RLMs are even sparser compared to the SC-RLMs. In addition, we observe that most models achieve log-likelihood values of a similar scale across the majority of cases, though the MP-MNL model exhibits significantly lower log-likelihood in some instances. In terms of the AIC_c , the MC-RLM consistently outperforms the MP-MNL model and SC-RLM, achieving lower AIC_c values, indicating better model fit. The in-sample and

Table 1 Estimation Results for Experiments Using the Cart Set as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM LL	MNL LL	MPMNL LL	MCRLM LL	SCRLM LL	ICM AIC_c	MNL AIC_c	MPMNL AIC_c	MCRLM AIC_c	SCRLM AIC_c	MCRLM Num Types	SCRLM Num Types
9-5209	Wampee Fruit Snacks	13	3	96.64	79.31	20.69	-4.16	-7.45	-437.49	-31.88	-42.5	48.32	54.91	936.8	72.91	96.76	4	5
9-5746	Candied Pomelo Snacks	36	3	92.54	77.78	22.22	-10.82	-22.68	-694.41	-75.5	-107.24	137.31	161.02	1526.37	161.65	225.13	5	5
17-3513	Seashell Decorations	58	3	96.39	79.59	20.41	-45.45	-70.22	-44.74	-190.82	-261.6	243.49	293.04	256.18	391.88	535.56	5	6
17-6551	Handmade Organic Soaps	9	4	95.98	58.82	41.18	-4.5	-15.93	-255.53	-21.66	-40.82	37.0	59.85	571.38	53.05	94.36	4	5
21-1432	Glass Door Hinges	40	3	92.26	75.31	24.69	-15.87	-38.08	-39.63	-83.18	-126.46	154.34	198.76	221.51	174.71	263.45	4	5
21-5587	Heat Shrink Tubing	26	3	86.59	77.97	22.03	-22.91	-27.12	-241.14	-72.25	-98.34	119.41	127.84	572.77	155.2	209.66	5	6
21-7230	Grooving Cutting Inserts	39	4	95.59	75.76	24.24	-3.64	-113.53	-3.64	-79.88	-123.68	129.84	349.63	156.2	168.14	257.94	4	5
22-1863	Beef Skewers & Meatballs	9	4	94.13	76.74	23.26	-15.81	-32.18	-14.39	-52.78	-85.06	52.53	85.27	64.15	111.91	181.04	3	5
22-7517	Marinated Mud Snails	25	3	96.12	85.11	14.89	-7.11	-13.78	-444.84	-60.13	-77.28	88.3	101.63	982.47	131.08	165.36	5	5
35-1542	Bi Shumin Book	10	4	76.36	69.23	30.77	0.0	-11.09	-207.23	-12.48	-30.09	51.43	73.61	684.47	29.75	75.17	2	5
35-1729	Computer Exam Software	45	4	78.83	68.12	31.88	-4.41	-58.37	-120.01	-75.38	-154.35	167.82	275.75	432.75	159.16	321.54	4	6
52-6505	Fountain Spray Nozzles	20	3	92.59	79.41	20.59	-4.61	-8.25	-644.73	-40.73	-51.67	73.23	80.5	1369.95	87.91	114.53	3	5
64-2877	PVC Cable Conduits	30	4	93.77	68.12	31.88	-21.16	-46.99	-18.14	-76.25	-128.82	127.12	178.78	142.28	163.09	270.49	5	6

out-of-sample performance results are presented in Table 2 and Table 3, respectively. For completeness, summary statistics across all case study experiments are reported in Section EC.5. Although the model was trained using a down-sampled dataset, the in-sample RMSE was calculated on the full original training set, ensuring evaluation against the actual data distribution. This approach maintains consistency with the real-world distribution, despite down-sampling during training.

In combination with estimation results in Table 1, it is evident that in category 35-1542, which includes books by a Chinese writer, the ICM perfectly estimates the data, achieving near-zero log-likelihood. This suggests that customer choices in this category are largely independent, leading to the ICM outperforming the other models. In contrast, for most other categories, the MC-RLM demonstrates superior in-sample RMSE in the majority of cases and achieves the lowest out-of-sample RMSE across all categories tested, except for 35-1542. While this appears inconsistent with the in-sample model fitting results reported in Table 1, it is important to distinguish between model fit and predictive accuracy. Log-likelihood value reflects how well a model explains the training data and often favors simpler models, such as the ICM, which are easier to estimate. In contrast, the predictive accuracy, as measured by out-of-sample RMSE, provides a more reliable assessment of a model's ability to generalize to unseen data, which is of primary interest in practical applications. The superior out-of-sample performance of the proposed MC-RLM further underscores the importance of capturing customers' intra-category multi-purchase behavior, which often reflects

interdependencies among products. Such behavioral patterns cannot be effectively modeled by frameworks that rely solely on independent choice assumptions. As summarized in Table EC.3, the MC-RLM achieves substantial reductions in test RMSE relative to all benchmark models. In particular, it outperforms the ICM by an average of 16.33%, with improvements ranging up to 33.58%. The results from experiments using the click set as the offer set follow a similar pattern to those using the cart set, with the MC-RLM consistently demonstrating higher predictive accuracy in most of the cases tested.

Interestingly, although the MP-MNL model accounts for multi-purchase behavior, it performs poorly, exhibiting the highest RMSE and lowest log-likelihood in most cases. This may be due to the high estimation complexity, as the standard maximum likelihood estimation requires evaluating all permutations of purchased items. In contrast, the MC-RLM, despite modeling complex preference structures, benefits from the efficient MCMD algorithm, making it more practical for real-world multi-purchase scenarios.

Table 2 In-Sample Performance of Models in Experiments Where the Cart Set Serves as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM Train RMSE	MNL Train RMSE	MPMNL Train RMSE	MCRLM Train RMSE	SCRLM Train RMSE	MCRLM Improv. ICM RMSE (%)	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.6253	0.6566	0.7071	0.5037	0.554	19.45	23.3	28.77	9.09
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.5651	0.5821	0.7311	0.498	0.5467	11.88	14.45	31.89	8.91
17-3513	Seashell Decorations	58	3	79.59	20.41	0.64	0.6352	0.6395	0.4804	0.5011	24.94	24.37	24.88	4.14
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.6396	0.6557	0.7256	0.5033	0.5887	21.31	23.24	30.63	14.5
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.5005	0.4816	0.5387	0.4921	0.5375	1.67	-2.18	8.64	8.45
21-5587	Heat Shrink Tubing	26	3	77.97	22.03	0.4752	0.4604	0.5472	0.4886	0.5258	-2.83	-6.14	10.7	7.06
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.4384	0.594	0.4383	0.5011	0.5824	-14.32	15.63	-14.32	13.96
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.5882	0.6177	0.591	0.5016	0.5677	14.72	18.79	15.12	11.64
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.5982	0.6076	0.6747	0.4952	0.5219	17.22	18.5	26.61	5.13
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.25	0.5863	0.7891	0.5	0.6889	-100.0	14.72	36.63	27.42
35-1729	Computer Exam Software	45	4	68.12	31.88	0.4294	0.549	0.4837	0.4987	0.6179	-16.15	9.17	-3.11	19.3
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.4796	0.4891	0.8514	0.5121	0.5499	-6.78	-4.71	39.85	6.88
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.6057	0.5558	0.6118	0.4999	0.5206	17.46	10.05	18.29	3.97

Table 3 Out-of-Sample Performance of Models in Experiments Where the Cart Set Serves as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM Test RMSE	MNL Test RMSE	MPMNL Test RMSE	MCRLM Test RMSE	SCRLM Test RMSE	MCRLM Improv. ICM RMSE (%)	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.618	0.7071	0.8864	0.5453	0.6706	11.77	22.89	38.49	18.7
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.7465	0.8003	0.7643	0.51	0.5702	31.67	36.27	33.27	10.55
17-3513	Seashell Decorations	58	3	79.59	20.41	0.763	0.751	0.769	0.5068	0.5132	33.58	32.52	34.1	1.25
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.6687	0.495	0.7189	0.4908	0.5941	26.6	0.85	31.73	17.39
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.691	0.658	0.7195	0.5235	0.5835	24.24	20.44	27.24	10.28
21-5587	Heat Shrink Tubing	26	3	77.97	22.03	0.6295	0.6155	0.6129	0.5283	0.6421	16.09	14.18	13.81	17.73
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.7011	0.6588	0.701	0.5265	0.5502	24.9	20.08	24.9	4.3
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.5706	0.5375	0.5703	0.5	0.538	12.37	6.98	12.32	7.06
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.6486	0.6042	0.6976	0.5536	0.5925	14.64	8.38	20.64	6.57
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.3333	0.7136	0.7197	0.5	0.7648	-50.0	29.94	30.53	34.62
35-1729	Computer Exam Software	45	4	68.12	31.88	0.6696	0.6985	0.6602	0.5194	0.647	22.44	25.65	21.33	19.72
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.7515	0.7534	0.8165	0.6086	0.6441	19.02	19.23	25.46	5.51
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.7211	0.6401	0.7097	0.5414	0.5525	24.92	15.42	23.71	2.0

These results highlight the importance of accounting for intra-category multi-purchase behavior. Ignoring multi-purchase behavior or merely adjusting data for single-purchase models leads to suboptimal performance, as demonstrated by the comparative results. Additionally, products within the same category often exhibit substitution effects, which the proposed MC-RLM effectively captures through its use of preference rankings. Together, these factors enable the MC-RLM to outperform the four benchmark models in most cases, demonstrating its strength in modeling both multi-purchase behavior and product substitution.

6.1.2 The Impact of Using Side Information in Estimation via BRP Rules

In this subsection, we evaluate the benefit of incorporating side information into model estimation via BRP rules. Specifically, we aim to assess whether models estimated using side information yield higher predictive accuracy than those estimated without it. Since BRP rules cannot be directly applied to the MNL, MP-MNL, or ICM models, our analysis focuses on the impact on the single- and multi-choice RLMs. Due to the lack of view data in the dataset, we implement only rules (R2) and (R3) from Definition 2.

We conduct two sets of experiments. In the first setting, only purchase information is used for estimation, with the complete product set treated as the offer set (estimation results are reported in Section EC.5.3). In the second setting, purchase, cart, and click data are used into model estimation via BRP rules, while the complete product set is again treated as the offer set. The corresponding estimation results are presented in Table 4⁷. In the prediction phase, both sets of estimated models are evaluated without access to cart or click information, ensuring a fair comparison. By comparing the prediction RMSE across the two settings, we evaluate whether incorporating side information during estimation enhances predictive performance. The results are reported in Table 5, where the last four columns explicitly indicate the improvement in predictive accuracy when the same model is estimated with side information versus without it.

Table 4 Estimation Results for Experiments Using Side Information in Model Estimation

CATE	Description	Num Prod.	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MCRLM Num Types	SCRLM Num Types	MCRLM LL	SCRLM LL	MCRLM AIC_c	SCRLM AIC_c
9-5209	Wampee Fruit Snacks	13	96.64	79.31	20.69	6	9	-107.76	-138.56	227.64	295.38
9-5746	Candied Pomelo Snacks	36	92.54	77.78	22.22	5	9	-201.66	-268.22	413.41	554.7
17-3513	Seashell Decorations	58	96.39	79.59	20.41	7	9	-602.41	-739.51	1218.85	1497.08
17-6551	Handmade Organic Soaps	9	95.98	58.82	41.18	4	5	-66.85	-106.94	141.81	224.06
21-1432	Glass Door Hinges	40	92.26	75.31	24.69	8	6	-244.63	-311.4	505.44	634.91
21-5587	Heat Shrink Tubing	26	86.59	77.97	22.03	5	5	-161.66	-209.06	333.49	428.29
21-7250	Grooving Cutting Inserts	39	95.59	75.76	24.24	6	8	-261.45	-343.34	534.98	702.81
22-1863	Beef Skewers & Meatballs	9	94.13	76.74	23.26	8	6	-167.8	-220.19	351.86	452.53
22-7517	Marinated Mud Snails	25	96.12	85.11	14.89	6	6	-194.87	-228.43	401.84	468.95
35-1542	Bi Shumin Book	10	76.36	69.23	30.77	2	5	-22.29	-46.51	48.88	104.61
35-1729	Computer Exam Software	45	78.83	68.12	31.88	9	8	-161.3	-261.18	341.33	538.94
52-6505	Fountain Spray Nozzles	20	92.59	79.41	20.59	6	5	-118.63	-133.98	249.5	278.13
64-2877	PVC Cable Conduits	30	93.77	68.12	31.88	7	8	-232.42	-316.98	478.97	650.12

As shown in Table 4, the MC-RLM consistently achieves higher log-likelihood values and lower corrected AIC scores than the SC-RLM across all tested categories, indicating superior in-sample fit. Table 5, together with Table EC.17, shows that using side information via BRP rules leads to substantial improvements in RMSE. In fact, Tables EC.12 and EC.14 show that when only transaction data is used, both RLMs perform the worst among the five models, likely due to the limited and noisy nature of the data, which may lead to overfitting. However, with sufficient data and the application of BRP rules, the performance of both RLMs improves significantly, consistently outperforming the other models in terms of predictive accuracy. On

⁷ BRP rules leverage side information from all customers, including those who did not make a purchase, allowing behavioral data to inform preference learning. As a result, no down-sampling is applied in either set of experiments in Section 6.1.2.

Table 5 Comparison of Prediction Results: Models Leveraging vs. Ignoring Side Information

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MCRLM Train RMSE	SCRLM Train RMSE	Train MCRLM Improv. SCRLM RMSE (%)	MCRLM Test RMSE	SCRLM Test RMSE	Test MCRLM Improv. SCRLM RMSE (%)	Train MCRLM RMSE Improv. With vs. Without Side Info (%)	Train SCRLM RMSE Improv. With vs. Without Side Info (%)	Test MCRLM RMSE Improv. With vs. Without Side Info (%)	Test SCRLM RMSE Improv. With vs. Without Side Info (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.0519	0.0519	0.0306	0.0721	0.0721	-0.0258	4.45	3.19	2.35	1.91
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.048	0.0482	0.4239	0.0529	0.0531	0.4061	11.71	8.34	8.59	5.89
17-3513	Seashell Decorations	58	3	79.59	20.41	0.0275	0.0275	0.1033	0.0253	0.0254	0.2377	5.95	3.71	5.28	2.43
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.0825	0.0831	0.7884	0.0861	0.0877	1.8715	5.45	3.51	4.83	2.54
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.0452	0.0453	0.0853	0.0583	0.0582	-0.1459	11.65	6.83	6.57	3.66
21-3587	Heat Shrink Tubing	26	3	77.97	22.03	0.0725	0.0734	1.1794	0.0788	0.0805	2.1	17.42	9.25	14.35	5.77
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.0382	0.0382	-0.0857	0.0317	0.0315	-0.8754	7.55	3.98	8.62	3.5
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.0922	0.0925	0.3545	0.0752	0.0752	-0.0659	9.4	5.38	12.33	6.26
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.043	0.043	0.1123	0.038	0.0379	-0.3892	6.9	3.22	7.26	2.97
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.1472	0.16	7.9858	0.2348	0.2612	10.128	29.14	20.19	19.61	12.52
35-1729	Computer Exam Software	45	4	68.12	31.88	0.0739	0.074	0.0075	0.085	0.0863	1.5094	28.14	22.42	22.96	16.32
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.063	0.0631	0.1234	0.0733	0.0735	0.334	8.23	8.04	4.0	3.55
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.0501	0.05	-0.1926	0.0559	0.0561	0.3635	11.1	6.11	8.67	4.28

average, the use of side information reduces test RMSE by 9.65% for MC-RLM and 5.51% for SC-RLM, with maximum improvements reaching 22.96% and 16.32%, respectively. These gains highlight the critical role of side information in enhancing model accuracy.

To further evaluate the performance of the proposed MC-RLM, we conducted an additional case study using data from Cainiao Network, which includes comprehensive viewing information. More details are provided in Section EC.6.

6.2 Assortment Optimization: Results from a Synthetic Experiment

We now conduct synthetic experiments to assess the performance of assortment optimization using the MC-RLM. The synthetic data is generated using a variant of the MP-MNL model, a special case of the proposed MC-RUM, as described in Section 3.2. In this variant, the random utility of product $n \in \mathbb{N}$ is defined as $U_n = V_n - \beta_n r_n + \epsilon_n$, where $\epsilon = (\epsilon_n)_{n \in \mathbb{N}^+}$ follow i.i.d. standard normal distributions instead of Gumbel distributions. The deterministic utility V_n is drawn from a normal distribution with a mean of 3 and a standard deviation of 1, product revenue r_n is uniformly sampled from the range $[1, 5]$, and the price sensitivity parameter β_n is generated as the negative absolute value of a standard normal variable. Each customer's intended purchase quantity is generated independently of the product utilities, following an exponential decay distribution with a decay rate of 0.6. Offer sets are randomly selected, with sizes ranging from $1/3$ to $2/3$ of the total product set $\mathbb{N}$, ensuring at least two products when $|\mathbb{N}| = 5$.

We conduct experiments across various sizes of $\mathbb{N}$, executing 10 independent runs for each size with different model parameters. In each run, we randomly generate 1,000 customers and split the data into 80% for training and 20% for testing. In this experiment, we aim to evaluate the impact of accounting for multi-purchase behavior in assortment optimization. To do so, we compare two sets of models: MC-RLM versus SC-RLM, and MP-MNL versus MNL. The average out-of-sample performance is reported in Table 6, while the estimation results and in-sample performance are provided in Section EC.7. Consistent with Section 6.1.1, the proposed MC-RLM outperforms the other four models in RMSE, confirming that accurately accounting for multi-purchase behavior improves predictive accuracy.

Table 6 Average Out-of-Sample Performance Comparison of Prediction Accuracy Across Different Models

Num Prod.	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL Test RMSE	MPMNL Test RMSE	MCRLM Test RMSE	SCRLM Test RMSE	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
5	57.01	42.99	0.4404	0.434	0.4212	0.464	4.34	2.47	9.25
10	41.41	58.59	0.3918	0.5295	0.3803	0.4098	2.91	28.24	7.24
15	38.52	61.48	0.3295	0.3775	0.3162	0.343	3.87	15.38	7.81
20	37.07	62.93	0.2986	0.3375	0.2904	0.3106	2.78	13.59	6.54
25	36.14	63.86	0.2855	0.3273	0.2733	0.2883	4.0	15.94	5.25
30	35.7	64.3	0.2563	0.2626	0.2474	0.2602	3.26	5.59	4.92

Building on the estimated models, we now evaluate the performance of assortment optimization across different models. In all experiments, we impose a cardinality constraint of $K = 4$, which is common in scenarios like the one shown in Figure 5. Since no established methods exist for calculating expected revenue under the MP-MNL variant, we conduct 10,000 customer choice simulations following the same generation procedure as described earlier. The average revenue is calculated based on the optimal cardinality-constrained assortments for each model. For both the RLMs, the optimal assortment is determined by solving MILP (3). For the MNL model, the optimal assortment is derived using the method proposed by Davis et al. (2013). For the MP-MNL model, we simply enumerate all the possible assortments and output the one has the highest expected revenue based on the model. The average results from ten repetitions for each product count setting are presented in Table 7, with details for each tested case provided in Table EC.24.

Table 7 Average Performance Comparison of Assortment Optimization Across Different Models

Num Products	MNL Avg. Revenue	MPMNL Avg. Revenue	MCRLM Avg. Revenue	SCRLM Avg. Revenue	MCRLM Improv. SCRLM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. MNL (%)
5	2.5797	3.7809	3.9944	3.8698	5.12	7.76	78.94
10	2.8699	3.5213	4.5219	4.3442	6.17	39.42	308.06
15	2.0784	3.929	4.9827	4.5233	13.89	38.04	715.02
20	1.9922	4.2668	4.8262	4.5145	8.65	13.61	3797.29
25	2.3412	3.3525	4.3844	4.1833	5.33	38.97	2338.48
30	2.2448	4.2774	4.5456	4.4185	3.16	6.27	7332.34

Comparing the average revenue generated by the MC-RLM and SC-RLM models, as well as the MP-MNL and MNL models, we find that accurately capturing intra-category multi-purchase behavior significantly improves revenue when such behavior is prevalent. Although the MNL model generally achieves a lower RMSE than the MP-MNL model, the latter yields substantially higher revenue. On average, the MP-MNL model generates 63.95% more revenue than the MNL model across all cases tested. The revenue gap between SC-RLM and MC-RLM is less pronounced; however, MC-RLM still achieves, on average, 5.42% higher revenue than SC-RLM. While MC-RLM yields higher expected revenue in the majority of cases, it does not always identify the revenue-maximizing assortment. Notably, failing to account for multi-purchase behavior—as in the MNL model—can result in extremely poor outcomes. In some cases, the MNL-based assortment generated revenue as low as 0.0088. These instances contribute to the relatively large improvement values for MC-RLM over MNL reported in the last column.

Overall, accurately modeling intra-category multi-purchase behavior leads to substantial improvements in assortment recommendation performance. Additional results under fixed intended purchase quantities are

provided in Section EC.7.1 to further examine the sensitivity of MC-RLM's performance to the distribution of intended purchase quantities. To further explore practical heuristics, we evaluate the performance of the revenue-ordered assortment heuristic in an unconstrained setting. Details can be found in Section EC.7.2. On average, the revenue achieved by the best revenue-ordered assortment under the MC-RLM model does not surpass that of the optimal solution obtained by solving MILP (3). However, in many of the tested cases, the heuristic yields higher revenue than the optimal assortment derived from the MP-MNL model.

7 Conclusions and Future Research Directions

In conclusion, this paper presents a comprehensive framework for understanding intra-category multi-choice behavior and optimizing assortment recommendations in e-commerce. The proposed multi-choice rank list model effectively captures both multi-purchase and substitution behaviors, providing a flexible and robust tool for assortment optimization. Through extensive numerical experiments and a case study using real-world data from Tmall, we demonstrate the superior predictive accuracy of the MC-RLM compared to existing models such as the independent choice model and multi-purchase multinomial logit model. Incorporating behavior-reveal-preference rules further enhances the model's performance, particularly in real-world applications. This research contributes valuable insights into multi-purchase behavior and provides practical solutions for assortment optimization in e-commerce settings.

Future research could focus on developing fast heuristics for real-time personalized product recommendations, enhancing the speed of updating processes in dynamic e-commerce environments. Additionally, extending the MC-RLM with market basket analysis to capture cross-category interactions presents a promising direction. By estimating preference rankings for multiple product categories and accounting for correlations between them, the model could offer deeper insights into cross-category purchasing behavior. For example, understanding how a customer's preference for hiking shoes may influence their choice of hiking clothing could provide e-commerce platforms with powerful tools for improving cross-category recommendations. This direction holds significant potential for advancing assortment planning strategies.

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E-Companion for *Intra-Category Multi-Choice Preferences Learning and Assortment Recommendation in E-Commerce*

EC.1 Proof of Statements

Proof of Proposition 1 Consider any two subset V and V' with $V \subseteq V' \subseteq N$.

1) When $n = 0$, we have

$$\begin{aligned} T_{0,V} &= \{(0, [0])\} \cup \{(q, \sigma) \in T : q > 0, \sigma(0) \leq \min_{k \in V} \sigma(k)\} \\ &\supseteq \{(0, [0])\} \cup \{(q, \sigma) \in T : q > 0, \sigma(0) \leq \min_{k \in V'} \sigma(k)\} \\ &= T_{0,V'}. \end{aligned}$$

Hence, $\mathbb{P}(0, V) = \pi_{0,V} = \sum_{(q,\sigma) \in T_{0,V}} \lambda_{q,\sigma} \geq \sum_{(q,\sigma) \in T_{0,V'}} \lambda_{q,\sigma} = \pi_{0,V'} = \mathbb{P}(0, V')$.

2) Consider any $n \in V$. For any customer type $(q, \sigma) \in T$, if $\sigma(n)$ ranks within the smallest q rankings among all products in V' , then it must also rank within the smallest q among all products in any subset $V \subseteq V'$. In other words, if a customer of type (q, σ) chooses product n when facing assortment V' , the same customer will also choose product n when facing any smaller assortment $V \subseteq V'$. While the inverse might not always hold. Hence, $\mathbb{P}(n, V) \geq \mathbb{P}(n, V')$. ■

EXAMPLE EC.1 (VIOLATION OF THE REGULARITY AXIOM IN THE BUNDLEMVL-2 MODEL). We provide a concrete example demonstrating that the BundleMVL-2 model can violate the regularity axiom. Consider a market with three products: A , B , and C . We analyze two nested assortments: $V = \{A, B\}$ and $V' = \{A, B, C\}$. Based on the definition of the BundleMVL-2 model, customers can purchase up to two items simultaneously, and the utility of a bundle includes potential interaction effects between items. We assign base utilities as $v_A = 0$, $v_B = 0$, and $v_C = 0$, and interaction terms as $\delta_{AC} = 1$, $\delta_{AB} = 0$, and $\delta_{BC} = 0$. The deterministic utility of a bundle S is given by: $V(S) = \sum_{i \in S} v_i + \sum_{\{i,j\} \subseteq S} \delta_{ij}$.

For assortment V , the utilities of the bundles are as follows: $V(\emptyset) = 0$, $V(\{A\}) = v_A = 0$, $V(\{B\}) = v_B = 0$, and $V(\{A, B\}) = v_A + v_B + \delta_{AB} = 0$. For assortment V' , the utilities of the bundles are as follows: $V(\emptyset) = 0$, $V(\{A\}) = 0$, $V(\{B\}) = 0$, $V(\{C\}) = 0$, $V(\{A, B\}) = 0$, $V(\{A, C\}) = v_A + v_C + \delta_{AC} = 1$, and $V(\{B, C\}) = 0$. Note that bundles larger than size two are not considered due to the BundleMVL-2 model's restriction.

In the BundleMVL-2 model, the probability of choosing a bundle S is proportional to $\exp(V(S))$.

For assortment V , the attractions of different subsets are: $\exp(V(\emptyset)) = e^0 = 1$, $\exp(V(\{A\})) = 1$, $\exp(V(\{B\})) = 1$, and $\exp(V(\{A, B\})) = 1$. The denominator is calculated as: $W(V) = 1 + 1 + 1 + 1 = 4$. Probability of purchasing product A is $\mathbb{P}(A, V) = \mathbb{P}(\{A\}, V) + \mathbb{P}(\{A, B\}, V) = \frac{\exp(V(\{A\}))}{W(V)} + \frac{\exp(V(\{A, B\}))}{W(V)} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

For assortment V' , the attractions of different subsets are: $\exp(V(\emptyset)) = 1$, $\exp(V(\{A\})) = 1$, $\exp(V(\{B\})) = 1$, $\exp(V(\{C\})) = 1$, $\exp(V(\{A, B\})) = 1$, $\exp(V(\{A, C\})) = e^1 = e$, and $\exp(V(\{B, C\})) =$

1. The denominator is calculated as: $W(V') = 1 + 1 + 1 + 1 + 1 + e + 1 = 6 + e$. Probability of purchasing product A is $\mathbb{P}(A, V') = \mathbb{P}(\{A\}, V') + \mathbb{P}(\{A, B\}, V') + \mathbb{P}(\{A, C\}, V') = \frac{1}{6+e} + \frac{1}{6+e} + \frac{e}{6+e} = \frac{1+1+e}{6+e}$.

Using $e \approx 2.71828$, we have: $\mathbb{P}(A, V') = \frac{1+1+2.71828}{6+2.71828} = \frac{4.71828}{8.71828} \approx 0.54137$, and $\mathbb{P}(A, V) = \frac{1}{2} = 0.5$. Thus, $\mathbb{P}(A, V') \approx 0.54137 > \mathbb{P}(A, V) = 0.5$. The regularity axiom states that for nested assortments $V \subseteq V'$ and any product n , it holds that $\mathbb{P}(n, V) \geq \mathbb{P}(n, V')$. However, in this example, we observe that: $\mathbb{P}(A, V') > \mathbb{P}(A, V)$, indicating a violation of the regularity axiom.

The violation occurs due to the positive interaction effect between products A and C (i.e., $\delta_{AC} = 1$). The inclusion of product C in the assortment V' increases the attractiveness of the bundle $\{A, C\}$, leading to a higher probability of purchasing product A when C is available. This effect outweighs the dilution typically expected when expanding assortments, resulting in an increase in the choice probability of A. ■

PROPOSITION EC.1. *The regularity axiom is satisfied by the multi-choice random utility model. Specifically, for any nested assortments $V \subseteq V' \subseteq \mathbb{N}$ and for any product $n \in V \cup \{0\}$, it holds that $\mathbb{P}(n, V) \geq \mathbb{P}(n, V')$.*

Proof of Proposition EC.1 Consider any two subset V and V' with $V \subseteq V' \subseteq \mathbb{N}$.

1) When $n = 0$, we have

$$\begin{aligned} \mathbb{P}(0, V) &= \pi_{0, V} \\ &= \mathbb{P}(Q = 0) + \mathbb{P}(Q > 0, U_0 > \max_{i \in V} U_i) \\ &\geq \mathbb{P}(Q = 0) + \mathbb{P}(Q > 0, U_0 > \max_{i \in V'} U_i) \\ &= \pi_{0, V'} \\ &= \mathbb{P}(0, V'). \end{aligned}$$

2) Consider any $n \in V$. For a customer with $Q = q$ and $\mathbf{U}^+ = \mathbf{u}^+$, if u_n is within the q -th largest utilities among all products in V' , then it must also be within the largest q utilities among all products in any subset $V \subseteq V'$. In other words, if this customer selects product n when presented with assortment V' , the same customer will also select product n when facing any smaller assortment $V \subseteq V'$. The converse, however, may not always apply. Hence, $\mathbb{P}(n, V) \geq \mathbb{P}(n, V')$. ■

Proof of Theorem 1 “ $\Leftarrow$ ” First, we show that if there exists a joint distribution over $(Q, \mathbf{U}^+)$ such that $\pi_{P, V} = \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k)$ holds for all non-empty subsets $V \subseteq \mathbb{N}$ and for all $P \subseteq V$, then a multi-choice rank list model can be defined accordingly.

For ease of our analysis, we denote $\mathbf{T}$ as the collection of all possible customer types. For each possible value q of Q , corresponding to its jointly distributed $\mathbf{U}^+$, we define

$$\lambda_{q, \sigma} = \mathbb{P}(Q = q, U_{\sigma^{-1}(1)} > U_{\sigma^{-1}(2)} > \cdots > U_{\sigma^{-1}(N+1)}).$$

Since $(Q, \mathbf{U}^+)$ follows a valid joint distribution, we have $\lambda_{q, \sigma} \in [0, 1]$ for all $(q, \sigma) \in \mathbf{T}$ and $\sum_{(q, \sigma) \in \mathbf{T}} \lambda_{q, \sigma} = 1$. Therefore, $(\lambda_{q, \sigma})_{(q, \sigma) \in \mathbf{T}}$ is a valid probability mass function.

For any non-empty subsets $V \subseteq N$ and for any $P \subseteq V$, we have

$$\begin{aligned}
\pi_{P,V} &= \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k) \\
&= \sum_{(q,\sigma) \in \left\{ (q,\sigma) \in T: q = |P|, \max_{i \in P} \sigma(i) < \min_{j \in V^+ \setminus P} \sigma(j) \right\}} \lambda_{q,\sigma} \\
&\quad + \sum_{(q,\sigma) \in \left\{ (q,\sigma) \in T: q > |P|, \max_{i \in P} \sigma(i) < \sigma(0), \sigma(0) < \min_{k \in V \setminus P} \sigma(k) \right\}} \lambda_{q,\sigma} \\
&= \sum_{(q,\sigma) \in T_{P,V}} \lambda_{q,\sigma}.
\end{aligned}$$

Hence, the choice probability $\pi_{P,V}$ is derived from a multi-choice rank list model with $(\lambda_{q,\sigma})_{(q,\sigma) \in T}$.

“ $\Rightarrow$ ” Consider a choice probability distribution π that is derived from a multi-choice rank list model with the probability mass function $(\lambda_{q,\sigma})_{(q,\sigma) \in T}$, where T refers to the collection of all possible customer types. We have, for any non-empty subsets $V \subseteq N$ and for any $P \subseteq V$, that $\pi_{P,V} = \sum_{(q,\sigma) \in T_{P,V}} \lambda_{q,\sigma}$.

We claim that there exists a joint distribution $(Q, \mathbf{U}^+)$ corresponding to a multi-choice random utility model such that $\pi_{P,V} = \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k)$ holds for all non-empty subsets $V \subseteq N$ and for all $P \subseteq V$.

For each pair $(q, \sigma) \in T$, we define $\mathbf{l}_\sigma = (\sigma(1), \sigma(2), \dots, \sigma(N+1))$, and let

$$\mathbb{P}(Q = q, \mathbf{U}^+ = -\mathbf{l}_\sigma) = \lambda_{q,\sigma}. \quad (\text{EC.1})$$

Since $(\lambda_{q,\sigma})_{(q,\sigma) \in T}$ is a valid probability mass function, we conclude that Equation (EC.1) constructs a valid joint distribution for Q and $\mathbf{U}^+$.

For any non-empty subsets $V \subseteq N$ and for any $P \subseteq V$, we have

$$\begin{aligned}
\pi_{P,V} &= \sum_{(q,\sigma) \in T} \lambda_{q,\sigma} \cdot \mathbb{I} \left(q = |P|, \max_{i \in P} \sigma(i) < \min_{j \in V^+ \setminus P} \sigma(j) \right) \\
&\quad + \sum_{(q,\sigma) \in T} \lambda_{q,\sigma} \cdot \mathbb{I} \left(q > |P|, \max_{i \in P} \sigma(i) < \sigma(0), \sigma(0) < \min_{k \in V \setminus P} \sigma(k) \right) \\
&= \sum_{(q,\sigma) \in T} \lambda_{q,\sigma} \cdot \mathbb{I} \left(q = |P|, \min_{i \in P} -\sigma(i) > \max_{j \in V^+ \setminus P} -\sigma(j) \right) \\
&\quad + \sum_{(q,\sigma) \in T} \lambda_{q,\sigma} \cdot \mathbb{I} \left(q > |P|, \min_{i \in P} -\sigma(i) > -\sigma(0), -\sigma(0) > \max_{k \in V \setminus P} -\sigma(k) \right) \\
&= \sum_{(q,\sigma) \in T} \mathbb{P}(Q = q, \mathbf{U}^+ = -\mathbf{l}_\sigma) \cdot \mathbb{I} \left(q = |P|, \min_{i \in P} -\sigma(i) > \max_{j \in V^+ \setminus P} -\sigma(j) \right) \\
&\quad + \sum_{(q,\sigma) \in T} \mathbb{P}(Q = q, \mathbf{U}^+ = -\mathbf{l}_\sigma) \cdot \mathbb{I} \left(q > |P|, \min_{i \in P} -\sigma(i) > -\sigma(0), -\sigma(0) > \max_{k \in V \setminus P} -\sigma(k) \right) \\
&= \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k).
\end{aligned}$$

Hence, such a joint distribution over $(Q, \mathbf{U}^+)$ exists such that $\pi_{P,V} = \mathbb{P}(Q = |P|, \min_{i \in P} U_i > \max_{j \in V^+ \setminus P} U_j) + \mathbb{P}(Q > |P|, \min_{i \in P} U_i > U_0, U_0 > \max_{k \in V \setminus P} U_k)$ holds for all possible pairs (P, V) . $\blacksquare$

PROPOSITION EC.2. For any non-empty $V \subseteq N$, we have $\sum_{P \subseteq V} \pi_{P,V}^{ICM} = 1$.

Proof of Proposition EC.2 Assume that $V = \{v_1, v_2, \dots, v_{n_V}\}$ and let $\bar{V}_k = V \setminus \{v_1, \dots, v_k\} = \{v_{k+1}, \dots, v_{n_V}\}$ for $k \in [n_V - 1]$, then we have

$$\begin{aligned}
 \sum_{P \subseteq V} \pi_{P,V}^{ICM} &= p_{v_1} \cdot \sum_{P \subseteq \bar{V}_1} \pi_{P,\bar{V}_1}^{ICM} + (1 - p_{v_1}) \cdot \sum_{P \subseteq \bar{V}_1} \pi_{P,\bar{V}_1}^{ICM} \\
 &= \sum_{P \subseteq \bar{V}_1} \pi_{P,\bar{V}_1}^{ICM} \\
 &= p_{v_2} \cdot \sum_{P \subseteq \bar{V}_2} \pi_{P,\bar{V}_2}^{ICM} + (1 - p_{v_2}) \cdot \sum_{P \subseteq \bar{V}_2} \pi_{P,\bar{V}_2}^{ICM} \\
 &= \sum_{P \subseteq \bar{V}_2} \pi_{P,\bar{V}_2}^{ICM} \\
 &= \dots \\
 &= \sum_{P \subseteq \bar{V}_{v_{n_V}-1}} \pi_{P,\bar{V}_{v_{n_V}-1}}^{ICM} \\
 &= p_{v_{n_V}} + (1 - p_{v_{n_V}}) \\
 &= 1. \quad \blacksquare
 \end{aligned}$$

Proof of Proposition 2 For an MC-RUM, let $\mathbb{P}(Q = N) = 1$, all U_i be independent of each other, and $U_0 = u_0$. Then, any product $i \in N$ will be chosen as long as its utility is greater than that of the outside option, i.e. $U_i > U_0 = u_0$. It is obvious that the probability of product i being chosen is $\mathbb{P}(U_i > u_0)$ and it has nothing to do with all other products. So, the demand for any product is independent of other products. That is, this special case of MC-RUM is exactly the independent choice model and $\pi_{P,V} = \mathbb{P}(Q = N, \min_{i \in P} U_i > u_0, u_0 > \max_{j \in V \setminus P} U_j) = \prod_{i \in P} \mathbb{P}(U_i > u_0) \cdot \prod_{j \in V \setminus P} \mathbb{P}(u_0 > U_j)$, where the marginal choice probability of product $i \in N$ is represented by $\mathbb{P}(U_i > u_0)$. $\blacksquare$

CLAIM EC.1. A multi-purchase multinomial logit model with $Q = N$ is not necessary an independent choice model. This is because the random multi-purchase multinomial logit model does not satisfy the conditions in Proposition 2, since U_0 is not deterministic.

Proof of Claim EC.1 We prove this claim by providing a counterexample. For a universe product set $N = \{1, 2, 3\}$, consider an MP-MNL model with $\mathbb{P}(Q = |N|) = \mathbb{P}(Q = 3) = 1$. Suppose the product attraction vector is $\mathbf{a} = (a_1, a_2, a_3)^\top = (2, 3, 4)^\top$, and the attraction for the outside option is set to be 1 by default. Then, given any non-empty assortment $V \subseteq N$, we have $\mathbb{P}_{MPMNL_3}(n, V) = \mathbb{P}(U_n > U_0) = \frac{a_n}{1+a_n} \triangleq p_n$ (this expression is derived by the definition of the MP-MNL model discussed in Bai et al. 2023). We can calculate that $p_1 = \frac{2}{3}$, $p_2 = \frac{3}{4}$, and $p_3 = \frac{4}{5}$.

Consider a case that the offer set $V = N$ and the purchase set $P = \{1, 2\}$. In this MP-MNL model, there are exactly two scenarios where a customer purchases both products $\{1, 2\}$. These occur when either $U_1 > U_2 > U_0 > U_3$ or $U_2 > U_1 > U_0 > U_3$, as the customer intends to purchase three products whose utilities exceed that of the outside option. If $U_3 > U_0$, the customer will also purchase product 3. By definition, we have

$$\begin{aligned}
 \pi_{P,V}^{MPMNL_3} &= \mathbb{P}(U_1 > U_2 > U_0) + \mathbb{P}(U_2 > U_1 > U_0) \\
 &= \frac{2}{1+(2+3+4)} \cdot \frac{3}{1+(3+4)} \cdot \frac{1}{1+(4)} + \frac{3}{1+(2+3+4)} \cdot \frac{2}{1+(2+4)} \cdot \frac{1}{1+(4)} \\
 &= \frac{9}{280}.
 \end{aligned}$$

However, by the definition of the ICM, we have

$$\begin{aligned}\pi_{p,v}^{ICM} &= p_1 \cdot p_2 \cdot (1 - p_3) \\ &= \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{5} \\ &= \frac{10}{10} \\ &\neq \pi_{p,v}^{MPMNL_3}.\end{aligned}$$

Hence, while the choice probabilities of each product is independent of the assortment, the MP-MNL model with $\mathbb{P}(Q = |\mathbb{N}|) = 1$ does not conform to the ICM. $\blacksquare$

Proof of Lemma 1 Since type 0 customers do not generate revenue, we discard this type and normalize the remaining appearance probabilities, λ , so that $\sum_{m=1}^M \lambda_m = 1$. This adjustment will not affect our decision, as it is equivalent to multiplying the original expected revenue by a positive constant. Without loss of generality, we rearrange the indices of customer types so that $h_1 \geq h_2 \geq \dots \geq h_M$.

Consider the revenue-ordered assortment $S_{h_m} = \{n \in \mathbb{N} : r_n \geq h_m\}$. Given that $h_1 \geq h_2 \geq \dots \geq h_M$, any interest list of customers from type $\{1, 2, \dots, m^*\}$ contains at least one product whose profit per unit is no less than h_{m^*} . Hence, any customer of these types can generate revenue of at least h_{m^*} when faced with assortment $S_{h_{m^*}}$. Now, we have

$$R(S_{h_{m^*}}) = \sum_{m=1}^M \lambda_m \cdot R_m(S_{h_{m^*}}) \geq h_{m^*} \cdot \sum_{m=1}^{m^*} \lambda_m. \quad (\text{EC.2})$$

To upper bound $\sum_{m=1}^M \lambda_m h_m$, we express it as $\sum_{m=1}^{d^*-1} \lambda_m h_m + \lambda_{d^*} h_{d^*} + \sum_{m=d^*+1}^M \lambda_m h_m$, where $d^* := \min \left\{ d \in [M] : \sum_{m=1}^d \lambda_m \geq \frac{1}{2} \cdot \frac{r_{\min}}{r_{\max}} \right\}$. According to the definition of d^* , we have $\sum_{m=1}^{d^*-1} \lambda_m h_m \leq \sum_{m=1}^{d^*-1} \lambda_m r_{\max} < r_{\max} \cdot \frac{1}{2} \cdot \frac{r_{\min}}{r_{\max}} = \frac{r_{\min}}{2}$, provided that $h_m \leq r_{\max}$ for all $m \in [M]$. According to the construction of m^* and d^* and Equation (EC.2), we have $\lambda_{d^*} h_{d^*} \leq R(S_{h_{m^*}})$. For $\sum_{m=d^*+1}^M \lambda_m h_m$, we have

$$\begin{aligned}\sum_{m=d^*+1}^M \lambda_m h_m &= \sum_{m=d^*+1}^M \frac{\lambda_m}{\sum_{i=1}^m \lambda_i} \cdot h_m \cdot \sum_{i=1}^m \lambda_i \\ &\leq \sum_{m=d^*+1}^M \frac{\lambda_m}{\sum_{i=1}^m \lambda_i} \cdot \left(h_{m^*} \cdot \sum_{i=1}^{m^*} \lambda_i \right) \\ &\leq \sum_{m=d^*+1}^M \frac{\lambda_m}{\sum_{i=1}^m \lambda_i} \cdot R(S_{h_{m^*}}) \\ &= \sum_{m=d^*+1}^M \left(\int_{\sum_{i=1}^{m-1} \lambda_i}^{\sum_{i=1}^m \lambda_i} \frac{1}{\sum_{i=1}^m \lambda_i} dx \right) \cdot R(S_{h_{m^*}}) \\ &\leq \sum_{m=d^*+1}^M \left(\int_{\sum_{i=1}^{m-1} \lambda_i}^{\sum_{i=1}^m \lambda_i} \frac{1}{x} dx \right) \cdot R(S_{h_{m^*}}) \\ &= \left(\int_{\sum_{i=1}^{d^*} \lambda_i}^1 \frac{1}{x} dx \right) \cdot R(S_{h_{m^*}}) \\ &\leq \left(\int_{\frac{1}{2} \cdot \frac{r_{\min}}{r_{\max}}}^1 \frac{1}{x} dx \right) \cdot R(S_{h_{m^*}}) \\ &= \ln \left(2 \cdot \frac{r_{\max}}{r_{\min}} \right) \cdot R(S_{h_{m^*}}),\end{aligned}$$

where the first inequality holds by the definition of m^* , and the second inequality holds by Equation (EC.2). The third inequality holds because $\frac{1}{x} \geq \frac{1}{\sum_{i=1}^m \lambda_i}$ always holds within the interval of integration, provided the monotonicity of $x \mapsto \frac{1}{x}$. The fourth inequality holds by the definition of d^* .

Thus, we have

$$\begin{aligned} \sum_{m=1}^M \lambda_m h_m &\leq \frac{r_{\min}}{2} + R(S_{h_{m^*}}) + \ln\left(2 \cdot \frac{r_{\max}}{r_{\min}}\right) R(S_{h_{m^*}}) \\ &= \frac{r_{\min}}{2} + R(S_{h_{m^*}}) \cdot \left(1 + \ln\left(2 \cdot \frac{r_{\max}}{r_{\min}}\right)\right) \\ &\leq R(S_{h_{m^*}}) \cdot \left(\frac{3}{2} + \ln\left(2 \cdot \frac{r_{\max}}{r_{\min}}\right)\right) \\ &\leq R(S_{h_{m^*}}) \cdot \frac{5}{2} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil, \end{aligned} \quad (\text{EC.3})$$

where the second inequality holds because $\sum_{m=1}^M \lambda_m = 1$ and $R(S_{h_{m^*}}) \geq h_{m^*} \cdot \sum_{m=1}^{m^*} \lambda_m \geq h_M \cdot \sum_{m=1}^M \lambda_m \geq r_{\min}$. ■

Proof of Theorem 2 Since type 0 customers do not generate revenue, we discard this type and normalize the remaining appearance probabilities, λ , so that $\sum_{m=1}^M \lambda_m = 1$.

We define R^* as the expected revenue derived from the optimal assortment S^* . We let $m^* = \arg \max_{m \in [M]} h_m \sum_{i=1}^m \lambda_i$ (selecting an arbitrary one in cases where the maximum value is achieved by multiple customer types $m \in [M]$) and consider a revenue-ordered assortment $S_{h_{m^*}}$, which includes all products with a profit of at least h_{m^*} .

For every customer of type $m \in [M]$, we specify a budget $B_m = q_m h_m$, where h_m refers to the highest profit in their interest list and q_m refers to the corresponding intended purchase quantity of type m . Now, we can upper bound the optimal expected revenue by

$$R^* \leq \sum_{m=1}^M \lambda_m B_m = \sum_{m=1}^M \lambda_m q_m h_m \leq q_{\max} \cdot \sum_{m=1}^M \lambda_m h_m \leq q_{\max} \cdot R(S_{h_{m^*}}) \cdot \left(\frac{5}{2} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right), \quad (\text{EC.4})$$

where the last inequality holds by Lemma 1. Equation (EC.4) implies that $S_{h_{m^*}}$ approximates the optimal expected revenue within factor $O\left(q_{\max} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right)$. Given that the expected revenue from $S_{h_{m^*}}$ cannot exceed that of the optimal revenue-ordered assortment, we conclude that the optimal revenue-ordered assortment approximates the optimal expected revenue within factor $O\left(q_{\max} \cdot \lceil \ln\left(\frac{r_{\max}}{r_{\min}}\right) \rceil\right)$. ■

Proof of Corollary 1 Instead of proving the original statement directly, we demonstrate that, after eliminating all products with profits less than $\frac{\lambda_{r_{\max}}}{2} \cdot r_{\max}$, there is still an assortment that can generate an expected revenue of at least $\frac{q_{\max}}{2} \cdot R^*$. This transformation ensures that the profits of the remaining products are within a factor of $\frac{\lambda_{r_{\max}}}{2}$ of each other, in which case the upper bound specified in Theorem 2 becomes $O\left(q_{\max} \cdot \lceil \ln\left(\frac{1}{\lambda_{r_{\max}}}\right) \rceil\right)$.

We let $\bar{S}$ denote the subset of products that are excluded, i.e., $\bar{S} = \left\{n \in [N] : r_n \leq \frac{\lambda_{r_{\max}}}{2} \cdot r_{\max}\right\}$, and let $\hat{S}$ denote the assortment consisting only of products with the highest profit. When some products are excluded from an assortment, the probability of a customer purchasing any of the remaining products can only

increase. Consequently, this also increases the expected revenue of the remaining assortment. For this reason, we now consider the contribution of $\bar{\mathbf{S}}$ to the expected revenue of the optimal assortment, which can be upper bounded by

$$\begin{aligned} \sum_{m=1}^M \lambda_m \cdot R_m(\bar{\mathbf{S}}) &\leq \sum_{m=1}^M \lambda_m \cdot q_m \cdot \frac{\lambda_{r_{\max}}}{2} \cdot r_{\max} \\ &\leq q_{\max} \cdot \frac{\lambda_{r_{\max}}}{2} \cdot r_{\max} \\ &\leq \frac{q_{\max}}{2} \cdot R^*, \end{aligned}$$

where the last inequality holds because $R^* \geq \lambda_{r_{\max}} \cdot r_{\max}$. According to Theorem 2, we have

$$R^* \leq R(\hat{\mathbf{S}}) \cdot \frac{q_{\max}}{2} \cdot \frac{5}{2} \cdot \lceil \ln\left(\frac{2}{\lambda_{r_{\max}}}\right) \rceil.$$

■

EC.2 Details of the Multi-Choice Market Discovery Algorithm with BRP Rules

Before discussing the details of the multi-choice market discovery algorithm, we explore some properties of Problem (MLLE) that motivates the algorithm. We introduce a set of new variables $\gamma_t = \sum_{(q,\sigma) \in \mathcal{T}_{P_t, C_t, K_t, V_t}} \lambda_{q,\sigma} = \mathbb{P}(P_t, C_t, K_t | V_t, \boldsymbol{\lambda})$ for all $t \in [T]$, representing the aggregate probability that a customer, after viewing V_t , clicks K_t , adds C_t to the shopping cart, and purchases P_t . Let $\boldsymbol{\gamma} = [\gamma_1, \gamma_2, \dots, \gamma_T]^\top$, and we can rewrite the log-likelihood function with respect to $\boldsymbol{\gamma}$ as $LL(\boldsymbol{\gamma}) = \sum_{t=1}^T \ln(\gamma_t)$. Let $M = |T| - 1$, denoting the number of possible customer types excluding $(q_0, \sigma_0) = (0, [0])$. Define matrix $A \in \{0, 1\}^{T \times (M+1)}$ with entries $a_{t,m+1} = 1$ if $(q_m, \sigma_m) \in \mathcal{T}_{P_t, C_t, K_t, V_t}$ and $a_{t,m+1} = 0$ otherwise. The matrix A has one row per customer and one column per customer type, with $a_{t,m+1} = 1$ when customer type (q_m, σ_m) is compatible with (P_t, C_t, K_t, V_t) for customer t . It is important to note that the behavior-reveal-preference rules are incorporated in the construction of matrix A . When no customer browsing histories are available, i.e., V , K , and C are all unobtainable, we can only use the transaction information, P , to construct matrix A . In this case, $a_{t,m+1} = 1$ if $P_t \subseteq \mathcal{I}_m$ and 0 otherwise.

Using matrix A , we can reformulate Problem (MLLE) with respect to $(\boldsymbol{\lambda}, \boldsymbol{\gamma})$:

$$\begin{aligned} &\max_{\boldsymbol{\lambda}, \boldsymbol{\gamma} \geq \mathbf{0}} LL(\boldsymbol{\gamma}) \\ &\text{s.t.} \quad \sum_{(q,\sigma) \in \mathcal{T}} \lambda_{q,\sigma} = 1, \\ &\quad A\boldsymbol{\lambda} = \boldsymbol{\gamma}. \end{aligned} \tag{EC.5}$$

Note that $\nabla LL(\boldsymbol{\gamma}) = \left[\frac{1}{\gamma_1}, \dots, \frac{1}{\gamma_T}\right]^\top$ and $\nabla^2 LL(\boldsymbol{\gamma}) = \text{diag}\left(-\frac{1}{\gamma_1^2}, \dots, -\frac{1}{\gamma_T^2}\right) \prec 0$, where $\text{diag}(\cdot)$ represents a diagonal matrix with the specified elements on the main diagonal. Hence, (EC.5) is a concave program with linear constraints and a bounded objective, $LL(\boldsymbol{\gamma}) \leq 0$ for all $\boldsymbol{\lambda}, \boldsymbol{\gamma} \geq \mathbf{0}$. Note that if $\boldsymbol{\lambda}^*$ is optimal for Problem (MLLE), then $(\boldsymbol{\lambda}^*, \boldsymbol{\gamma}^*)$, where $\gamma_t^* = \sum_{(q,\sigma) \in \mathcal{T}_{P_t, C_t, K_t, V_t}} \lambda_{q,\sigma}^*$, is an optimal solution to Problem (EC.5).

The Lagrangian dual problem of Problem (EC.5) is defined as:

$$\min_{\beta \in \mathbb{R}, \mu \in \mathbb{R}^T} \phi(\beta, \mu), \quad (\text{EC.6})$$

where $\phi(\beta, \mu) = \sup_{\lambda, \gamma \geq 0} \{LL(\gamma) + \beta (\sum_{(q,\sigma) \in T} \lambda_{q,\sigma} - 1) + \mu^\top (A\lambda - \gamma)\}$. We then have the following proposition:

PROPOSITION EC.3. *The strong duality holds for the primal problem (EC.5) and its dual problem (EC.6), i.e., they have the same finite optimal objective value. Additionally, (λ^*, γ^*) is an optimal solution to the primal problem (EC.5) and (β^*, μ^*) is an optimal solution to dual problem (EC.6) if and only if $(\lambda^*, \gamma^*, \beta^*, \mu^*)$ is a saddle point of the Lagrangian function, $g(\lambda, \gamma, \beta, \mu) = LL(\gamma) + \beta (\sum_{(q,\sigma) \in T} \lambda_{q,\sigma} - 1) + \mu^\top (A\lambda - \gamma)$, i.e., $g(\lambda^*, \gamma^*, \beta, \mu) \geq g(\lambda^*, \gamma^*, \beta^*, \mu^*) \geq g(\lambda, \gamma, \beta^*, \mu^*)$ for all $\lambda, \gamma \geq 0$, $\beta \in \mathbb{R}$, and $\mu \in \mathbb{R}^T$.*

This proposition follows from the strong duality theorem (see, e.g., Theorem 6.2.4 in Bazaraa et al. 2006) and is useful in designing the estimation algorithm.

As mentioned earlier, the number of possible customer types is factorial in the number of alternatives. Therefore, when N becomes large, it is infeasible to directly solve Problem (MLLE) or Problem (EC.5) for estimating the multi-choice rank list model. Thus, a feasible estimation approach is needed. In what follows, we introduce the multi-choice market discovery algorithm, a natural extension of the market discovery algorithm by van Ryzin and Vulcano (2015), for the rank list model in the multi-purchase setting. The key idea is to construct an initial feasible set of customer types and then iteratively add customer types that improve the model's likelihood until no new types are identified or a stopping criterion is met.

EC.2.1 Construct an Initial Feasible Set of Customer Types

Although the entire set of possible customer types, T , is very large, we can start with a subset, $T_0 \subseteq T$, that can still explain the historical data. If some marketing surveys or insights are available, we can apply them directly in constructing the initial set. If not, we provide a straightforward way to construct T_0 , which uses the smallest number of types that can explain all customer behaviors.

To distinguish between ordered and unordered sets, we call the ordered set a list, denoted as $[\cdot]$. For any set S , $[S]$ denotes mapping S to a list with elements ordered alphabetically by default (in practice, if timestamps of all actions are available, we can order the elements in increasing order of their timestamps, with earlier actions appearing first). We define the “ $\oplus$ ” operation between two lists as follows: if $[A] = [a_1, a_2, \dots, a_m]$ and $[B] = [b_1, b_2, \dots, b_n]$, then $[A] \oplus [B] = [a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n]$. We now present the algorithm for constructing an initial feasible set of customer types, outlined in Algorithm 2. The idea of the algorithm is to start with the customers purchasing the most products and gradually introduce new customer types to $\{(0, [0])\}$ if the current customer cannot be explained by the existing types. To help understand Algorithm 2, we have some remarks: (i) from Step 2 to Step 9, we separate the T customers into $q_{\max} + 1$ groups according

Algorithm 2: Construct an Initial Feasible Set of Customer Types

Input : $\mathbb{N} = \{1, 2, \dots, N\}$, $(P_t, C_t, K_t, V_t)_{t \in [T]}$
Output: T_0

```

1  $q_{max} \leftarrow \max \{|P_1|, |P_2|, \dots, |P_T|\}$ ;
2  $CT_q \leftarrow \emptyset \forall q \in \{0, 1, \dots, q_{max}\}$ ;
3 for  $q \in \{0, 1, \dots, q_{max}\}$  do
4   for  $t \in \{1, \dots, T\}$  do
5     if  $|P_t| = q$  then
6        $CT_q \leftarrow CT_q \cup \{t\}$ ;
7     end
8   end
9 end
10  $T_0 \leftarrow \{(q_0, \sigma_0)\}$  with  $(q_0, \sigma_0) = (0, [0])$ ;
11 for  $q = q_{max}$  down to 1 do
12   for  $t \in CT_q$  do
13     if  $(P_t, C_t, K_t, V_t)$  cannot be explained by any customer type in  $T_0$  then
14        $\sigma_{new} \leftarrow [P_t] \oplus [0] \oplus [C_t \setminus P_t] \oplus [K_t \setminus C_t] \oplus [V_t \setminus K_t]$ 
15     end
16      $T_0 \leftarrow T_0 \cup \{(q, \sigma_{new})\}$ ;
17   end
18 end

```

to the number of products they actually purchased; (ii) from Step 10 to Step 18, we construct the simplest customer types that can explain the transactions and the corresponding customer behavior. In addition, T_0 is a subset of T , and for any customer, at least one type can be found in T_0 to explain the corresponding behavior. At this stage, all types excluded from T_0 are assumed to have zero probability of arrival.

EC.2.2 Formulating the Restricted Maximum Log-Likelihood Estimation Problem

We denote $\tilde{T}$ as the current collection of customer types being considered (initially, $\tilde{T} = T_0$) and $\tilde{\lambda}$ as the corresponding probabilities of arrival. Additionally, we let $\tilde{M} = |\tilde{T}| - 1$ denote the number of customer types excluding $(0, [0])$. The log-likelihood function restricted to the current set of customer types $\tilde{T}$ is defined as $LL(\tilde{\lambda}) = \sum_{t=1}^T \ln \left(\mathbb{P}(P_t, C_t, K_t | V_t, \tilde{\lambda}) \right) = \sum_{t=1}^T \ln \left(\sum_{(q, \sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\lambda}_{q, \sigma} \right)$, where $\tilde{T}_{P_t, C_t, K_t, V_t}$ denotes as collection of the customer types in $\tilde{T}$ that can explain (P_t, C_t, K_t, V_t) . It worth noting that the vector of appearance probabilities, $\tilde{\lambda}$, maps to $\tilde{T}$, and the appearance probabilities of customer types excluded from $\tilde{\lambda}$ are set to zero. Accordingly, Problem (MLLE) restricted to the current set of customer types $\tilde{T}$, denoted as (rMLLE), is defined as follows:

$$\begin{aligned}
 \text{(rMLLE)} : \max_{\tilde{\lambda} \geq 0} & LL(\tilde{\lambda}) \\
 \text{s.t.} & \sum_{(q, \sigma) \in \tilde{T}} \tilde{\lambda}_{q, \sigma} = 1.
 \end{aligned} \tag{EC.7}$$

It is evident that Problem (rMLLE) is a concave constrained maximization problem for any given collection of customer types $\tilde{T}$. Therefore, the restricted maximum log-likelihood estimation problem can be solved (the optimal solution is not necessarily unique) using standard nonlinear optimization methods or off-the-shelf solvers such as IPOPT, or through an efficient expectation-maximization (EM) method developed by van Ryzin and Vulcano (2017).

EC.2.3 Introducing New Customer Types to Increase Likelihood

Once we have obtained an optimal solution of $\tilde{\lambda}$ to Problem (rMLLE) with respect to $\tilde{T}$, we can construct a feasible solution to Problem (MLLE) with respect to the entire set of possible customer types, T , by setting the corresponding λ to zero for all types in $T \setminus \tilde{T}$.

Given $\tilde{T}$ and the corresponding $\tilde{\lambda}$, we define $\tilde{\gamma} \in \mathbb{R}^T$ with $\tilde{\gamma}_t = \sum_{(q,\sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\lambda}_{q,\sigma} = \mathbb{P}(P_t, C_t, K_t | V_t, \tilde{\lambda})$ for all $t \in [T]$, and define $\tilde{A} \in \{0, 1\}^{T \times (\tilde{M}+1)}$ with entries $\tilde{a}_{t,m+1} = 1$ if $(q_m, \sigma_m) \in \tilde{T}_{P_t, C_t, K_t, V_t}$ and $\tilde{a}_{t,m+1} = 0$ otherwise. Then, we can formulate restricted maximum log-likelihood estimation problem with respect to $(\tilde{\lambda}, \tilde{\gamma})$:

$$\begin{aligned} \max_{\tilde{\lambda}, \tilde{\gamma} \geq 0} \quad & LL(\tilde{\lambda}) \\ \text{s.t.} \quad & \sum_{(q,\sigma) \in \tilde{T}} \tilde{\lambda}_{q,\sigma} = 1, \\ & \tilde{A}\tilde{\lambda} = \tilde{\gamma}. \end{aligned} \tag{EC.8}$$

Similar to Problem (rMLLE), Problem (EC.8) is also a concave constrained maximization problem with bounded objective $LL(\tilde{\lambda}) \leq 0$ for all $\tilde{\lambda}, \tilde{\gamma} \geq 0$. If $\tilde{\lambda}^*$ is optimal for Problem (rMLLE), then $(\tilde{\lambda}^*, \tilde{\gamma}^*)$, where $\tilde{\gamma}_t^* = \sum_{(q,\sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\lambda}_{q,\sigma}^*$, is an optimal solution to Problem (EC.8).

Similar to the discussion in Proposition EC.3, the strong duality also holds for Problem (EC.8). Let $g_{\tilde{T}}(\tilde{\lambda}, \tilde{\gamma}, \tilde{\beta}, \tilde{\mu})$ be the Lagrangian function restricted to the current set of customer types, $\tilde{T}$. Suppose that $(\tilde{\lambda}^*, \tilde{\gamma}^*)$ is a primal optimal solution and $(\tilde{\beta}^*, \tilde{\mu}^*)$ is a dual optimal solution. We define $G_{\tilde{T}}(\lambda, \gamma) = g_{\tilde{T}}(\lambda, \gamma, \tilde{\beta}^*, \tilde{\mu}^*)$. Because strong duality holds, the Karush-Kuhn-Tucker (KKT) conditions imply the following two statements:

- $\frac{\partial G_{\tilde{T}}(\lambda, \gamma)}{\partial \gamma_t} |_{(\lambda, \gamma) = (\tilde{\lambda}^*, \tilde{\gamma}^*)} = \frac{1}{\tilde{\gamma}_t^*} - \tilde{\mu}_t^* = 0$ for all $t \in [T]$, i.e., $\tilde{\mu}_t^* = \frac{1}{\tilde{\gamma}_t^*} \geq 1$ for all $t \in [T]$;
- $\frac{\partial G_{\tilde{T}}(\lambda, \gamma)}{\partial \lambda_{q,\sigma}} |_{(\lambda, \gamma) = (\tilde{\lambda}^*, \tilde{\gamma}^*)} = \tilde{\beta}^* + \sum_{t: (q,\sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\mu}_t^* = 0$ for all $(q, \sigma) \in \tilde{T}$, i.e., $-\tilde{\beta}^* = \sum_{t: (q,\sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\mu}_t^*$.

Because $\tilde{A}\tilde{\lambda}^* = \tilde{\gamma}^*$, then we have $(\tilde{\mu}^{*\top} \tilde{A})\tilde{\lambda}^* = \tilde{\mu}^{*\top} \tilde{\gamma}^* = T$ and $(\tilde{\mu}^{*\top} \tilde{A})\tilde{\lambda}^* = \sum_{(q,\sigma) \in \tilde{T}} (\sum_{t: (q,\sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\mu}_t^*) \tilde{\lambda}_{q,\sigma}^* = \sum_{(q,\sigma) \in \tilde{T}} (-\tilde{\beta}^*) \tilde{\lambda}_{q,\sigma}^* = (-\tilde{\beta}^*) \sum_{(q,\sigma) \in \tilde{T}} \tilde{\lambda}_{q,\sigma}^* = T$. Therefore, $\tilde{\beta}^* = -T$.

Now, considering the entire collection of customer types T . We define $G(\lambda, \gamma) = g(\lambda, \gamma, \tilde{\beta}^*, \tilde{\mu}^*)$. Note that if $\frac{\partial G(\lambda, \gamma)}{\partial \lambda_{q,\sigma}} \leq 0$ for all $(q, \sigma) \in T \setminus \tilde{T}$, then $(\tilde{\lambda}^*, \tilde{\gamma}^*, \tilde{\beta}^*, \tilde{\mu}^*)$ is a saddle point of $g(\lambda, \gamma, \beta, \mu)$. According to Proposition EC.3, $(\tilde{\lambda}^*, \tilde{\gamma}^*)$ is optimal to the primal problem (EC.5). Otherwise, there exists some $(q, \sigma) \in T \setminus \tilde{T}$ such that

$$\frac{\partial G(\lambda, \gamma)}{\partial \lambda_{q,\sigma}} = \sum_{t: (q,\sigma) \in \tilde{T}_{P_t, C_t, K_t, V_t}} \tilde{\mu}_t^* > T, \tag{EC.9}$$

customer type (q, σ) gives a direction to improve the Lagrangian dual function. Equation (EC.9) can be interpreted as follows. If a new customer type (q, σ) is added, then there is a perturbation on $\tilde{\lambda}$, which shifts ε probability mass to customer type (q, σ) and reduces the probability mass of the current set of customer types by $1 - \varepsilon$ such that the total probability remains 1. That is, $\varepsilon + \sum_{(q, \sigma) \in \tilde{\mathcal{T}}} (1 - \varepsilon) \tilde{\lambda}_{q, \sigma} = \varepsilon + (1 - \varepsilon) \sum_{(q, \sigma) \in \tilde{\mathcal{T}}} \tilde{\lambda}_{q, \sigma} = 1$. Since for any $t \in [T]$, $\tilde{\mu}_t^*$ gives the marginal change in the log-likelihood with respect to the changes in the probability $\tilde{\gamma}_t^*$, and the reduction of probability of all the current type by $(1 - \varepsilon)$ induces the reduction of the sum of any combination of $\lambda_{q, \sigma}$ (each γ_t is a combination of $\lambda_{q, \sigma}$) by the same factor. Then considering the Taylor expansion of $LL(\gamma)$ around $\tilde{\gamma}^*$, $LL(\gamma) = \sum_{t=1}^T \sum_{n=0}^{\infty} \frac{LL^{(n)}(\tilde{\gamma})}{n!} (-\varepsilon \tilde{\gamma}_t^*)^n = LL(\tilde{\gamma}^*) - \varepsilon T + O(\varepsilon^2)$, we find that the log-likelihood decreases by $\varepsilon \sum_{t=1}^T \tilde{\mu}_t^* \tilde{\gamma}_t^* = \varepsilon T$. Because of the introduction of the new customer type (q, σ) , the log-likelihood increases by $\varepsilon \sum_{t: (q, \sigma) \in \mathcal{T}_{P_t, \mathcal{C}_t, \mathcal{K}_t, \mathcal{V}_t}} \tilde{\mu}_t^*$. Therefore, Equation (EC.9) suggests that if the current collection of customer types is not optimal, there should be a new customer type (q, σ) that will increase the total likelihood.

Thus, we can solve Problem (EC.5) in an iterative way: starting from the initial feasible set of customer types $\mathcal{T}_0$, we check whether new customer types could be added to increase the likelihood. If no new types can increase the likelihood, the current customer types are optimal for Problem (EC.5). If new types can increase the likelihood, we add the new customer type, set the updated set of customer types as the new initial set, and restart the process. Since the possible customer types are finite, we can find the optimal (λ^*, γ^*) in a limited number of steps.

In fact, for any given q , checking whether (EC.9) holds is equivalent to solving the following optimization problem:

$$\begin{aligned} \max_{\omega} \quad & \sum_{t \in [T]} \tilde{\mu}_t \omega_t \\ \text{s.t.} \quad & \omega_t = 0 \end{aligned} \quad \text{for all } t \in [T] \text{ with } |\mathcal{P}_t| > q, \quad (\text{EC.10a})$$

$$\omega_t \leq \mathbb{I} \left(\begin{array}{l} \sigma(0) < \min_{v \in \mathcal{V}_t} \sigma(v) \\ \max_{c \in \mathcal{C}_t} \sigma(c) < \min_{k \in \{\mathcal{K}_t \setminus \mathcal{C}_t\}} \sigma(k) \\ \max_{k \in \mathcal{K}_t} \sigma(k) < \min_{v \in \{\mathcal{V}_t \setminus \mathcal{K}_t\}} \sigma(v) \\ \min_{v \in \mathcal{V}_t^+} \sigma(v) > q \end{array} \right) \quad \text{for all } t \in [T] \text{ with } \mathcal{P}_t = \emptyset, \quad (\text{EC.10b})$$

$$\omega_t \leq \mathbb{I} \left(\begin{array}{l} \max_{p \in \mathcal{P}_t} \sigma(p) < \min_{c \in \{\mathcal{C}_t^+ \setminus \mathcal{P}_t\}} \sigma(c) \\ \max_{c \in \mathcal{C}_t} \sigma(c) < \min_{k \in \{\mathcal{K}_t \setminus \mathcal{C}_t\}} \sigma(k) \\ \max_{k \in \mathcal{K}_t} \sigma(k) < \min_{v \in \{\mathcal{V}_t \setminus \mathcal{K}_t\}} \sigma(v) \\ \min_{v \in \{\mathcal{V}_t^+ \setminus \mathcal{P}_t\}} \sigma(v) > q \end{array} \right) \quad \text{for all } t \in [T] \text{ with } \mathcal{P}_t \neq \emptyset, |\mathcal{P}_t| = q, \quad (\text{EC.10c})$$

$$\omega_t \leq \mathbb{I} \left(\begin{array}{l} \sigma(0) < \min_{v \in \{V_t \setminus P_t\}} \sigma(v) \\ \max_{p \in P_t} \sigma(p) < \min_{c \in \{C_t^+ \setminus P_t\}} \sigma(c) \\ \max_{c \in C_t} \sigma(c) < \min_{k \in \{K_t \setminus C_t\}} \sigma(k) \\ \max_{k \in K_t} \sigma(k) < \min_{v \in \{V_t \setminus K_t\}} \sigma(v) \\ \min_{v \in \{V_t^+ \setminus P_t\}} \sigma(v) > q \end{array} \right) \quad \text{for all } t \in [T] \text{ with } P_t \neq \emptyset, |P_t| < q, \quad (\text{EC.10d})$$

where $\omega \in \{0, 1\}^T$ are binary variables, with $\omega_t = 1$ if the new customer type can explain $\tilde{T}_{P_t, C_t, K_t, V_t}$ and $\omega_t = 0$ otherwise. For a given q , if the objective value of Problem (EC.10) is greater than T , then there exists a new type with IPQ being q that can increase the likelihood.

Define binary variables $\mathbf{x} \in \{0, 1\}^{(N+1) \times (N+1)}$ with $x_{ij} = 1$ if product i is preferred to product j and $x_{ij} = 0$ otherwise (according to the Noncoincidence Assumption, two products will not have the same preference). Then, we can further reformulate Problem (EC.10) as the following mixed-integer linear programming:

$$\begin{aligned} \max_{\mathbf{x}, \omega} \quad & \sum_{t \in [T]} \tilde{\mu}_t \omega_t \\ \text{s.t.} \quad & x_{ij} + x_{ji} = 1 \quad \forall i, j, 0 \leq i < j \leq N, \end{aligned} \quad (\text{EC.11a})$$

$$x_{ij} + x_{jk} + x_{ki} \leq 2 \quad \forall i, j, k, i \neq j \neq k, 0 \leq i, j, k \leq N, \quad (\text{EC.11b})$$

$$\sum_{i=1}^N x_{i,0} \geq q, \quad (\text{EC.11c})$$

$$\omega_t = 0 \quad \forall t \in [T] \text{ with } |P_t| > q, \quad (\text{EC.11d})$$

$$\omega_t \leq x_{0,v} \quad \forall v \in V_t, t \in [T] \text{ with } P_t = \emptyset, \quad (\text{EC.11e})$$

$$\omega_t \leq x_{c,k} \quad \forall c \in C_t, k \in \{K_t \setminus C_t\}, t \in [T] \text{ with } P_t = \emptyset, \quad (\text{EC.11f})$$

$$\omega_t \leq x_{k,v} \quad \forall k \in K_t, v \in \{V_t \setminus K_t\}, t \in [T] \text{ with } P_t = \emptyset, \quad (\text{EC.11g})$$

$$q\omega_t \leq \sum_{u \in N \setminus V_t} x_{u,v} \quad \forall v \in V_t^+, t \in [T] \text{ with } P_t = \emptyset, \quad (\text{EC.11h})$$

$$\omega_t \leq x_{0,v} \quad \forall v \in \{V_t \setminus P_t\}, t \in [T] \text{ with } P_t \neq \emptyset, |P_t| < q, \quad (\text{EC.11i})$$

$$\omega_t \leq x_{p,c} \quad \forall p \in P_t, c \in \{C_t^+ \setminus P_t\}, t \in [T] \text{ with } P_t \neq \emptyset, |P_t| \leq q, \quad (\text{EC.11j})$$

$$\omega_t \leq x_{c,k} \quad \forall c \in C_t, k \in \{K_t \setminus C_t\}, t \in [T] \text{ with } P_t \neq \emptyset, |P_t| \leq q, \quad (\text{EC.11k})$$

$$\omega_t \leq x_{k,v} \quad \forall k \in K_t, v \in \{V_t \setminus K_t\}, t \in [T] \text{ with } P_t \neq \emptyset, |P_t| \leq q, \quad (\text{EC.11l})$$

$$q\omega_t \leq \sum_{p \in P_t} x_{p,v} + \sum_{u \in N \setminus V_t} x_{u,v} \quad \forall v \in V_t^+ \setminus P_t, t \in [T] \text{ with } P_t \neq \emptyset, |P_t| \leq q, \quad (\text{EC.11m})$$

$$x_{ij}, \omega_t \in \{0, 1\} \quad \forall i, j, 0 \leq i, j \leq N, t \in [T]. \quad (\text{EC.11n})$$

Here, we have some remarks on the MILP formulation: (i) Constraint (EC.11a) ensures that either product i is preferred to product j or product j is preferred to product i ; (ii) Constraint (EC.11b) ensures that among every three products, they follow a linear order; (iii) Constraint (EC.11c) prevents the outside option from ranking before the q -th position (otherwise, the preference ranking contradicts the quantity of distinct

products that a customer intends to choose); (iv) Constraint (EC.11d) to (EC.11m) ensure that the new customer type should be able to explain (P_t, C_t, K_t, V_t) with t be the one that the new customer type should be added to $\tilde{T}_{P_t, C_t, K_t, V_t}$; (v) Constraint (EC.11h) can actually be covered by Constraint (EC.11e); similarly, Constraint (EC.11m) can be covered by Constraint (EC.11i). The reason for retaining these two constraints is to keep consistency with Equation (EC.10) and to make it clearer to understand, but they need not be included in the actual estimation process.

If the optimal objective value of Problem (EC.11) is greater than T , then we have identified a new customer type that can increase the likelihood. Specifically, the new type is (q, σ_{new}) , where $\sigma_{new}(i) = \sum_{j \in N^+, j \neq i} x_{ji} + 1$ for all $i \in N^+$.

EC.2.4 Implementing the Multi-Choice Market Discovery Algorithm

An important feature of the multi-choice rank list model is that each customer type contains information not only on product preferences but also on the number of products the customer intends to purchase. Consider customers of type (q, σ) with $q \geq 1$. They intend to buy q products and will do so if enough items on their preference lists are available. However, if not enough products of interest are offered, they will buy fewer than q products. This indicates that customer types with larger q can explain a wider range of customer choice behaviors. Specifically, the purchase of $q' < q$ products can potentially be explained by a customer type with an intended purchase quantity of q . For illustration, consider the following example.

EXAMPLE EC.2. Consider a situation with $N = \{1, 2, 3, 4\}$. Customer A viewed $\{1, 2, 3\}$, clicked $\{1, 2, 3\}$, added $\{1, 2\}$ to the shopping cart, and purchased $\{1\}$; customer B viewed $\{1, 2, 3, 4\}$, clicked $\{1, 2, 3, 4\}$, added $\{1, 2, 4\}$ to the shopping cart, and purchased $\{1, 4\}$. Their underlying customer types are not attainable, but we find that $(2, \sigma) = (2, [4, 1, 2, 3, 0])$ may explain the purchase behavior of both customer A and customer B since customer A did not have a look at product 4 and there are chances that customer A actually prefers 4 than all other products. In other words, if customer A once has a look at product 4, then she will add it to the cart and purchase it.

According to this property of the multi-choice rank list model, to use as few customer types as possible to reasonably explain all the transactions and customer behavior, we conduct the multi-choice market discovery algorithm in the decreasing order of the intended purchase quantity. For any given IPQ, the proposed algorithm resembles the market discovery algorithm by van Ryzin and Vulcano (2015).

Combining all the discussions above, we present the Multi-Choice Market Discovery Algorithm, outlined in Algorithm 1. To further clarify, we provide the following remarks: (i) In Step 8, it is not necessary to solve MILP (EC.11) to optimality; a feasible solution with an objective value greater than T is sufficient to identify a new customer type that can increase the likelihood. (ii) Discovering all new customer types that can increase the likelihood can be very time-consuming. Therefore, we can apply a log-likelihood

ratio test, as used in van Ryzin and Vulcano (2015), to check whether a new customer type statistically significantly increases the likelihood. Specifically, for the set of customer types, $\tilde{T}$, at any iteration, we have the corresponding likelihood $LL(\tilde{\lambda})$; for the consecutive iteration after adding one new customer type, we have the increased likelihood $LL(\lambda_{new})$. We then conduct a hypothesis test where the null hypothesis is that the probability mass of the newly added customer type is zero, and the alternative hypothesis is that it is not zero. According to Theorem 17.5 in Greene (2003), the larger sample distribution $-2(LL(\tilde{\lambda}) - LL(\lambda_{new}))$ follows a χ^2 -distribution with one degree of freedom, and the critical value of the statistic at the 5% significance level is 3.842. Thus, the criteria for breaking to next q in Step 9 can be replaced with whether the log-likelihood ratio test is passed.

EC.3 Estimation Methods for Benchmark Models

Here, we briefly discuss how the multinomial logit model, the independent choice model, and the multi-purchase multinomial logit model can be estimated by maximum likelihood estimation. For all these three models, it is challenging to incorporate the behavior-revealed preference rules into model estimation as rank list models do. Therefore, we consider only the offer set, which comprises the products displayed on the mobile screen in our case study.

Estimation of the Classic Single-Purchase Multinomial Logit Model

Since the MNL model is not designed to handle multi-purchase behavior, the dataset is adjusted by decomposing each multi-purchase event into multiple single-purchase instances (details are provided in Section 6.1). The adjusted dataset is denoted as $\{(p_t, C_t, K_t, V_t)\}_{t \in [T']}$.

For MNL, let $\mathbf{v} = [v_1, v_2, \dots, v_N]^\top$ be the mean utilities of the N products, then the choice probability of product $i \in V$ can be represented as $\mathbb{P}(i|V, \mathbf{v}) = \frac{\exp(v_i)}{1 + \sum_{j \in V} \exp(v_j)}$, and the corresponding no purchase probability is $\mathbb{P}(0|V, \mathbf{v}) = \frac{1}{1 + \sum_{j \in V} \exp(v_j)}$. Same as the FC-RLM, the MNL cannot be used to explain cases with multi-purchase, so we also used the same adjusted data, $\{(p_t, V_t)\}_{t \in [T']}$, to estimate the MNL. Thus, the maximum log-likelihood estimator can be calculated by solving the following problem:

$$\max_{\mathbf{v}} \sum_{t=1}^{T'} \ln(\mathbb{P}(i_t | V_t, \mathbf{v}))$$

using many off-the-shelf solvers, like IPOPT (Interior Point OPTimizer).

Estimation of the Independent Choice Model

For the ICM, let $p_i = \mathbb{P}(U_i > 0) \forall i \in N$ and $\mathbf{p} = [p_1, p_2, \dots, p_N]^\top$. Given T distinct customers and their corresponding (P_t, V_t) , we have the log-likelihood function with respect to $\mathbf{p}$: $LL(\mathbf{p}) = \sum_{t=1}^T \ln(\mathbb{P}(P_t | V_t, \mathbf{p})) = \sum_{t=1}^T \prod_{i \in P} p_i \cdot \prod_{j \in V \setminus P} (1 - p_j)$. Then, the maximum log-likelihood estimate, $\hat{\mathbf{p}}$, is an optimal solution of the following problem:

$$\begin{aligned} \max_{\mathbf{p}} \quad & LL(\mathbf{p}) \\ \text{s.t.} \quad & 0 \leq p_i \leq 1 \quad \forall i \in N, \end{aligned}$$

and it can also be solved by many off-the-shelf solvers.

Estimation of the Multi-Purchase Multinomial Logit Model

For the multi-purchase multinomial-logit model, we present a different approach to represent choice probabilities compared to Bai et al. (2023). While their approach focuses on the choice probability of each single alternative, ours emphasizes the choice probability of each subset. Similar to the classic single-purchase multinomial logit model, let $\mathbf{v} = [v_1, v_2, \dots, v_N]^\top$ represent the mean utilities of the N alternatives, and let $v_0 = 1$ represents the mean utility of the outside option without loss of generality. Additionally, we assume that a random customer has an intended purchase quantity $q \in \{0, 1, 2, \dots, q_{\max}\}$ with probability α_q , where $\sum_{q \in \{0, 1, 2, \dots, q_{\max}\}} \alpha_q = 1$. Given a set $S = \{s_1, s_2, \dots, s_j\}$, we let $\beta \in B_S$ denote a permutation of elements in S , where B_S represents the collection of all permutations. Particularly, β can be regarded as an ordered list $[\beta_1, \beta_2, \dots, \beta_j]$. Then, given a non-empty offer set $V \subseteq N$, the choice probability of selecting $P = \{n_1, n_2, \dots, n_{|P|}\} \subseteq V$ is represented as follows:

$$\begin{aligned} \mathbb{P}(P \mid V, \mathbf{v}) &= \alpha_{|P|} \cdot \sum_{\beta \in B_P} \frac{e^{v_{\beta_1}}}{1 + \sum_{n \in V} e^{v_n}} \cdot \frac{e^{v_{\beta_2}}}{1 + \sum_{n \in V \setminus \{\beta_1\}} e^{v_n}} \cdots \frac{e^{v_{\beta_{|P|}}}}{1 + \sum_{n \in V \setminus \{\beta_1, \beta_2, \dots, \beta_{|P|-1}\}} e^{v_n}} \\ &\quad + \left(\sum_{q > |P|} \alpha_q \right) \cdot \sum_{\beta \in B_P} \frac{e^{v_{\beta_1}}}{1 + \sum_{n \in V} e^{v_n}} \cdot \frac{e^{v_{\beta_2}}}{1 + \sum_{n \in V \setminus \{\beta_1\}} e^{v_n}} \cdots \frac{e^{v_{\beta_{|P|}}}}{1 + \sum_{n \in V \setminus \{\beta_1, \beta_2, \dots, \beta_{|P|-1}\}} e^{v_n}} \cdot \frac{1}{1 + \sum_{n \in V \setminus P} e^{v_n}} \\ &= \left(\alpha_{|P|} + \frac{\sum_{q > |P|} \alpha_q}{1 + \sum_{n \in V \setminus P} e^{v_n}} \right) \cdot \sum_{\beta \in B_P} \frac{e^{v_{\beta_1}}}{1 + \sum_{n \in V} e^{v_n}} \cdot \frac{e^{v_{\beta_2}}}{1 + \sum_{n \in V \setminus \{\beta_1\}} e^{v_n}} \cdots \frac{e^{v_{\beta_{|P|}}}}{1 + \sum_{n \in V \setminus \{\beta_1, \beta_2, \dots, \beta_{|P|-1}\}} e^{v_n}}. \end{aligned}$$

To this end, we can estimate the multi-purchase multinomial-logit model by solving the following maximum log-likelihood estimation problem:

$$\max_{\boldsymbol{\alpha}, \mathbf{v}} \sum_{t=1}^T \ln(\mathbb{P}(P \mid V, \mathbf{v})),$$

where $\boldsymbol{\alpha} = [\alpha_0, \alpha_1, \dots, \alpha_{q_{\max}}]$.

EC.4 Efficiency Analysis of the MILP Formulation for Assortment Recommendation

To rigorously evaluate the computational efficiency of the MILP formulation (3), we conduct a comprehensive set of synthetic experiments. These experiments examine the impact of key parameters, including the number of products N , the number of customer types $|T|$ (referring only to those with positive arrival probabilities, as others do not affect efficiency), and the maximum intended purchase quantity $q_{\max}$, on the time required to find an optimal solution. In each experiment, one parameter is varied while the others are kept constant to assess its effect on computational time. Each test case is repeated across 100 simulations to ensure robustness.

In each simulation, prices for the N products are drawn from a uniform distribution over the interval $[1, 10)$. The multi-choice rank list model includes $|T|$ customer types, explicitly incorporating the no-purchase type $(0, [0])$, which does not affect the optimal assortment computation. For each customer type,

the intended purchase quantity q is uniformly sampled from $\{1, 2, \dots, q_{\max}\}$. The preference ranking σ is a random permutation of $\{0, 1, \dots, N\}$, ensuring that the number of products ranked above the outside option is at least equal to the intended purchase quantity. Choice probabilities, λ , are randomly generated and normalized so that their sum equals one. The MILPs are solved using Gurobi v10.

First, we examine the impact of varying the number of products on the time required to solve the optimization problem. For this set of experiments, the number of customer types is held constant at $|T| = 20$ and the maximum IPQ is set at $q_{\max} = 3$. The results are illustrated in Figures EC.1 and EC.2. Notice that

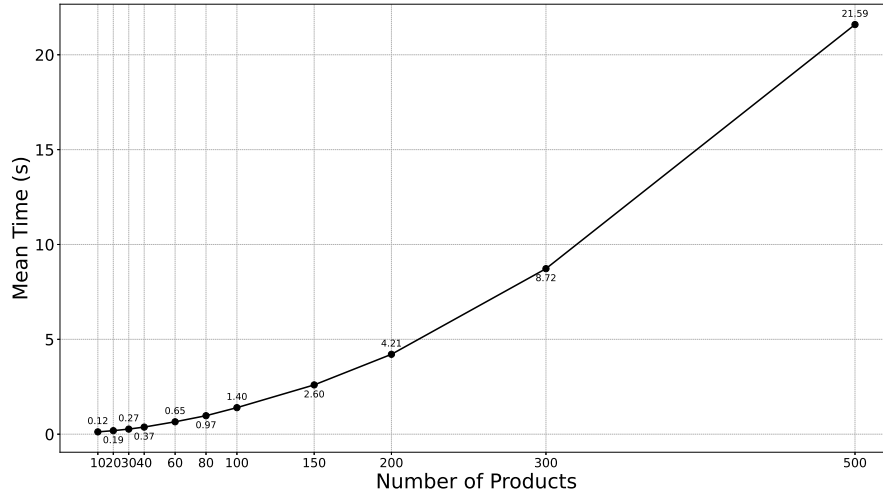


Figure EC.1 Mean Time for Solving MILP VS. Number of Products with $|T| = 20$ and $q_{\max} = 3$

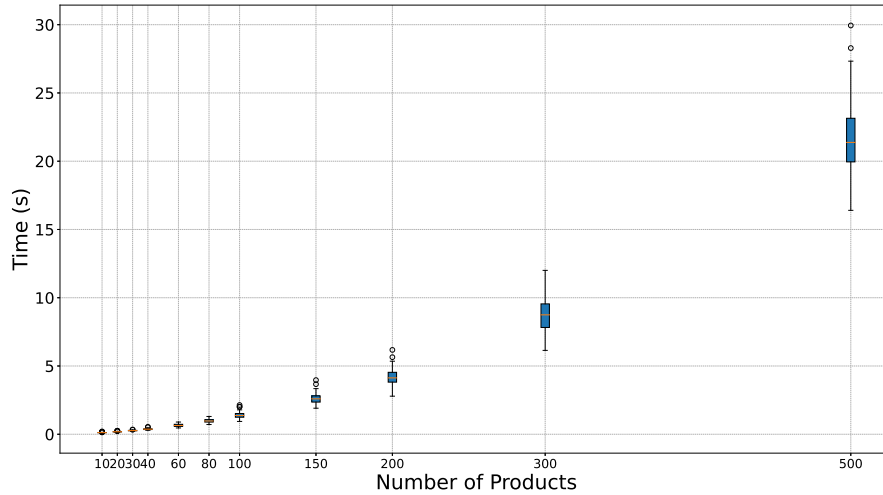


Figure EC.2 Distribution of Time for Solving MILP VS. Number of Products with $|T| = 20$ and $q_{\max} = 3$

the mean computational time exhibits a rising trend as the number of products increases, a result consistent with expectations when holding both the number of customer types and the maximum IPQ constant.

Additionally, our findings reveal an increase in the variance of the computational time required to solve the MILP as the number of products escalates.

Next, we focus on understanding how the computational time is influenced by changes in the number of customer types. In this set of experiments, the number of products is fixed at $N = 60$ and the maximum IPQ remains constant at $q_{max} = 3$. The results are illustrated in Figures EC.3 and EC.4. It is observed that

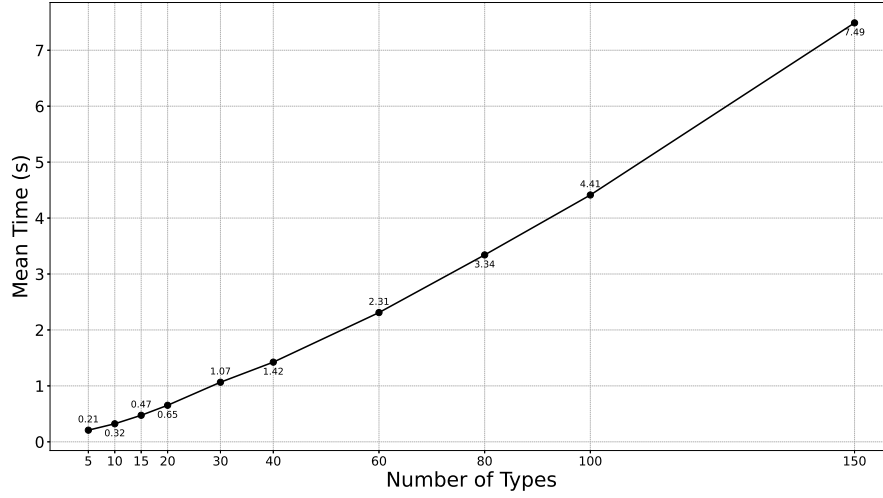


Figure EC.3 Mean Time for Solving MILP VS. Number of Customer Types with $N = 60$ and $q_{max} = 3$

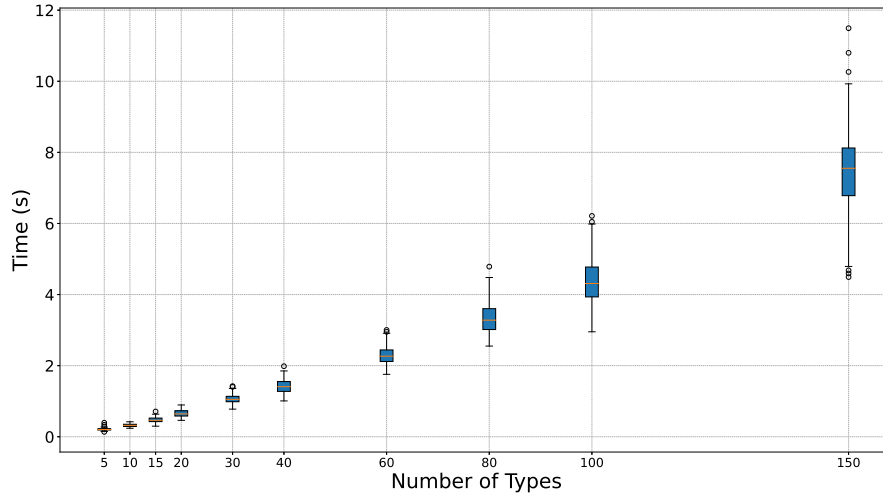


Figure EC.4 Distribution of Time for Solving MILP VS. Number of Customer Types with $N = 60$ and $q_{max} = 3$

the computational time also tends to increase as the number of customer types escalates, while keeping the number of products and the maximum IPQ fixed. This observation aligns with theoretical expectations. In addition, we observe a corresponding increase in the variance of computational times, suggesting that the complexity of the MILP formulation scales with the diversity of customer types.

Finally, we examine the effects of varying the maximum IPQ on the computational efficiency of the optimization problem. Here, the number of products and customer types are held constant at $N = 60$ and $|T| = 60$, respectively. The results are illustrated in Figures EC.5 and EC.6. Similarly, a noticeable uptick

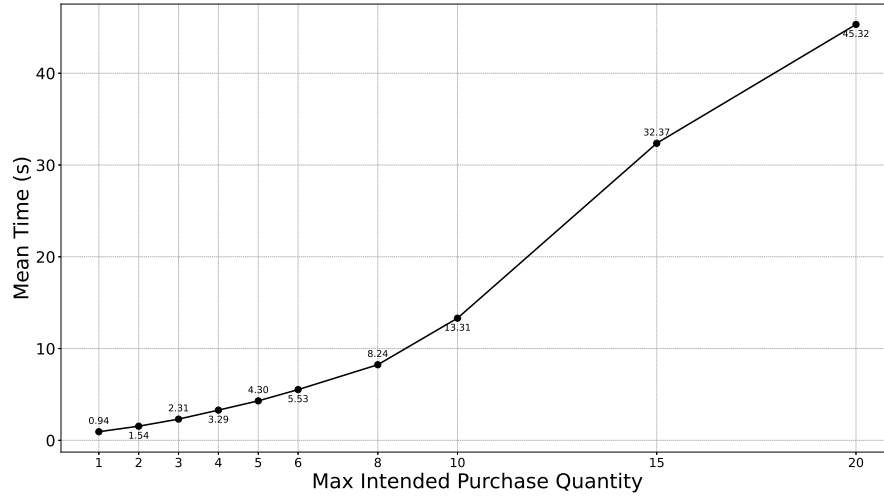


Figure EC.5 Mean Time for Solving MILP VS. Max IPQ with $N = 60$ and $|T| = 60$

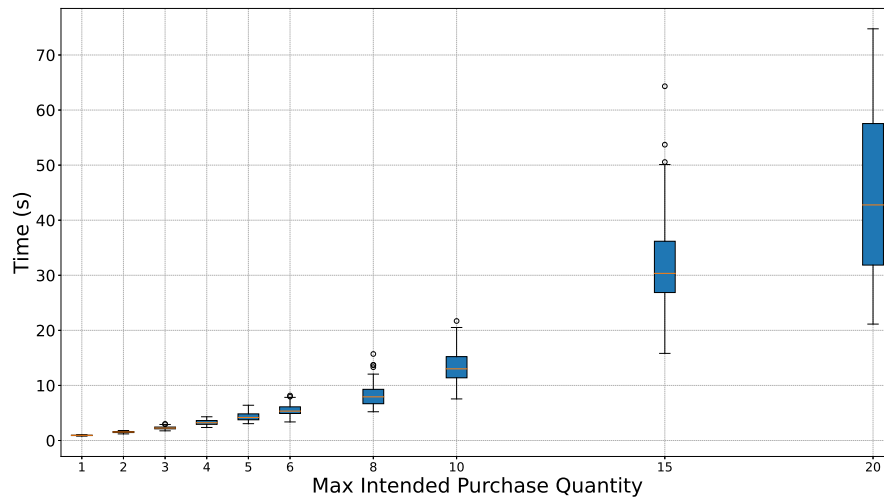


Figure EC.6 Distribution of Time for Solving MILP VS. Max IPQ with $N = 60$ and $|T| = 60$

in both the mean and variance of the computational time is evident as the maximum IPQ is incremented, while maintaining constant levels for the number of products and customer types.

In summary, as expected, increasing any of the parameters, the number of products, customer types, or the maximum intended purchase quantity, results in longer computational times for solving the MILP to obtain an optimal solution.

EC.5 Supporting Materials for Case Study with Tmall Dataset

EC.5.1 Supplementary Results for Experiments Where the Cart Set Serves as the Offer Set

This subsection provides summary statistics corresponding to Table 1, Table 2, and Table 3, offering a consolidated overview of the experimental results.

Table EC.1 Summary Statistics Corresponding to Table 1

	mean	std	min	median	max
Num Prod.	27.69	15.46	9.0	26.0	58.0
Max Purch. Qty	3.46	0.52	3.0	3.0	4.0
No Purch. Prop. (%)	91.37	6.69	76.36	93.77	96.64
Single Purch. Prop. Norm. (%)	74.71	6.89	58.82	76.74	85.11
Multi Purch. Prop. Norm. (%)	25.29	6.89	14.89	23.26	41.18
ICM LL	-12.34	12.31	-45.45	-7.11	0.0
MNL LL	-35.82	30.52	-113.53	-27.12	-7.45
MPMNL LL	-243.53	241.71	-694.41	-207.23	-3.64
MCRLM LL	-67.15	43.98	-190.82	-72.25	-12.48
SCRLM LL	-102.15	61.86	-261.6	-98.34	-30.09
ICM AIC_c	110.01	59.4	37.0	119.41	243.49
MNL AIC_c	156.97	97.33	54.91	127.84	349.63
MPMNL AIC_c	609.02	474.46	64.15	571.38	1526.37
MCRLM AIC_c	143.11	88.84	29.75	155.2	391.88
SCRLM AIC_c	216.23	123.59	75.17	209.66	535.56
MPMNL Time (s)	0.27	0.23	0.06	0.18	0.85
MCRLM Time (s)	7.41	10.02	0.72	3.17	37.36
SCRLM Time (s)	9.52	15.51	0.53	3.31	54.34
MCRLM Num Types	4.08	0.95	2.0	4.0	5.0
SCRLM Num Types	5.31	0.48	5.0	5.0	6.0

Table EC.2 Summary Statistics Corresponding to Table 2

	mean	std	min	median	max
ICM Train RMSE	0.53	0.11	0.25	0.57	0.64
MNL Train RMSE	0.57	0.06	0.46	0.59	0.66
MPMNL Train RMSE	0.64	0.12	0.44	0.64	0.85
MCRLM Train RMSE	0.5	0.01	0.48	0.5	0.51
SCRLM Train RMSE	0.56	0.05	0.5	0.55	0.69
MCRLM Improv. ICM RMSE (%)	-0.88	32.82	-100.0	11.88	24.94
MCRLM Improv. MNL RMSE (%)	12.25	10.58	-6.14	14.72	24.37
MCRLM Improv. MPMNL RMSE (%)	19.58	15.87	-14.32	24.88	39.85
MCRLM Improv. SCRLM RMSE (%)	10.8	6.69	3.97	8.91	27.42

Table EC.3 Summary Statistics Corresponding to Table 3

	mean	std	min	median	max
ICM Test RMSE	0.65	0.11	0.33	0.67	0.76
MNL Test RMSE	0.66	0.09	0.5	0.66	0.8
MPMNL Test RMSE	0.72	0.08	0.57	0.72	0.89
MCRLM Test RMSE	0.53	0.03	0.49	0.52	0.61
SCRLM Test RMSE	0.6	0.07	0.51	0.59	0.76
MCRLM Improv. ICM RMSE (%)	16.33	21.08	-50.0	22.44	33.58
MCRLM Improv. MNL RMSE (%)	19.45	10.34	0.85	20.08	36.27
MCRLM Improv. MPMNL RMSE (%)	25.96	7.76	12.32	25.46	38.49
MCRLM Improv. SCRLM RMSE (%)	11.98	9.37	1.25	10.28	34.62

EC.5.2 Supplementary Results for Experiments Where the Click Set Serves as the Offer Set

This subsection presents the results of experiments using the click set as the offer set, which exhibit patterns and insights similar to those in Section 6.1.1, where the cart set is used as the offer set. Table EC.4, Table EC.6, and Table EC.8 report detailed estimation results, in-sample performance comparisons, and out-of-sample performance comparisons, respectively. Corresponding summary statistics are provided in Table EC.5, Table EC.7, and Table EC.9, offering a consolidated view of the experimental outcomes. These findings further reinforce the conclusions from the main text, particularly the superior predictive accuracy of the proposed MC-RLM in most cases.

Table EC.4 Estimation Results for Experiments Using the Click Set as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM LL	MNL LL	MPMNL LL	MCRLM LL	SCRLM LL	ICM AIC_c	MNL AIC_c	MPMNL AIC_c	MCRLM AIC_c	SCRLM AIC_c	MCRLM Num Types	SCRLM Num Types
9-5209	Wampee Fruit Snacks	13	3	96.64	79.31	20.69	-36.82	-38.25	-81.33	-41.67	-53.24	113.64	116.49	224.49	97.89	121.02	6	6
9-5746	Candied Pomelo Snacks	36	3	92.54	77.78	22.22	-80.59	-86.85	-81.09	-96.9	-125.37	276.85	289.38	299.71	206.72	265.99	6	7
17-3513	Seashell Decorations	58	3	96.39	79.59	20.41	-193.84	-193.54	-197.21	-251.76	-319.02	540.28	539.68	561.11	515.88	654.66	6	8
17-6551	Handmade Organic Soaps	9	4	95.98	58.82	41.18	-27.08	-34.05	-172.7	-28.97	-52.34	82.15	96.11	405.7	67.69	120.69	4	6
21-1432	Glass Door Hinges	40	3	92.26	75.31	24.69	-84.47	-94.88	-88.34	-110.23	-153.37	291.54	312.36	318.93	233.23	319.49	6	6
21-5587	Heat Shrink Tubing	26	3	86.59	77.97	22.03	-80.21	-81.09	-95.7	-91.01	-120.43	234.02	235.78	281.9	192.73	253.84	5	6
21-7250	Grooving Cutting Inserts	39	4	95.59	75.76	24.24	-101.88	-100.91	-99.69	-112.9	-151.65	326.32	324.39	348.29	238.62	318.39	6	7
22-1863	Beef Skewers & Meatballs	9	4	94.13	76.74	23.26	-72.61	-77.02	-72.04	-80.57	-107.74	166.13	174.94	179.46	179.43	228.76	8	6
22-7517	Mammated Mud Snails	25	3	96.12	85.11	14.89	-61.27	-59.02	-101.64	-82.13	-97.63	196.62	192.11	296.08	179.81	210.81	7	7
35-1542	Br Shumin Book	10	4	76.36	69.23	30.77	-4.92	-16.01	-166.4	-12.48	-31.85	61.28	83.46	602.79	29.75	78.7	2	5
35-1729	Computer Exam Software	45	4	78.83	68.12	31.88	-92.31	-125.49	-153.07	-102.29	-174.01	343.63	409.98	498.88	222.07	360.86	8	6
52-6505	Fountain Spray Nozzles	20	3	92.59	79.41	20.59	-54.03	-49.49	-52.65	-60.93	-66.08	172.06	162.97	185.8	135.57	143.35	6	5
64-2877	PVC Cable Conduits	30	4	93.77	68.12	31.88	-93.86	-103.24	-90.86	-105.2	-149.72	272.53	291.28	287.72	227.89	314.59	8	7

Table EC.5 Summary Statistics Corresponding to Table EC.4

	mean	std	min	median	max
Num Prod.	27.69	15.46	9.0	26.0	58.0
Max Purch. Qty	3.46	0.52	3.0	3.0	4.0
No Purch. Prop. (%)	91.37	6.69	76.36	93.77	96.64
Single Purch. Prop. Norm. (%)	74.71	6.89	58.82	76.74	85.11
Multi Purch. Prop. Norm. (%)	25.29	6.89	14.89	23.26	41.18
ICM LL	-75.68	45.59	-193.84	-80.21	-4.92
MNL LL	-81.53	46.1	-193.54	-81.09	-16.01
MPMNL LL	-111.75	44.85	-197.21	-95.7	-52.65
MCRLM LL	-90.54	58.16	-251.76	-91.01	-12.48
SCRLM LL	-123.27	74.04	-319.02	-120.43	-31.85
ICM AIC_c	236.7	128.49	61.28	234.02	540.28
MNL AIC_c	248.38	131.71	83.46	235.78	539.68
MPMNL AIC_c	345.45	135.46	179.46	299.71	602.79
MCRLM AIC_c	194.41	117.34	29.75	192.73	515.88
SCRLM AIC_c	260.86	148.47	78.7	253.84	654.66
MPMNL Time (s)	0.52	0.91	0.06	0.23	3.51
MCRLM Time (s)	18.93	37.83	0.81	4.82	141.3
SCRLM Time (s)	11.98	18.36	0.4	3.7	65.8
MCRLM Num Types	6.0	1.68	2.0	6.0	8.0
SCRLM Num Types	6.31	0.85	5.0	6.0	8.0

Table EC.6 In-Sample Performance of Models in Experiments Where the Click Set Serves as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM Train RMSE	MNL Train RMSE	MPMNL Train RMSE	MCRLM Train RMSE	SCRLM Train RMSE	MCRLM Improv. ICM RMSE (%)	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.4288	0.4409	0.4802	0.4027	0.399	6.07	8.67	16.13	-0.93
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.4979	0.4784	0.4916	0.3868	0.3887	22.33	19.16	21.33	0.51
17-3513	Seashell Decorations	58	3	79.59	20.41	0.4237	0.4332	0.4151	0.3514	0.3173	17.06	18.89	15.35	-10.76
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.5227	0.539	0.3738	0.4282	0.3841	18.06	20.55	-14.57	-11.5
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.5356	0.5167	0.5039	0.4004	0.3694	25.24	22.51	20.54	-8.38
21-5587	Heat Shrink Tubing	26	3	77.97	22.03	0.4527	0.4724	0.4591	0.4042	0.4037	10.71	14.45	11.97	-0.11
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.4072	0.3867	0.4011	0.3602	0.3263	11.55	6.87	10.19	-10.4
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.4558	0.4615	0.4509	0.4131	0.3921	9.38	10.5	8.39	-5.35
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.4249	0.4503	0.4351	0.3327	0.333	21.68	26.1	23.52	0.08
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.3988	0.4969	0.5197	0.5	0.5474	-25.39	-0.62	3.79	8.66
35-1729	Computer Exam Software	45	4	68.12	31.88	0.4886	0.4821	0.4811	0.4348	0.4456	11.03	9.82	9.63	2.44
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.3682	0.3759	0.3696	0.3152	0.3104	14.41	16.16	14.73	-1.53
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.4771	0.4664	0.4776	0.4087	0.3784	14.32	12.37	14.41	-8.01

Table EC.7 Summary Statistics Corresponding to Table EC.6

	mean	std	min	median	max
ICM Train RMSE	0.45	0.05	0.37	0.45	0.54
MNL Train RMSE	0.46	0.05	0.38	0.47	0.54
MPMNL Train RMSE	0.45	0.05	0.37	0.46	0.52
MCRLM Train RMSE	0.4	0.05	0.32	0.4	0.5
SCRLM Train RMSE	0.38	0.06	0.31	0.38	0.55
MCRLM Improv. ICM RMSE (%)	12.03	12.56	-25.39	14.32	25.24
MCRLM Improv. MNL RMSE (%)	14.26	7.31	-0.62	14.45	26.1
MCRLM Improv. MPMNL RMSE (%)	11.95	9.7	-14.57	14.41	23.52
MCRLM Improv. SCRLM RMSE (%)	-3.48	6.1	-11.5	-1.53	8.66

Table EC.8 Out-of-Sample Performance of Models in Experiments Where the Click Set Serves as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM Test RMSE	MNL Test RMSE	MPMNL Test RMSE	MCRLM Test RMSE	SCRLM Test RMSE	MCRLM Improv. ICM RMSE (%)	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.4596	0.4689	0.494	0.4111	0.4208	10.56	12.33	16.78	2.31
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.5199	0.5008	0.5137	0.398	0.4063	23.45	20.53	22.53	2.05
17-3513	Seashell Decorations	58	3	79.59	20.41	0.4489	0.4563	0.4424	0.3603	0.3239	19.75	21.05	18.57	-11.22
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.5236	0.527	0.3851	0.4316	0.3942	17.56	18.1	-12.09	-9.5
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.5685	0.5358	0.536	0.44	0.4089	22.6	17.88	17.91	-7.59
21-5587	Heat Shrink Tubing	26	3	77.97	22.03	0.4902	0.5019	0.5075	0.4308	0.4305	12.11	14.15	15.1	-0.08
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.424	0.4226	0.4284	0.3722	0.3351	12.2	11.91	13.12	-11.08
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.4453	0.4473	0.4392	0.4038	0.3797	9.32	9.73	8.08	-6.34
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.421	0.458	0.4415	0.3346	0.3368	20.52	26.95	24.22	0.65
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.4454	0.6499	0.7005	0.5	0.6534	-12.27	23.06	28.62	23.48
35-1729	Computer Exam Software	45	4	68.12	31.88	0.5414	0.4646	0.5003	0.4344	0.4499	19.76	6.5	13.18	3.46
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.3788	0.3821	0.3826	0.3341	0.3309	11.81	12.55	12.68	-0.98
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.5288	0.497	0.5239	0.4293	0.3948	18.81	13.61	18.05	-8.75

Table EC.9 Summary Statistics Corresponding to Table EC.8

	mean	std	min	median	max
ICM Test RMSE	0.48	0.06	0.38	0.46	0.57
MNL Test RMSE	0.49	0.06	0.38	0.47	0.65
MPMNL Test RMSE	0.48	0.08	0.38	0.49	0.7
MCRLM Test RMSE	0.41	0.05	0.33	0.41	0.5
SCRLM Test RMSE	0.41	0.09	0.32	0.39	0.65
MCRLM Improv. ICM RMSE (%)	14.32	9.32	-12.27	17.56	23.45
MCRLM Improv. MNL RMSE (%)	16.03	5.81	6.5	14.15	26.95
MCRLM Improv. MPMNL RMSE (%)	15.13	9.81	-12.09	16.78	28.62
MCRLM Improv. SCRLM RMSE (%)	-1.81	9.33	-11.22	-0.98	23.48

EC.5.3 Supplementary Results for Experiments Where the Complete Set Serves as the Offer Set

This subsection presents the results of experiments conducted under the setting where only purchase information is available and the complete set is used as the offer set. These experiments serve as a baseline for comparison against models that incorporate side information via BRP rules during the estimation process. Table EC.10, Table EC.12, and Table EC.14 report detailed estimation results, in-sample performance comparisons, and out-of-sample performance comparisons, respectively. The corresponding summary statistics are presented in Table EC.11, Table EC.13, and Table EC.15.

Table EC.10 Estimation Results for Experiments Using the Complete Set as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM LL	MNL LL	MPMNL LL	MCRLM LL	SCRLM LL	ICM AIC_c	MNL AIC_c	MPMNL AIC_c	MCRLM AIC_c	SCRLM AIC_c	MPMNL Solve Time (s)	MCRLM Solve Time (s)	SCRLM Solve Time (s)	MCRLM Num Types	SCRLM Num Types
9-5209	Wampee Fruit Snacks	13	3	96.64	79.31	20.69	-165.52	-165.29	-460.52	-90.56	-119.73	357.57	357.12	955.96	187.15	247.52	27.12	1.45	1.71	3	4
9-5746	Candied Pomelo Snacks	36	3	92.54	77.78	22.22	-412.76	-411.19	-1128.28	-175.77	-236.77	901.7	898.55	2341.72	355.57	481.6	90.63	6.51	6.94	2	4
17-3513	Seashell Decorations	38	3	96.39	79.39	20.41	-1165.67	-1163.54	-2832.22	-323.45	-673.03	2449.49	2445.23	5790.88	1052.9	1358.08	690.7	25.2	52.94	3	6
17-6551	Handmade Organic Soaps	9	4	95.98	58.82	41.18	-127.39	-127.29	-322.37	-58.28	-99.81	273.33	273.13	674.03	122.64	209.8	17.66	1.4	1.05	3	5
21-1432	Glass Door Hinges	40	3	92.26	75.31	24.69	-497.27	-495.52	-1358.54	-213.35	-286.55	1078.66	1075.16	2810.07	430.72	583.18	118.33	7.48	8.31	2	5
21-5587	Heat Shrink Tubing	26	3	86.59	77.97	22.03	-334.41	-331.75	-1059.19	-136.46	-185.01	725.13	719.83	2184.18	279.0	382.27	38.99	2.94	4.2	3	6
21-7250	Grooving Cutting Inserts	39	4	95.59	75.76	24.24	-517.39	-516.31	-1266.44	-223.13	-308.2	1115.47	1113.33	2624.31	452.28	628.46	131.21	9.11	10.5	3	6
22-1863	Beef Skewers & Meatballs	9	4	94.13	76.74	23.26	-265.87	-265.13	-828.94	-135.24	-199.54	350.05	348.58	1686.61	274.5	409.19	0.53	1.32	1.63	2	5
22-7517	Marinated Mud Snails	25	3	96.12	85.11	14.89	-332.39	-331.64	-921.04	-166.62	-203.38	716.16	714.66	1901.94	337.24	416.83	123.05	2.67	3.59	2	5
35-1542	Bi Shumin Book	10	4	76.36	69.23	30.77	-61.68	-61.46	-168.96	-22.29	-44.75	150.02	149.59	385.07	51.18	101.08	3.11	1.16	0.59	3	5
35-1729	Computer Exam Software	45	4	78.83	68.12	31.88	-476.93	-473.51	-1220.38	-131.48	-239.29	1063.21	1056.36	2565.15	269.05	490.91	98.51	13.16	31.32	3	6
52-6505	Fountain Spray Nozzles	20	3	92.59	79.41	20.59	-194.53	-193.58	-644.73	-98.95	-116.48	431.49	429.58	1338.68	203.97	241.06	46.97	1.33	1.38	3	4
64-2877	PVC Cable Conduits	30	4	93.77	68.12	31.88	-466.31	-463.06	-1220.38	-200.53	-290.43	994.79	992.3	2513.73	407.09	590.93	180.26	5.41	5.92	3	5

Table EC.11 Summary Statistics Corresponding to Table EC.10

	mean	std	min	median	max
Num Prod.	27.69	15.46	9.0	26.0	58.0
Max Purch. Qty	3.46	0.52	3.0	3.0	4.0
No Purch. Prop. (%)	91.37	6.69	76.36	93.77	96.64
Single Purch. Prop. Norm. (%)	74.71	6.89	58.82	76.74	85.11
Multi Purch. Prop. Norm. (%)	25.29	6.89	14.89	23.26	41.18
ICM LL	-386.01	278.43	-1165.67	-334.41	-61.68
MNL LL	-384.71	277.87	-1163.54	-331.75	-61.46
MPMNL LL	-1033.23	662.19	-2832.22	-1059.19	-168.96
MCRLM LL	-167.39	122.47	-523.45	-136.46	-22.29
SCRLM LL	-231.0	155.28	-673.03	-203.38	-44.75
ICM AIC_c	831.31	585.19	150.02	725.13	2449.49
MNL AIC_c	828.72	584.04	149.59	719.83	2445.23
MPMNL AIC_c	2136.33	1351.25	385.07	2184.18	5790.88
MCRLM AIC_c	340.25	244.84	51.18	279.0	1052.9
SCRLM AIC_c	472.38	311.14	101.08	416.83	1358.08
MPMNL Time (s)	120.54	180.21	0.53	90.63	690.7
MCRLM Time (s)	6.09	6.84	1.16	2.94	25.2
SCRLM Time (s)	10.01	15.23	0.59	4.2	52.94
MCRLM Num Types	2.69	0.48	2.0	3.0	3.0
SCRLM Num Types	5.08	0.76	4.0	5.0	6.0

Table EC.12 In-Sample Performance of Models in Experiments Where the Complete Set Serves as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM Train RMSE	MNL Train RMSE	MPMNL Train RMSE	MCRLM Train RMSE	SCRLM Train RMSE	MCRLM Improv. ICM RMSE (%)	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.0527	0.0527	0.0528	0.0543	0.0536	-3.17	-3.17	-2.97	-1.32
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.0504	0.0504	0.0505	0.0544	0.0526	-7.93	-7.93	-7.71	-3.39
17-3513	Seashell Decorations	58	3	79.59	20.41	0.0282	0.0282	0.0282	0.0292	0.0286	-3.76	-3.76	-3.68	-2.38
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.0847	0.0847	0.085	0.0873	0.0861	-3.07	-3.07	-2.6	-1.32
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.0475	0.0475	0.0476	0.0512	0.0486	-7.68	-7.68	-7.38	-5.22
21-5587	Heat Shrink Tubing	26	3	77.97	22.03	0.0792	0.0792	0.0796	0.0878	0.0809	-10.84	-10.84	-10.29	-8.55
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.0392	0.0392	0.0393	0.0413	0.0398	-5.34	-5.34	-5.21	-3.86
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.0949	0.0949	0.0955	0.1018	0.0978	-7.19	-7.19	-6.58	-4.09
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.044	0.044	0.0441	0.0462	0.0444	-4.95	-4.95	-4.81	-3.96
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.18	0.18	0.1811	0.2077	0.2005	-15.44	-15.39	-14.7	-3.62
35-1729	Computer Exam Software	45	4	68.12	31.88	0.0847	0.0847	0.0852	0.1028	0.0954	-21.36	-21.35	-20.65	-7.81
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.0658	0.0658	0.066	0.0687	0.0686	-4.37	-4.37	-3.97	-0.05
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.052	0.052	0.0521	0.0564	0.0533	-8.42	-8.42	-8.16	-5.83

Table EC.13 Summary Statistics Corresponding to Table EC.12

	mean	std	min	median	max
ICM Train RMSE	0.07	0.04	0.03	0.05	0.18
MNL Train RMSE	0.07	0.04	0.03	0.05	0.18
MPMNL Train RMSE	0.07	0.04	0.03	0.05	0.18
MCRLM Train RMSE	0.08	0.05	0.03	0.06	0.21
SCRLM Train RMSE	0.07	0.04	0.03	0.05	0.2
MCRLM Improv. ICM RMSE (%)	-7.96	5.31	-21.36	-7.19	-3.07
MCRLM Improv. MNL RMSE (%)	-7.96	5.3	-21.35	-7.19	-3.07
MCRLM Improv. MPMNL RMSE (%)	-7.59	5.15	-20.65	-6.58	-2.6
MCRLM Improv. SCRLM RMSE (%)	-3.95	2.47	-8.55	-3.86	-0.05

Table EC.14 Out-of-Sample Performance of Models in Experiments Where the Complete Set Serves as the Offer Set

CATE	Description	Num Prod.	Max Purch. Qty	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	ICM Test RMSE	MNL Test RMSE	MPMNL Test RMSE	MCRLM Test RMSE	SCRLM Test RMSE	MCRLM Improv. ICM RMSE (%)	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
9-5209	Wampee Fruit Snacks	13	3	79.31	20.69	0.0729	0.0729	0.073	0.0738	0.0735	-1.22	-1.22	-1.08	-0.45
9-5746	Candied Pomelo Snacks	36	3	77.78	22.22	0.0543	0.0543	0.0544	0.0579	0.0564	-6.48	-6.48	-6.39	-2.56
17-3513	Seashell Decorations	58	3	79.59	20.41	0.0256	0.0256	0.0256	0.0267	0.026	-4.2	-4.2	-4.18	-2.61
17-6551	Handmade Organic Soaps	9	4	58.82	41.18	0.0881	0.0881	0.0886	0.0905	0.09	-2.64	-2.64	-2.16	-0.54
21-1432	Glass Door Hinges	40	3	75.31	24.69	0.0597	0.0597	0.0598	0.0624	0.0604	-4.58	-4.58	-4.42	-3.29
21-5587	Heat Shrink Tubing	26	3	77.97	22.03	0.0833	0.0833	0.0836	0.092	0.0854	-10.4	-10.4	-10.01	-7.69
21-7250	Grooving Cutting Inserts	39	4	75.76	24.24	0.032	0.032	0.032	0.0347	0.0326	-8.31	-8.31	-8.32	-6.27
22-1863	Beef Skewers & Meatballs	9	4	76.74	23.26	0.0776	0.0776	0.0778	0.0858	0.0802	-10.49	-10.51	-10.31	-6.93
22-7517	Marinated Mud Snails	25	3	85.11	14.89	0.0385	0.0385	0.0386	0.041	0.0391	-6.32	-6.32	-6.23	-4.9
35-1542	Bi Shumin Book	10	4	69.23	30.77	0.2751	0.2761	0.2759	0.2921	0.2986	-6.17	-5.79	-5.87	2.18
35-1729	Computer Exam Software	45	4	68.12	31.88	0.0932	0.0932	0.0936	0.1103	0.1031	-18.35	-18.36	-17.92	-6.98
52-6505	Fountain Spray Nozzles	20	3	79.41	20.59	0.0736	0.0736	0.0737	0.0764	0.0762	-3.71	-3.71	-3.57	-0.2
64-2877	PVC Cable Conduits	30	4	68.12	31.88	0.0574	0.0574	0.0575	0.0612	0.0586	-6.71	-6.71	-6.5	-4.44

Table EC.15 Summary Statistics Corresponding to Table EC.14

	mean	std	min	median	max
ICM Test RMSE	0.08	0.06	0.03	0.07	0.28
MNL Test RMSE	0.08	0.06	0.03	0.07	0.28
MPMNL Test RMSE	0.08	0.06	0.03	0.07	0.28
MCRLM Test RMSE	0.08	0.07	0.03	0.07	0.29
SCRLM Test RMSE	0.08	0.07	0.03	0.07	0.3
MCRLM Improv. ICM RMSE (%)	-6.89	4.4	-18.35	-6.32	-1.22
MCRLM Improv. MNL RMSE (%)	-6.86	4.41	-18.36	-6.32	-1.22
MCRLM Improv. MPMNL RMSE (%)	-6.69	4.35	-17.92	-6.23	-1.08
MCRLM Improv. SCRLM RMSE (%)	-3.44	3.09	-7.69	-3.29	2.18

EC.5.4 Supplementary Results for Experiments Where Side Information Is Used in Estimating RLMS

This subsection presents summary statistics corresponding to Table 4 and Table 5, providing a consolidated overview of the experimental results in which side information is incorporated into the estimation of both single- and multi-choice RLMS.

Table EC.16 Summary Statistics Corresponding to Table 4

	mean	std	min	median	max
Num Prod.	27.69	15.46	9.0	26.0	58.0
No Purch. Prop. (%)	91.37	6.69	76.36	93.77	96.64
Single Purch. Prop. Norm. (%)	74.71	6.89	58.82	76.74	85.11
Multi Purch. Prop. Norm. (%)	25.29	6.89	14.89	23.26	41.18
MCRLM Num Types	6.08	1.85	2.0	6.0	9.0
SCRLM Num Types	6.85	1.68	5.0	6.0	9.0
MCRLM LL	-195.67	140.53	-602.41	-167.8	-22.29
SCRLM LL	-255.72	170.15	-739.51	-228.43	-46.51
MCRLM AIC_c	403.69	282.63	48.88	351.86	1218.85
SCRLM AIC_c	525.42	342.06	104.61	468.95	1497.08
MCRLM Time (s)	18.94	32.02	0.87	7.05	120.05
SCRLM Time (s)	24.3	54.76	0.41	4.61	203.4

Table EC.17 Summary Statistics Corresponding to Table 5

	mean	std	min	median	max
MCRLM Train RMSE	0.06	0.03	0.03	0.05	0.15
SCRLM Train RMSE	0.07	0.03	0.03	0.05	0.16
Train MCRLM Improv. SCRLM RMSE (%)	0.84	2.18	-0.19	0.11	7.99
MCRLM Test RMSE	0.07	0.05	0.03	0.07	0.23
SCRLM Test RMSE	0.08	0.06	0.03	0.07	0.26
Test MCRLM Improv. SCRLM RMSE (%)	1.19	2.83	-0.88	0.33	10.13
Train MCRLM RMSE Improv. With vs. Without Side Info (%)	12.08	8.1	4.45	9.4	29.14
Train SCRLM RMSE Improv. With vs. Without Side Info (%)	8.01	6.26	3.19	6.11	22.42
Test MCRLM RMSE Improv. With vs. Without Side Info (%)	9.65	6.13	2.35	8.59	22.96
Test SCRLM RMSE Improv. With vs. Without Side Info (%)	5.51	4.26	1.91	3.66	16.32

EC.6 Case Study with Cainiao Network

To demonstrate the predictive performance of the multi-choice rank list model and the benefits of incorporating BRP rules, we present a supplementary case study using real-world data from Cainiao Network. The Cainiao mobile app features a program called *Get Gift Money Every Day*, where customers receive a small amount of money that can be redeemed by making a purchase within the app. In addition to regular advertisements, customers can search for specific products, with the search results page displaying available gift money at the top, as shown in Figure 3.

The data records what products are exposed⁸ to customers and customers' subsequent behaviors, including clicking, adding to the shopping cart, and purchasing, which exactly correspond to the four behaviors used in our model. We utilize the data recorded on two random consecutive weekdays (November 24th to 25th, 2021) to conduct our experiments. Since we study the multi-choice behavior over substitutable products, we conduct experiments separately over products of different categories. These categories belong to many different types, including toys, stationery, plants, data cables, storage bags, etc. For data sensitivity, we would only present the analog of the category ID (i.e., A to L) instead of its explicit category name or ID.

In this case study, we benchmark the proposed MC-RLM against the classic first-choice rank list model (FC-RLM), the independent choice model (ICM), and the MNL model. For each category, the data is randomly split into training and test sets with a 4:1 ratio to evaluate the predictive performance of each model. For performance metric, in addition to RMSE, we also compute the $RMSE_i$ to measure prediction error for individual customers. For a customer with offer set V , it is defined as: $RMSE_i = \sqrt{\frac{\sum_{v \in V} (v_i - \hat{v}_i)^2}{|V|}}$. Notably, down-sampling is not employed in this case study.

EC.6.1 The Benefit of Considering the Multi-Choice Behavior

To show the benefit of considering customers' multi-choice behavior, we compare MC-RLM and the ICM with the first-choice RLM (FC-RLM) and MNL. Note that both the MNL model and the ICM do not easily adapt to incorporate the BRP rules. It would be unfair to estimate MNL and ICM using incomplete data while estimating MC-RLM and FC-RLM using the entire dataset. Therefore, we construct two sub-experiments to compare the four models. In the first sub-experiment, we ignore the viewing and clicking information and consider the add-to-cart set as the choice set. This setting is similar to the one used in Aouad et al. (2019), where the products that are clicked are treated as the consideration set. In the second sub-experiment, we ignore the clicking and adding-to-cart information and only use the viewing and purchasing

⁸ It is doable if we record all items that have been exposed on the screen of smartphones, and it is acceptable to assume that customers have viewed these products owing to the screen size. While in terms of e-commerce websites on computers' screens, some adjustments might apply.

data for the experiment. As discussed earlier, in this setting, the view set acts as the offer set, which is consistent with most assortment optimization literature (e.g., van Ryzin and Vulcano 2015).

To make a comprehensive comparison of the modeling power, we selected 11 different categories that contain multi-purchase behaviors. To perform the train-test split, we used ten sets of random seeds. If a category had an excessive number of products, we randomly selected 30, 40, or 50 products, which had a purchase history, for the experiments. As mentioned earlier, we conducted two sets of experiments. In the first sub-experiment, we focused on customers' choice decisions after they added products to their shopping cart, and the average results are presented in Table EC.18. We present the analysis of this experiment

Table EC.18 Prediction Results of Using Cart Information

Analog Category ID	Num of Products	MC Mean RMSE	FC Mean RMSE	MNL Mean RMSE	ICM Mean RMSE	Mean RMSE Improvement from FC to MC	Mean RMSE Improvement from MNL to MC	Mean RMSE Improvement from ICM to MC	MC mean types	FC mean types	MC AIC_c	FC AIC_c	MNL AIC_c
A	30	0.3785	0.5107	0.3995	0.3960	25.88%	5.24%	4.41%	5.9	7.6	110.23	195.90	144.61
B	30	0.4049	0.5468	0.4557	0.7329	25.96%	11.15%	44.73%	5.5	6.5	61.62	86.97	129.12
C	40	0.5350	0.6492	0.5719	0.9508	17.59%	6.46%	43.74%	7.7	9.2	76.58	99.80	203.61
D	30	0.4660	0.5848	0.4879	0.6101	20.31%	4.48%	23.62%	5.8	7.7	69.39	103.28	120.84
E	50	0.4904	0.5846	0.5923	0.7507	16.11%	17.20%	34.67%	8.9	13.5	119.77	195.16	170.06
F	40	0.4269	0.5372	0.5075	0.7082	20.53%	15.88%	39.72%	6.4	9.5	72.80	135.97	221.41
G	67	0.1625	0.4353	0.4644	0.8061	62.66%	65.01%	79.84%	4.5	10.1	39.92	119.23	347.79
H	55	0.4330	0.6145	0.6345	0.9115	29.53%	31.75%	52.49%	7.1	12.4	68.08	147.47	434.54
I	50	0.4619	0.5489	0.5889	0.7035	15.84%	21.57%	34.34%	7.0	10.7	126.24	175.56	190.13
J	33	0.4679	0.6136	0.5330	0.9079	23.74%	12.22%	48.47%	4.6	8.1	48.96	73.81	141.29
K	49	0.3501	0.5435	0.5681	0.6866	35.58%	38.38%	49.01%	6.0	13.1	55.19	169.89	182.22

together with that of the second sub-experiment below.

To better illustrate the predictive performance, we present the results of a specific experiment with purchases made from the products added to the shopping cart in *category L*, applying random seeds 12, 26, 95. The dataset consists of 33 products and 1336 transaction histories (customers). Within the dataset, there are 985 customers choosing the outside option, 297 customers purchasing one product, 51 customers purchasing two products, and 3 customers purchasing three products. In the training set, these numbers are 798, 227, 42, and 2 respectively, and in the test set, the numbers are 187, 70, 9, and 1. Moreover, in the training set, 789, 299, 47, and 4 customers added 0, 1, 2, and 3 products to the shopping cart, while in the test set, 186, 69, 10, and 2 customers added 0, 1, 2, and 3 products to the shopping cart. This dataset indicates that the products added to the shopping cart are a strong indicator of the intended purchase. Thus, we can presume that all four models would perform well. The results are shown in Figure EC.7. The RMSE values for the MC-RLM, FC-RLM, MNL, and ICM models are 0.2030, 0.5042, 0.3397, and 0.3112, respectively. Note that the vertical axis represents the RMSE for each individual test transaction (customer), not for the entire test set. The test transactions are separated based on the ground truth purchase quantity. In the upper left subplot of Figure EC.7, the MC-RLM clearly outperforms the other three models, though the MNL and ICM also perform well in most cases. Since most products added to the cart are eventually purchased, it is not surprising that all four models exhibit near-zero RMSE in the upper right subplot. The lower left subplot shows that, aside from a few outliers, the MNL performs best in predicting single-product purchases. This

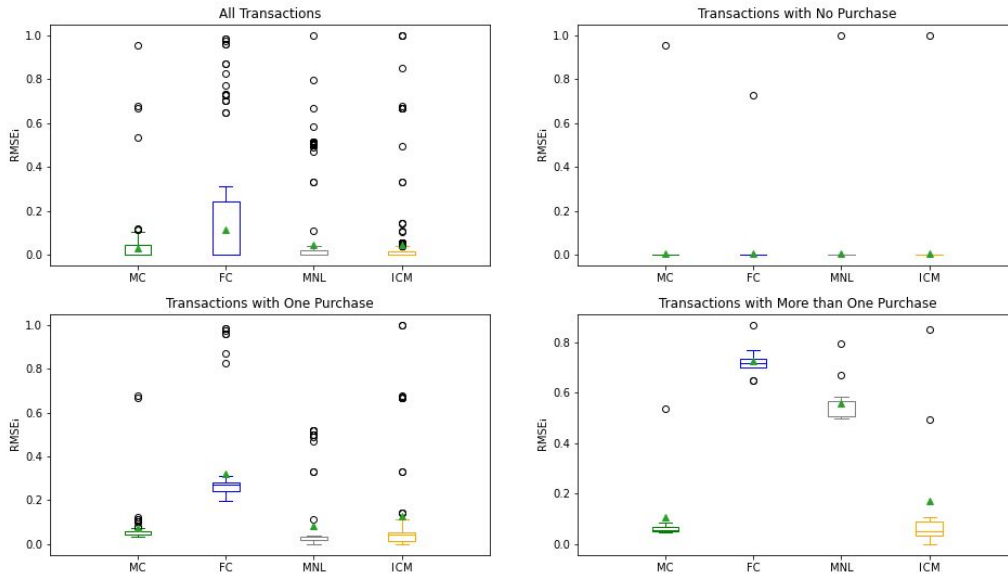


Figure EC.7 Detailed RMSE Comparison with Cart Information

is expected, as the MNL model is specifically designed for single-purchase scenarios, and most historical purchases in this dataset are single-product transactions, leading to strong performance. However, the MC-RLM achieves a lower average RMSE and, surprisingly, outperforms the FC-RLM. This suggests that while the MC-RLM is primarily designed for multi-purchase scenarios, its structure does not compromise its ability to handle single-purchase cases. In contrast, the FC-RLM's performance is impacted by the presence of multi-purchases and the adjustments made to the dataset to fit its single-purchase assumption during estimation. Finally, the lower right subplot focuses on transactions where more than one product is purchased. As expected, the models that account for multi-choice behavior, particularly the MC-RLM, perform the best, with the MC-RLM showing the lowest mean RMSE.

In the second sub-experiment, we use only the viewing and purchasing information to focus on customers' choice decisions over the entire offer sets. To ensure fairness, we use the same train-test split as in the previous sub-experiment. The average results are presented in Table EC.19.

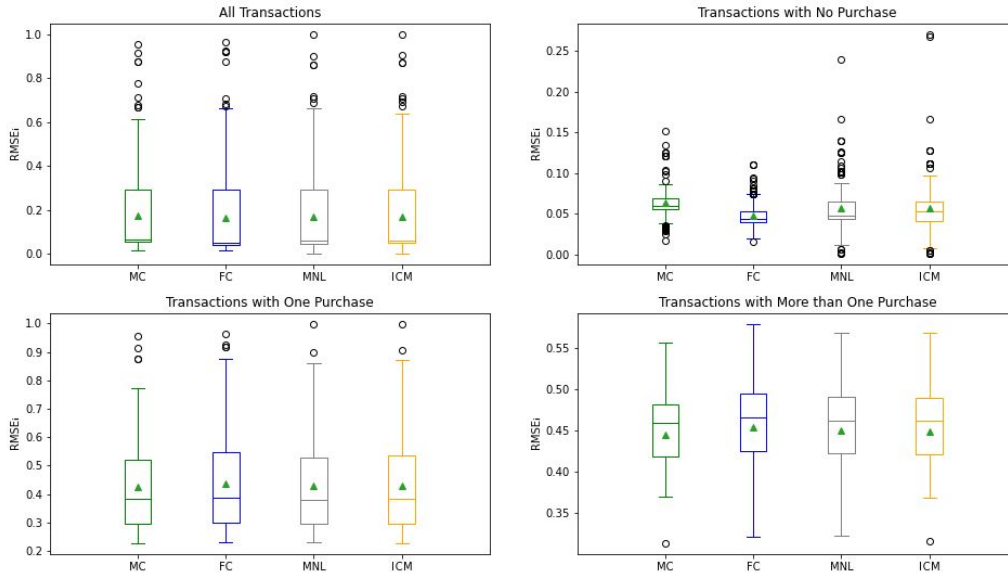
Table EC.18 and Table EC.19 demonstrate that the MC-RLM consistently outperforms the FC-RLM, the MNL model, and the ICM in most cases, particularly in the first sub-experiment where the MC-RLM has significantly lower RMSEs than the other three models. The resulting customer types are generally the most sparse, as indicated by the number of possible customer types and the corrected AIC values. Overall, the MC-RLM exhibits higher predictive accuracy than the other models when multi-purchases occur. Additionally, the classic FC-RLM frequently performs worse than MNL, a finding consistent with the empirical results reported in van Ryzin and Vulcano (2015). However, as a generalization of the FC-RLM,

Table EC.19 Prediction Results of Using View Information

Analog Category ID	Num of Products	MC Mean RMSE	FC Mean RMSE	MNL Mean RMSE	ICM Mean RMSE	Mean RMSE Improvement from FC to MC	Mean RMSE Improvement from MNL to MC	Mean RMSE Improvement from ICM to MC	MC mean types	FC mean types	MC AIC_c	FC AIC_c	MNL AIC_c
A	30	0.3367	0.3429	0.3203	0.3294	1.87%	-5.11%	-2.23%	19.7	18.7	496.85	560.60	457.88
B	30	0.5084	0.5121	0.5327	0.5913	0.73%	4.57%	14.03%	10.5	11.7	157.83	177.67	197.60
C	40	0.4675	0.4874	0.4809	0.5591	4.08%	2.79%	16.38%	14.2	14.7	170.59	196.98	278.26
D	30	0.4609	0.5222	0.5096	0.4836	11.74%	9.56%	4.68%	10.7	11.7	156.05	190.60	211.82
E	50	0.2719	0.2721	0.2795	0.2866	0.08%	2.70%	5.13%	31.2	33.6	452.40	508.22	433.34
F	40	0.4767	0.5139	0.5159	0.5593	7.25%	7.59%	14.77%	13.5	15.8	215.85	279.06	304.85
G	67	0.4405	0.4411	0.4904	0.5068	0.13%	10.17%	13.09%	22.0	26.5	370.17	401.89	559.64
H	55	0.5247	0.5531	0.5473	0.6544	5.14%	4.13%	19.82%	15.8	23.7	208.64	310.79	494.17
I	50	0.3223	0.3261	0.3242	0.3395	1.16%	0.58%	5.06%	24.0	28.6	473.19	518.25	469.10
J	33	0.5009	0.5036	0.5179	0.5752	0.54%	3.30%	12.93%	12.3	14.0	166.21	178.65	240.77
K	49	0.3307	0.3356	0.3260	0.3453	1.44%	-1.47%	4.23%	24.6	30.4	351.31	468.57	395.54

the proposed MC-RLM overcomes some of the FC-RLM's shortcomings and provides improved prediction performance.

We also conducted an experiment using purchase records from items viewed in *category L*, applying the same random seeds as the previous experiment. The results are shown in Figure EC.8. — The RMSE val-

**Figure EC.8** Detailed RMSE Comparison with View Information

ues for MC-RLM, FC-RLM, MNL, and ICM are 0.1977, 0.1982, 0.1975, and 0.2001, respectively. Nearly half of the customers viewed more than ten products before making a purchase, introducing significant noise into the prediction process. As a result, there is little noticeable difference in performance across the four models, except that the MC-RLM performs slightly better in predicting multi-product purchases. We will further discuss Figure EC.8 in the next subsection, where we demonstrate the benefit of incorporating the proposed BRP rules.

EC.6.2 The Value of Incorporating Behavior Information

Besides capturing customers' multi-choice behavior, another feature of our model is that we combine the BRP rules. In this subsection, we examine the value of the BRP rules. Specifically, we conduct the experiments using the full behavior information recorded. Again, precisely the same random seeds as the above two sets of experiments are used to train-test split the data.

Now, we have the average results of the experiments over different categories, which are shown in Table EC.20. Similar to the results in the last subsection, in almost all the cases, the proposed MC-RLM

Table EC.20 Prediction Results of Using Full Behavior Information

Analogue Category ID	Num of Products	MC Mean RMSE	FC Mean RMSE	MNL Mean RMSE	ICM Mean RMSE	Mean RMSE Improvement from FC to MC	Mean RMSE Improvement from MNL to MC	Mean RMSE Improvement from ICM to MC	MC mean types	FC mean types	MC AIC_c	FC AIC_c	MNL AIC_c
A	30	0.2848	0.2848	0.3203	0.3294	0.0%	11.08%	13.52%	13.8	11.4	428.70	456.06	457.88
B	30	0.4277	0.4319	0.5327	0.5913	0.98%	19.72%	27.67%	8.7	9.5	144.16	157.17	197.60
C	40	0.4392	0.4502	0.4809	0.5591	2.44%	8.67%	21.44%	13.5	13.5	162.21	182.99	278.26
D	30	0.4246	0.4982	0.5096	0.4836	14.76%	16.68%	12.18%	9.9	12.3	150.41	191.66	211.82
E	50	0.2524	0.2531	0.2795	0.2866	0.26%	9.68%	11.93%	21.4	21.5	380.14	415.40	433.34
F	40	0.3712	0.4024	0.5159	0.5593	7.77%	28.05%	33.64%	11.2	10.6	179.90	219.86	304.85
G	67	0.3610	0.3740	0.4904	0.5068	0.13%	10.17%	13.09%	15.8	18.8	370.17	401.89	559.64
H	55	0.4662	0.5104	0.5473	0.6544	8.67%	14.82%	28.76%	13.0	19.9	183.31	273.54	494.17
I	50	0.2990	0.2899	0.3242	0.3395	-3.13%	7.78%	11.93%	17.7	16.5	414.58	412.01	469.10
J	33	0.4483	0.4547	0.5179	0.5752	1.40%	13.45%	22.07%	10.0	11.5	148.82	151.43	240.77
K	49	0.3110	0.3138	0.3260	0.3453	0.88%	4.58%	9.94%	20.1	17.0	309.59	369.94	395.54

outperforms the FC-RLM, the MNL model, and the ICM. Compared the results in Table EC.20 with that in Table EC.19, we have Table EC.21. We find that when making use of the full behavior series, both the

Table EC.21 Comparison Between Only Using View Information and Using Full Behavior Information

Analogue Category ID	Num of Products	MC Mean RMSE with Partial Info	MC Mean RMSE with Full Info	Mean RMSE Improvement with Full Info under MC	FC Mean RMSE with Partial Info	FC Mean RMSE with Full Info	Mean RMSE Improvement with Full Info under FC
A	30	0.3367	0.2848	15.40%	0.3429	0.2848	16.94%
B	30	0.5084	0.4277	15.87%	0.5121	0.4319	15.66%
C	40	0.4675	0.4392	6.05%	0.4874	0.4502	7.623%
D	30	0.4609	0.4246	7.87%	0.5222	0.4982	4.61%
E	50	0.2719	0.2524	7.17%	0.2721	0.2531	7.01%
F	40	0.4767	0.3712	22.14%	0.5139	0.4024	21.70%
G	67	0.4405	0.3610	18.04%	0.4411	0.3741	15.19%
H	55	0.5247	0.4662	11.15%	0.5531	0.5104	7.71%
I	50	0.3223	0.2990	7.24%	0.3261	0.2899	11.10%
J	33	0.5009	0.4483	10.50%	0.5036	0.4547	9.72%
K	49	0.3307	0.3110	5.96%	0.3356	0.3138	6.49%

MC-RLM and the FC-RLM become much better than MNL and ICM. Additionally, the AIC_c of both RLMs is reduced to some extent, which means although fewer types are used to capture the customer population, the likelihood of the models is improved. Comparing all three sets of experiments, we conclude that by incorporating the BRP rules, both the MC-RLM and the FC-RLM improve significantly compared to not using any middle behavior information. Even though the FC-RLM works extremely poorly when predicting customers' choices from the shopping cart, the BRP rules help it beat the MNL and the ICM. This finding

demonstrates that the internal complicated preference relationship revealed by the full behavior information helps to improve the modeling accuracy significantly.

Again, we present the result of a specific experiment with full behavior records in *category L* applying the same random seeds as those in the previous subsection. The results are shown in Figure EC.9. The RMSE

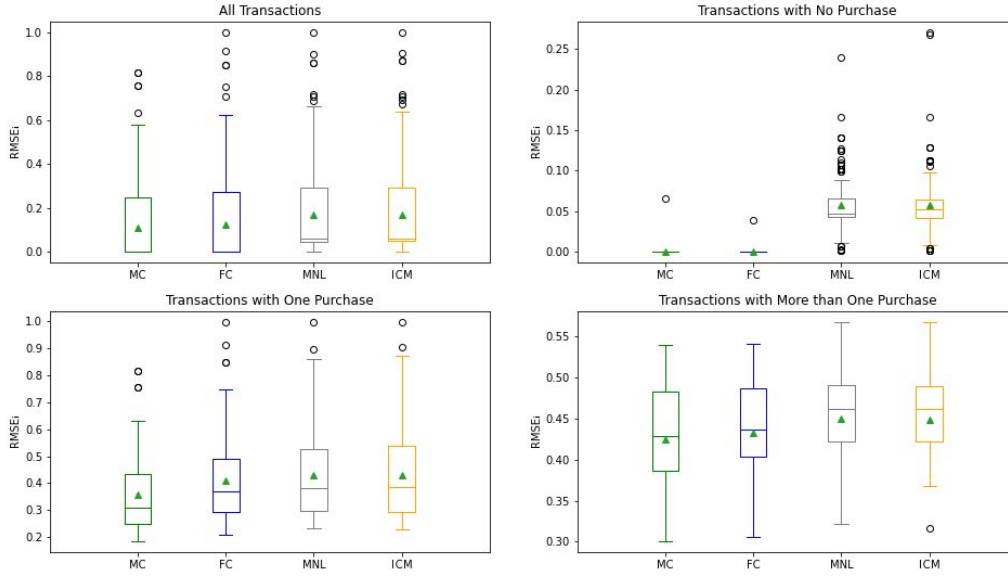


Figure EC.9 Detailed RMSE Comparison with Full Behavior Information

values for MC-RLM, FC-RLM, MNL, and ICM are 0.1647, 0.1833, 0.1975, and 0.2001, respectively, indicating that the proposed MC-RLM outperforms the other three models. The upper left subplot shows the average prediction performance across all models, which is consistent with the RMSE comparison. The upper right subplot shows that, unlike the MNL and ICM models, which make predictions based on the offer set (i.e., view set), both RLM models achieve nearly zero RMSE, largely benefiting from the use of behavioral data, particularly shopping cart information. If no product is added to the shopping cart, the customer cannot make a purchase. The lower right subplot reveals that for customers purchasing two or more products, the MC-RLM exhibits a lower predictive error. Surprisingly, the lower left subplot shows that even for single-product purchases, the MC-RLM still achieves a lower RMSE than the other models. This further demonstrates that adapting the classic rank list model to account for multi-purchase behavior does not compromise its ability to predict single-purchase scenarios. Comparing these results to Figure EC.7 and Figure EC.8, we observe that while cart information is a strong indicator of customer purchases, it does not guarantee highly accurate predictions. This highlights that the comprehensive preference relationships, revealed through full behavioral data, play a crucial role in improving predictive accuracy.

Moreover, it's worth noting that many studies in e-commerce data analysis, such as Jiang et al. (2020), Feldman et al. (2022), typically employ resampling of the training set to mitigate the impact of the large number of non-purchasing customers. However, we find that incorporating the BRP rules reduce the dependence on resampling and that its estimation performance is not greatly impacted. This is likely because the inclusion of behavior information enhances the learning of customer preferences, thereby reducing estimation error.

EC.7 Supporting Materials for Synthetic Experiments

The average estimation results, grouped by the total number of products $|N|$, are shown in Table EC.22. It is

Table EC.22 Average Estimation Results Across Different Models

Num Prod.	Max Purch. Qty	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL LL	MPMNL LL	MCRLM LL	SCRLM LL	MNL AIC_c	MPMNL AIC_c	MCRLM AIC_c	SCRLM AIC_c	MCRLM Num Para	SCRLM Num Para
5	2.0	22.05	57.01	42.99	-1093.66	-2620.58	-1136.74	-1372.01	2197.4	5257.34	2296.03	2756.53	11.1	6.2
10	2.0	20.08	41.41	58.59	-1524.64	-14775.69	-1593.23	-2050.84	3069.56	29577.84	3232.84	4121.95	22.5	10.0
15	2.0	17.18	38.52	61.48	-2339.73	-9941.65	-1868.15	-2358.95	4710.07	19920.17	3808.52	4745.39	34.5	13.5
20	2.0	16.96	37.07	62.93	-1906.83	-9085.65	-1952.0	-2460.52	3854.74	18218.72	3983.48	4951.66	37.8	15.0
25	2.0	16.83	36.14	63.86	-4176.88	-11617.59	-2176.61	-2660.88	8405.44	23293.28	4453.98	5360.94	47.3	19.1
30	2.0	15.86	35.7	64.3	-3102.91	-6255.46	-2295.12	-2819.88	6268.24	12579.85	4709.67	5682.11	55.4	20.6

noteworthy that, unlike the results presented in Section 6.1.1, the estimated MC-RLMs do not always feature fewer customer types compared to the SC-RLMs. This discrepancy may stem from differences between the simulated choice structure and the true choice structure in the real-world dataset. Although the MC-RLMs may not achieve the highest log-likelihood values on the training data compared to the other models, they consistently outperform the SC-RLMs in log-likelihood. A similar trend is observed with the AIC_c , where the MC-RLMs consistently show lower AIC_c values than the SC-RLMs.

Table EC.23 summarizes the average in-sample performance of predictive accuracy, grouped by different total numbers of products $|N|$.

Table EC.23 Average In-Sample Performance Comparison of Prediction Accuracy Across Different Models

Num Prod.	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL Train RMSE	MPMNL Train RMSE	MCRLM Train RMSE	SCRLM Train RMSE	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
5	57.01	42.99	0.4402	0.4416	0.425	0.4629	3.46	3.33	8.21
10	41.41	58.59	0.391	0.5308	0.3787	0.4104	3.19	28.75	7.84
15	38.52	61.48	0.3299	0.3766	0.3154	0.3424	4.24	15.23	7.86
20	37.07	62.93	0.2963	0.3364	0.2877	0.3091	2.9	13.99	6.91
25	36.14	63.86	0.2835	0.3256	0.2709	0.2853	4.17	16.19	5.08
30	35.7	64.3	0.2574	0.2636	0.248	0.2611	3.51	5.74	5.07

Table EC.24 presents the detailed assortment optimization results for each test case across different models, where the category name indicates the format 'num_product_random_seed' (e.g., '5pro_11').

Table EC.24 Detailed Performance Comparison of Assortment Optimization Across Different Models

Category	Num Products	MNL Revenue	MPMNL Revenue	MCRLM Revenue	SCRLM Revenue	MCRLM Improv. SCRLM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. MNL (%)
5pro_11	5	4.4172	4.8072	4.9762	4.9039	1.47	3.51	12.65
5pro_12	5	3.3826	5.0568	5.0688	5.0045	1.29	0.24	49.85
5pro_16	5	1.9363	2.668	2.6726	2.3258	14.92	0.17	38.03
5pro_22	5	0.8772	1.7927	2.3153	1.8361	26.1	29.15	163.94
5pro_24	5	3.4168	4.6109	4.5945	4.571	0.52	-0.35	34.47
5pro_28	5	3.8725	5.2257	5.3922	5.3538	0.72	3.19	39.24
5pro_3	5	1.2636	2.3162	2.808	2.7808	0.98	21.23	122.22
5pro_5	5	1.3836	4.4693	4.4611	4.2947	3.87	-0.18	222.43
5pro_7	5	3.5262	3.8793	4.6475	4.6786	-0.66	19.8	31.8
5pro_9	5	1.7214	2.9829	3.0082	2.9492	2.0	0.85	74.75
10pro_13	10	3.0618	3.0578	3.3723	3.3771	-0.14	10.28	10.14
10pro_16	10	4.211	4.0847	4.662	4.8022	-2.92	14.13	10.71
10pro_19	10	2.0934	2.0546	3.5849	2.9211	22.72	74.48	71.24
10pro_20	10	3.373	3.3606	4.4557	3.9803	11.94	32.59	32.1
10pro_22	10	2.4358	4.8097	4.5162	4.4412	1.69	-6.1	85.41
10pro_23	10	2.5419	5.1325	4.934	4.9309	0.06	-3.87	94.1
10pro_25	10	0.1904	1.6762	2.9746	2.3557	26.27	77.46	1462.6
10pro_26	10	5.6162	4.101	5.955	6.0805	-2.06	45.21	6.03
10pro_30	10	0.3527	2.1519	4.8896	4.7481	2.98	127.22	1286.52
10pro_9	10	4.8231	4.784	5.8743	5.8045	1.2	22.79	21.8
15pro_1	15	1.0412	4.8847	5.9306	5.9261	0.07	21.41	469.59
15pro_10	15	4.438	2.9106	5.3412	3.1644	68.79	83.51	20.35
15pro_15	15	1.7214	5.2257	5.2347	4.5236	15.72	0.17	204.09
15pro_18	15	0.1001	3.9855	4.0751	4.0027	1.81	2.25	3972.08
15pro_23	15	1.7384	3.5233	4.2028	4.3173	-2.65	19.29	141.75
15pro_26	15	0.706	1.9625	4.1194	3.7449	10.0	109.91	483.52
15pro_27	15	4.801	5.1574	5.1519	5.3259	-3.27	-0.11	7.31
15pro_5	15	5.2725	2.7704	6.3326	6.2587	1.18	128.58	20.11
15pro_7	15	0.6063	5.2324	5.2615	5.0488	4.21	0.56	767.75
15pro_9	15	0.359	3.6376	4.1773	2.921	43.01	14.83	1063.7
20pro_10	20	3.8646	5.2936	6.3991	6.0458	5.84	20.88	65.58
20pro_11	20	1.219	4.2805	4.1839	3.3116	26.34	-2.26	243.23
20pro_16	20	0.0812	4.0723	5.7014	5.739	-0.66	40.0	6921.58
20pro_19	20	0.0172	3.8593	5.0862	4.6154	10.2	31.79	29395.01
20pro_22	20	5.5056	5.3207	5.3599	4.8294	10.98	0.74	-2.65
20pro_23	20	0.5232	3.6069	4.3126	4.3956	-1.89	19.56	724.23
20pro_26	20	3.6455	4.3059	4.9317	4.7321	4.22	14.53	35.28
20pro_3	20	1.2849	2.2735	2.4074	1.9352	24.4	5.89	87.37
20pro_30	20	0.912	4.5564	4.7992	4.6395	3.44	5.33	426.25
20pro_9	20	2.8692	5.0994	5.0805	4.9012	3.66	-0.37	77.07
25pro_12	25	3.1098	2.3448	3.7445	3.547	5.57	59.7	20.41
25pro_13	25	0.7967	4.1526	4.3216	4.565	-5.33	4.07	442.41
25pro_15	25	0.2886	4.7015	5.3567	5.3277	0.55	13.94	1756.21
25pro_18	25	2.0209	2.8219	3.0649	2.8844	6.26	8.61	51.66
25pro_19	25	0.4819	1.9962	3.0119	2.744	9.76	50.88	525.02
25pro_21	25	3.9902	5.1284	5.3507	4.5713	17.05	4.34	34.1
25pro_26	25	4.0224	2.903	5.6904	5.7234	-0.58	96.02	41.47
25pro_27	25	2.76	2.136	4.0467	3.583	12.94	89.45	46.62
25pro_4	25	0.0171	2.4062	3.5257	3.499	0.76	46.53	20470.16
25pro_5	25	5.9248	4.9342	5.7313	5.3886	6.36	16.16	-3.27
30pro_10	30	2.8559	5.6409	5.7207	5.6883	0.57	1.42	100.31
30pro_11	30	3.312	3.078	3.1023	3.1801	-2.45	0.79	-6.33
30pro_13	30	0.4124	4.276	4.2954	4.2823	0.31	0.45	941.48
30pro_15	30	3.3846	3.7632	4.8691	4.8599	0.19	29.39	43.86
30pro_17	30	0.0088	4.3667	5.2804	3.7964	39.09	20.93	60228.35
30pro_23	30	2.7016	4.5629	4.574	4.4455	2.89	0.24	69.31
30pro_28	30	3.4104	3.3477	3.3743	3.3899	-0.46	0.79	-1.06
30pro_29	30	3.2485	4.3571	4.2911	4.3296	-0.89	-1.51	32.09
30pro_5	30	0.0322	3.8381	3.8365	4.3755	-12.32	-0.04	11817.03
30pro_9	30	3.0821	5.5433	6.1127	5.8373	4.72	10.27	98.33

EC.7.1 Supplementary Experiments with Fixed Intended Purchase Quantities

This subsection presents supplementary experiments in which the ground-truth choice data is generated under a fixed intended purchase quantity. This setting allows us to isolate the effect of model structure on revenue performance, independent of differences in modeling the purchase quantity distribution.

Following the setup in Section 6.2, we generate synthetic data using a variant of the MP-MNL model. In this variant, the random utility of product $n \in \mathbb{N}$ is defined as $U_n = V_n - \beta_n r_n + \epsilon_n$, where $\epsilon = (\epsilon_n)_{n \in \mathbb{N}^+}$ follows an i.i.d. standard normal distribution, instead of the Gumbel distribution used previously. The deterministic utility V_n is drawn from a normal distribution with mean 3 and standard deviation 1; product revenues r_n are uniformly sampled from the range $[1, 5]$; and the price sensitivity β_n is generated as the negative absolute value of a standard normal variable. Unlike the setup in Section 6.2, where the intended purchase quantity follows an exponential decay distribution, here we fix the intended purchase quantity to a constant value for each customer, independent of the product utilities. It is worth noting that even with a fixed intended purchase quantity, the actual number of products purchased by each customer may still vary, as it depends on the number of products whose utility exceeds that of the outside option. If fewer products satisfy this condition than the intended purchase quantity, the customer purchases all such available products.

We conduct two sets of experiments: one with the intended purchase quantity fixed at 2, and another with it fixed at 3.

EC.7.1.1 Experiments with Intended Purchase Quantity Fixed at Two

In this subsection, we present the results of experiments where the intended purchase quantity is fixed at 2 for the ground-truth customer choices. Table EC.25 summarizes how different models fit the synthetic customer choice data. Table EC.26 and Table EC.27 report the in-sample and out-of-sample predictive accuracies, respectively. Table EC.28 summarizes the constrained ($K = 4$) assortment optimization performance, while Table EC.29 provides the detailed results for each individual test case.

We find that, consistent with the results in Section 6.2, the proposed MC-RLM model achieves lower RMSE and higher revenue on average across all cases tested.

Table EC.25 Average Estimation Results Across Different Models with IPQ Fixed at 2

Num. Prod.	Max Purch. Qty	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL LL	MPMNL LL	MCRLM LL	SCRLM LL	MNL AIC_c	MPMNL AIC_c	MCRLM AIC_c	SCRLM AIC_c	MCRLM Num Para	SCRLM Num Para
5	2.0	16.77	43.99	56.01	-1021.64	-4326.18	-795.78	-1518.38	2053.35	8668.55	1610.0	3049.07	9.1	6.1
10	2.0	3.03	12.59	87.41	-1685.99	-16172.19	-1323.4	-2417.84	3392.26	32370.84	2686.64	4858.43	19.4	11.2
15	2.0	0.64	2.83	97.17	-2185.7	-17227.49	-1773.45	-3015.28	4402.01	34491.85	3615.71	6059.5	32.9	14.2
20	2.0	0.27	1.39	98.61	-2375.32	-18015.79	-1968.75	-3138.5	4791.71	36079.0	4026.1	6313.06	41.9	17.6
25	2.0	0.28	1.13	98.87	-2520.13	-18372.34	-2116.7	-3249.09	5091.94	36802.8	4333.77	6540.33	47.1	20.5
30	2.0	0.16	0.64	99.36	-2696.44	-18384.06	-2240.62	-3412.8	5455.3	36837.06	4608.02	6871.99	58.5	22.5

Table EC.26 Average In-Sample Prediction Accuracy Across Different Models with IPQ Fixed at 2

Num Prod.	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL Train RMSE	MPMNL Train RMSE	MCRLM Train RMSE	SCRLM Train RMSE	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
5	43.99	56.01	0.4271	0.4219	0.3535	0.4529	15.99	10.76	21.09
10	12.59	87.41	0.4287	0.6104	0.3658	0.4469	14.76	38.66	18.25
15	2.83	97.17	0.3811	0.4605	0.3418	0.3952	10.41	24.53	13.58
20	1.39	98.61	0.3488	0.4302	0.3178	0.3601	8.95	25.42	11.78
25	1.13	98.87	0.3183	0.4065	0.2942	0.3257	7.61	27.63	9.72
30	0.64	99.36	0.2906	0.3585	0.2688	0.2992	7.55	24.8	10.22

Table EC.27 Average Out-of-Sample Prediction Accuracy Across Different Models with IPQ Fixed at 2

Num Prod.	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL Test RMSE	MPMNL Test RMSE	MCRLM Test RMSE	SCRLM Test RMSE	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
5	43.99	56.01	0.4234	0.4142	0.3441	0.4491	17.4	11.78	22.54
10	12.59	87.41	0.4304	0.61	0.3709	0.4486	13.96	37.78	17.48
15	2.83	97.17	0.3808	0.4578	0.3433	0.3949	9.95	23.65	13.14
20	1.39	98.61	0.3486	0.4325	0.3176	0.3595	8.95	25.8	11.69
25	1.13	98.87	0.3172	0.4059	0.2924	0.3252	7.85	27.96	10.12
30	0.64	99.36	0.2913	0.3594	0.2706	0.3002	7.21	24.52	9.94

Table EC.28 Average Constrained Assortment Optimization Performance Across Different Models with IPQ at 2

Num Products	MNL Avg. Revenue	MPMNL Avg. Revenue	MCRLM Avg. Revenue	SCRLM Avg. Revenue	MCRLM Improv. SCRLM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. MNL (%)
5	2.8013	4.693	5.3246	5.1039	4.47	38.4	173.88
10	3.1512	4.3688	6.9502	6.6437	5.57	116.7	251.82
15	2.9955	6.0415	7.7819	7.1227	10.85	41.53	70653.9
20	3.0899	5.9876	7.5612	6.9521	10.28	31.02	438.86
25	2.3283	5.1327	6.5388	6.067	9.95	40.18	2503.77
30	4.3719	6.4836	7.5026	6.7592	11.06	21.11	16935.39

Table EC.29 Detailed Constrained Assortment Optimization Performance Across Different Models with IPQ at 2

Category	Num Products	MNL Revenue	MPMNL Revenue	MCRLM Revenue	SCRLM Revenue	MCRLM Improv. SCRLM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. MNL (%)
5pro_10	5	3.2128	6.3157	6.2944	6.1985	1.55	-0.34	95.92
5pro_12	5	0.7729	4.5954	5.4004	5.398	0.05	17.52	598.7
5pro_13	5	3.1487	4.9699	4.9837	5.0292	-0.9	0.28	58.28
5pro_2	5	3.2662	6.1003	6.1081	4.7139	29.58	0.13	87.01
5pro_22	5	3.2562	6.2604	6.2742	6.2832	-0.14	0.22	92.69
5pro_23	5	0.3193	0.7124	2.1552	2.1325	1.06	202.54	575.08
5pro_24	5	3.4042	4.5432	5.7066	5.7091	-0.04	25.61	67.64
5pro_6	5	4.0354	6.302	6.3022	5.7081	10.41	0.0	56.17
5pro_7	5	3.6397	2.0915	4.9753	4.8961	1.62	137.88	36.7
5pro_9	5	2.9575	5.0391	5.0463	4.9708	1.52	0.14	70.63
10pro_11	10	4.2275	4.2562	8.1192	8.1167	0.03	90.76	92.06
10pro_15	10	1.8094	4.8788	4.8637	4.1781	16.41	-0.31	168.81
10pro_16	10	3.5452	3.5175	5.7094	5.7048	0.08	62.31	61.05
10pro_17	10	0.8889	0.92	5.4869	5.4789	0.15	496.4	517.26
10pro_2	10	4.955	8.6654	8.732	8.7135	0.21	0.77	76.23
10pro_23	10	4.2612	4.3296	7.5247	7.5312	-0.09	73.8	76.59
10pro_25	10	2.7862	1.7906	6.0116	5.0723	18.52	235.72	115.76
10pro_3	10	0.5662	3.0456	7.62	6.726	13.29	150.2	1245.79
10pro_4	10	4.6243	5.4158	8.3644	8.1282	2.91	54.44	80.88
10pro_7	10	3.8477	6.8686	7.0701	6.7876	4.16	2.93	83.75
15pro_1	15	4.7276	3.5037	8.9952	9.1176	-1.34	156.74	90.27
15pro_14	15	2.7587	7.8787	7.9278	7.5864	4.5	0.62	187.38
15pro_17	15	3.2074	3.2248	5.6483	5.5585	1.62	75.15	76.1
15pro_18	15	3.9568	6.621	7.8486	6.5882	19.13	18.54	98.36
15pro_2	15	3.7985	7.8267	8.4308	8.4253	0.06	7.72	121.95
15pro_20	15	4.7184	4.5736	7.636	6.2601	21.98	66.96	61.84
15pro_23	15	4.3229	8.5764	8.7138	8.7292	-0.18	1.6	101.58
15pro_25	15	0.381	4.7095	7.4631	7.404	0.8	58.47	1858.71
15pro_28	15	0.001	5.3699	6.8148	5.6987	19.59	26.91	703642.35
15pro_9	15	2.0828	8.1307	8.3407	5.8588	42.36	2.58	300.46
20pro_12	20	1.4646	3.2959	5.2527	4.9843	5.39	59.37	258.65
20pro_15	20	0.4856	4.6985	6.5276	5.4677	19.38	38.93	1244.34
20pro_16	20	1.6053	5.5393	7.0695	6.8563	3.11	27.63	340.39
20pro_17	20	4.1421	4.6938	7.9991	7.6671	4.33	70.42	93.12
20pro_21	20	0.3196	5.8652	6.8185	6.2136	9.73	16.25	2033.56
20pro_24	20	4.6658	6.3472	6.3509	5.2118	21.86	0.06	36.12
20pro_26	20	4.7358	7.9745	9.2248	9.3247	-1.07	15.68	94.79
20pro_29	20	4.4412	5.6823	9.6033	9.1793	4.62	69.0	116.23
20pro_3	20	4.4191	7.6448	8.5777	8.5696	0.09	12.2	94.1
20pro_8	20	4.6194	8.1347	8.1881	6.0468	35.41	0.66	77.25
25pro_13	25	3.4337	4.6991	4.943	4.1152	20.12	5.19	43.95
25pro_14	25	0.0548	3.7579	7.879	7.8854	-0.08	109.66	14277.23
25pro_16	25	2.795	5.1924	6.7334	6.7255	0.12	29.68	140.91
25pro_17	25	4.5459	7.1128	8.8025	8.3767	5.08	23.76	93.64
25pro_19	25	0.7627	5.6018	6.9122	5.9229	16.7	23.39	806.27
25pro_24	25	4.4414	5.9939	6.4889	6.4999	-0.17	8.26	46.1
25pro_29	25	2.4521	4.623	5.2349	3.7184	40.78	13.24	113.49
25pro_4	25	0.0863	1.7502	4.6315	4.3329	6.89	164.62	5264.38
25pro_7	25	0.1425	4.9044	6.1051	5.8657	4.08	24.48	4184.12
25pro_9	25	4.5689	7.6917	7.6578	7.2279	5.95	-0.44	67.61
30pro_15	30	4.839	4.9758	7.4575	6.7999	9.67	49.88	54.11
30pro_17	30	4.2724	6.8183	8.6855	7.2696	19.48	27.38	103.29
30pro_19	30	3.8357	8.4615	9.0997	8.1795	11.25	7.54	137.24
30pro_2	30	5.6955	7.7673	7.7957	6.4653	20.58	0.37	36.88
30pro_22	30	6.6296	8.3172	8.7207	7.1743	21.55	4.85	31.54
30pro_24	30	5.2366	4.7454	7.3253	7.4102	-1.15	54.37	39.89
30pro_26	30	6.6888	7.8655	7.2281	5.7829	24.99	-8.1	8.06
30pro_27	30	0.004	6.401	6.7426	6.8186	-1.11	5.34	168747.6
30pro_3	30	4.2286	6.0864	6.3697	6.3863	-0.26	4.66	50.63
30pro_8	30	2.2889	3.3977	5.6007	5.3053	5.57	64.84	144.69

EC.7.1.2 Experiments with Intended Purchase Quantity Fixed at Three

In this subsection, we present the results of experiments where the intended purchase quantity is fixed at 3 for the ground-truth customer choices. Table EC.30 summarizes how different models fit the synthetic customer choice data. Table EC.31 and Table EC.32 report the in-sample and out-of-sample predictive accuracies, respectively. Table EC.33 summarizes the constrained ($K = 4$) assortment optimization performance, while Table EC.34 provides the detailed results for each individual test case.

We find that, overall, the results resemble the structure observed under the setting with the intended purchase quantity fixed at 2. One notable difference is that, for cases with 10 products, the MP-MNL model achieves slightly better average predictive accuracy (lower RMSE) than MC-RLM, which may also contribute to its comparable revenue performance relative to MC-RLM in these cases. Nevertheless, MC-RLM still achieves higher revenue on average when there are 10 products.

In addition, compared to Table EC.28, we observe that the average revenue achieved by each model increases as the intended purchase quantity grows, and the relative improvement of MC-RLM over the benchmark models becomes even more pronounced. More importantly, we note that ignoring multi-purchase behavior and simply applying the classic MNL model can result in extremely poor revenue performance in many cases. These findings further highlight the importance of accurately modeling multi-purchase behavior when multi-purchases are prevalent.

Moreover, we observe that the overall structure of the results in Sections EC.7.1.1 and EC.7.1.2 remains consistent with those obtained when intended purchase quantities follow an exponential decay distribution, as presented in Section 6.2. This consistency indicates that the performance of MC-RLM is not sensitive to the specific distribution of customers' intended purchase quantities.

Table EC.30 Average Estimation Results Across Different Models with IPQ Fixed at 3

Num Prod.	Max Purch. Qty	No Purch. Prop. (%)	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL LL	MPMNL LL	MCRLM LL	SCRLM LL	MNL AIC_c	MPMNL AIC_c	MCRLM AIC_c	SCRLM AIC_c	MCRLM Num Para	SCRLM Num Para
5	3.0	7.84	35.71	64.29	-1245.7	-2819.32	-708.35	-1729.87	2501.48	5656.87	1436.58	3472.05	9.8	6.1
10	3.0	2.71	12.34	87.66	-2454.76	-1413.34	-1203.45	-3362.97	4929.79	2855.22	2454.6	6749.52	23.1	11.6
15	3.0	0.65	2.56	97.44	-3521.58	-11001.7	-1918.93	-4721.41	7073.78	22042.37	3941.62	9476.79	48.5	16.6
20	3.0	0.58	2.28	97.72	-3794.35	-16953.52	-2205.02	-4873.15	7629.78	33956.59	4547.42	9784.45	62.9	18.6
25	3.0	0.3	1.23	98.76	-4113.74	-18281.27	-2484.61	-5180.63	8279.16	36622.79	5147.71	10402.77	79.9	20.2
30	3.0	0.23	0.48	99.52	-4534.52	-18085.09	-2799.65	-5623.27	9131.47	36241.29	5843.51	11295.88	105.4	23.9

Table EC.31 Average In-Sample Prediction Accuracy Across Different Models with IPQ Fixed at 3

Num Prod.	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL Train RMSE	MPMNL Train RMSE	MCRLM Train RMSE	SCRLM Train RMSE	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
5	35.71	64.29	0.4784	0.3633	0.3166	0.4958	32.89	8.87	35.54
10	12.34	87.66	0.5153	0.3343	0.3372	0.5239	34.5	-1.2	35.62
15	2.56	97.44	0.4768	0.4617	0.3511	0.4837	26.42	21.28	27.47
20	2.28	97.72	0.4357	0.4694	0.3374	0.4446	22.68	26.86	24.22
25	1.23	98.76	0.4011	0.4622	0.3218	0.4073	19.81	30.15	21.03
30	0.48	99.52	0.3709	0.4271	0.3074	0.3758	17.14	27.69	18.23

Table EC.32 Average Out-of-Sample Prediction Accuracy Across Different Models with IPQ Fixed at 3

Num Prod.	Single Purch. Prop. Norm. (%)	Multi Purch. Prop. Norm. (%)	MNL Test RMSE	MPMNL Test RMSE	MCRLM Test RMSE	SCRLM Test RMSE	MCRLM Improv. MNL RMSE (%)	MCRLM Improv. MPMNL RMSE (%)	MCRLM Improv. SCRLM RMSE (%)
5	35.71	64.29	0.4777	0.3702	0.321	0.4957	32.01	9.49	34.75
10	12.34	87.66	0.518	0.3314	0.3345	0.5267	35.33	-1.33	36.47
15	2.56	97.44	0.478	0.4634	0.3544	0.4852	25.89	20.87	26.98
20	2.28	97.72	0.4351	0.4692	0.3376	0.4441	22.52	26.86	24.1
25	1.23	98.76	0.4033	0.465	0.326	0.4103	19.23	29.71	20.59
30	0.48	99.52	0.3714	0.4279	0.3085	0.3768	16.97	27.56	18.16

Table EC.33 Average Constrained Assortment Optimization Performance Across Different Models with IPQ at 3

Num Products	MNL Avg. Revenue	MPMNL Avg. Revenue	MCRLM Avg. Revenue	SCRLM Avg. Revenue	MCRLM Improv. SCRLM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. MNL (%)
5	2.3433	7.3144	7.7533	6.9921	11.79	7.76	6318.39
10	2.1563	9.1538	9.3131	8.4737	10.21	1.67	335572.54
15	2.2458	9.6575	10.9241	10.4382	4.66	14.56	5060.57
20	2.1129	8.2077	10.2423	9.2884	11.0	107.89	26117.63
25	2.5336	7.2538	10.3931	9.6388	9.19	62.62	1081.73
30	2.1711	8.4771	10.857	10.3264	5.6	277.85	68829.67

Table EC.34 Detailed Constrained Assortment Optimization Performance Across Different Models with IPQ at 3

Category	Num Products	MNL Revenue	MPMNL Revenue	MCRLM Revenue	SCRLM Revenue	MCRLM Improv. SCRLM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. MNL (%)
5pro_16	5	0.0083	3.7956	4.6788	4.7196	-0.87	23.27	56565.12
5pro_17	5	4.6243	10.1654	10.1554	9.4489	7.48	-0.1	119.61
5pro_19	5	3.6267	6.817	6.8494	5.6129	22.03	0.47	88.86
5pro_21	5	1.5314	5.7347	5.7587	4.2956	34.06	0.42	276.03
5pro_22	5	0.3578	6.9453	9.7471	6.9289	40.67	40.34	2624.06
5pro_25	5	2.1478	6.8584	6.8615	6.8636	-0.03	0.05	219.47
5pro_26	5	2.4295	4.8693	5.5021	5.4379	1.18	12.99	126.47
5pro_28	5	0.249	7.4205	7.4185	7.439	-0.28	-0.03	2879.89
5pro_5	5	4.3996	11.6093	11.6316	10.4451	11.36	0.19	164.38
5pro_9	5	4.0586	8.928	8.93	8.729	2.3	0.02	120.02
10pro_11	10	2.5253	7.1279	7.4568	7.1116	4.85	4.61	195.29
10pro_13	10	1.5554	11.5224	11.5272	10.0305	14.92	0.04	641.11
10pro_14	10	0.0473	7.499	7.5008	7.4947	0.08	0.02	15750.93
10pro_18	10	3.4098	10.4015	10.5559	9.6972	8.86	1.48	209.58
10pro_2	10	2.9972	7.6975	7.8282	5.9523	31.52	1.7	161.18
10pro_23	10	4.3829	11.2131	11.7591	11.6842	0.64	4.87	168.29
10pro_24	10	1.6997	8.6646	8.668	8.0633	7.5	0.04	409.98
10pro_25	10	0.0005	5.8979	5.9024	5.8105	1.58	0.08	1200499.31
10pro_27	10	0.0005	10.5797	10.5818	9.1673	15.43	0.02	2137560.18
10pro_8	10	4.944	10.9346	11.3506	9.7251	16.71	3.8	129.58
15pro_1	15	0.6645	10.5773	10.5852	9.4394	12.14	0.07	1492.99
15pro_12	15	0.0236	9.2224	9.5431	9.1637	4.14	3.48	40277.06
15pro_14	15	5.2742	10.6604	10.6915	10.2049	4.77	0.29	102.71
15pro_17	15	0.7355	10.8835	10.8787	10.1289	7.4	-0.04	1379.16
15pro_18	15	0.6848	9.5067	9.701	9.4187	3.0	2.04	1316.58
15pro_20	15	4.1791	11.7661	13.2006	12.6211	4.59	12.19	215.87
15pro_21	15	4.2978	8.3288	12.674	11.6575	8.72	52.17	194.9
15pro_29	15	0.0	8.2569	11.1816	10.8105	3.43	35.42	
15pro_4	15	4.8891	8.524	11.8331	11.8537	-0.17	38.82	142.03
15pro_9	15	1.709	8.8494	8.9519	9.0837	-1.45	1.16	423.81
20pro_10	20	3.5909	11.6089	11.8343	10.0275	18.02	1.94	229.56
20pro_11	20	3.8623	7.3579	8.8449	9.2232	-4.1	20.21	129.0
20pro_13	20	0.0067	8.2355	10.3606	8.0895	28.07	25.8	153914.99
20pro_14	20	3.3252	9.5849	10.4101	9.7444	6.83	8.61	213.06
20pro_17	20	0.0134	7.2741	9.7065	9.1573	6.0	33.44	72217.63
20pro_18	20	0.1373	7.7487	9.9281	7.4437	33.38	28.13	7130.84
20pro_23	20	4.0516	7.6842	9.8356	9.8873	-0.52	28.0	142.76
20pro_27	20	0.0	0.9044	9.2944	8.8316	5.24	927.73	
20pro_30	20	5.1455	11.5332	11.7196	10.9579	6.95	1.62	127.76
20pro_6	20	0.996	10.1454	10.489	9.5217	10.16	3.39	953.1
25pro_10	25	4.3982	8.2246	12.0106	11.9943	0.14	46.03	173.08
25pro_13	25	4.5446	10.3514	12.458	11.6426	7.0	20.35	174.13
25pro_14	25	0.3698	2.4559	7.1221	5.6299	26.5	189.99	1826.12
25pro_15	25	1.3433	6.5178	10.4837	10.2843	1.94	60.85	680.43
25pro_19	25	0.4258	11.2174	13.0805	13.0618	0.14	16.61	2971.95
25pro_20	25	4.9373	10.29	12.2751	10.2393	19.88	19.29	148.62
25pro_26	25	3.4979	5.4741	9.1707	8.9657	2.29	67.53	162.18
25pro_27	25	3.8568	7.3158	10.5329	9.7403	8.14	43.98	173.1
25pro_5	25	1.7432	3.1265	7.4322	7.0881	4.85	137.72	326.34
25pro_8	25	0.2187	7.5645	9.3653	7.7414	20.98	23.81	4181.32
30pro_1	30	3.3841	0.386	10.3952	10.1854	2.06	2592.86	207.17
30pro_10	30	5.7786	8.0416	9.9969	9.9928	0.04	24.31	73.0
30pro_19	30	0.0037	10.0051	11.7923	9.0329	30.55	17.86	314967.34
30pro_2	30	0.579	11.2972	11.4584	11.602	-1.24	1.43	1878.83
30pro_20	30	4.7701	10.577	10.822	10.8068	0.14	2.32	126.87
30pro_21	30	4.4207	12.0587	12.0853	10.332	16.97	0.22	173.38
30pro_3	30	0.0845	11.1822	11.8538	10.6651	11.15	6.01	13928.69
30pro_4	30	2.6719	6.5178	10.0977	11.0459	-8.58	54.93	277.92
30pro_7	30	0.0034	6.1889	9.7067	9.9262	-2.21	56.84	286656.44
30pro_8	30	0.0148	8.5161	10.3622	9.6744	7.11	21.68	70007.03

EC.7.2 Supplementary Experiments on Unconstrained Assortment Optimization

In addition to the constrained assortment optimization experiments presented in Section 6.2, we conduct supplementary experiments under an unconstrained setting, where the performance of the revenue-ordered assortment heuristic can be evaluated. The experimental setup follows that section exactly, except that no cardinality constraint is imposed on the assortment. As prior results have shown, ignoring customer intra-category multi-purchase behavior can lead to significant revenue loss. Therefore, in the experiments presented in this section, we exclude both the SC-RLM and the MNL choice model from comparison. As an additional benchmark, we consider the strategy of offering the complete set N , which is equivalent to applying the ICM when all products have positive revenue. Since this model assumes independent choice across products and the assortment is unconstrained, offering all products with positive revenue is optimal under the ICM.

Average results over 10 repetitions for each product count setting are reported in Table EC.35. The estimated model used in these experiments is exactly the same as that reported in Section 6.2; however, the 10,000 simulated customer choices used for evaluation may differ from those in the earlier experiments due to resampling.

Table EC.35 Average Performance Comparison of Unconstrained Assortment Optimization Across Different Models

Num Products	Complete Set Avg. Revenue	MPMNL Avg. Revenue	MCRLM Avg. Revenue	RO Avg. Revenue	MCRLM Improv. ICM (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. RO (%)
5	3.3721	3.8037	3.9331	3.7304	16.76	3.81	5.72
10	3.7182	3.5261	4.4438	3.8741	21.4	37.62	17.13
15	4.1418	4.7252	4.8023	4.3114	17.42	2.78	12.5
20	3.8527	4.2113	4.4543	4.184	15.03	5.52	6.22
25	3.4531	3.903	4.0803	3.6206	22.03	8.59	15.22

In Table EC.35, “RO” refers to the best revenue-ordered assortment under the MC-RLM model. On average, the revenue achieved by this heuristic does not exceed that of the optimal solution obtained by solving MILP (3). However, in many of the tested cases (see Table EC.36 for details), it yields higher revenue than the optimal assortment derived from the MP-MNL model. Moreover, the naive strategy of offering all products generally does not achieve satisfactory performance.

Table EC.36 Detailed Performance Comparison of Unconstrained Assortment Optimization Across Different Models

Category	Num Products	Complete Set Revenue	MPMNL Revenue	MCRLM Revenue	RO Revenue	MCRLM Improv. Complete Set (%)	MCRLM Improv. MPMNL (%)	MCRLM Improv. RO (%)
5pro_11	5	4.0711	4.8992	4.9394	4.9178	21.33	0.82	0.44
5pro_12	5	3.3718	5.0566	5.0588	5.0728	50.04	0.05	-0.28
5pro_16	5	2.6574	2.6491	2.6589	2.6628	0.06	0.37	-0.15
5pro_22	5	1.8032	1.7865	1.7903	1.7924	-0.71	0.21	-0.12
5pro_24	5	4.49	4.6082	4.5631	4.5174	1.63	-0.98	1.01
5pro_28	5	5.2707	5.3091	5.4401	5.3542	3.21	2.47	1.6
5pro_3	5	2.2797	2.3807	2.74	2.7107	20.19	15.09	1.08
5pro_5	5	3.2911	4.4458	4.4968	3.8062	36.64	1.15	18.15
5pro_7	5	3.9184	3.901	4.6565	3.8885	18.84	19.37	19.75
5pro_9	5	2.5676	3.0011	2.987	2.5811	16.34	-0.47	15.73
10pro_13	10	2.5253	3.0855	3.3731	2.5445	33.57	9.32	32.56
10pro_16	10	3.8673	4.0744	4.7735	4.4098	23.43	17.16	8.25
10pro_19	10	3.0012	2.0625	3.5974	3.2984	19.86	74.42	9.06
10pro_20	10	3.5376	3.3875	3.8039	3.5722	7.53	12.29	6.49
10pro_22	10	3.8163	4.7867	4.4844	3.8359	17.51	-6.32	16.91
10pro_23	10	4.3276	5.1036	4.9835	4.8488	15.16	-2.35	2.78
10pro_25	10	2.3419	1.689	2.9991	2.3123	28.07	77.57	29.7
10pro_26	10	5.5922	4.1168	6.011	5.7259	7.49	46.01	4.98
10pro_30	10	3.383	2.1353	5.0629	3.394	49.66	137.1	49.17
10pro_9	10	4.7895	4.8201	5.3487	4.7994	11.68	10.97	11.45
15pro_1	15	5.1613	5.2431	5.3952	5.1144	4.53	2.9	5.49
15pro_10	15	3.5008	4.5935	4.6677	3.5549	33.33	1.62	31.3
15pro_15	15	4.4615	5.2247	5.1843	4.4775	16.2	-0.77	15.78
15pro_18	15	3.1349	4.0849	3.9727	3.9394	26.73	-2.75	0.84
15pro_23	15	3.4652	3.4887	4.1398	3.4479	19.47	18.66	20.07
15pro_26	15	3.5373	3.5291	4.05	3.5329	14.49	14.76	14.64
15pro_27	15	4.0548	4.9841	5.3495	4.0389	31.93	7.33	32.45
15pro_5	15	6.0075	6.0418	6.1908	5.9584	3.05	2.47	3.9
15pro_7	15	4.4171	4.5214	4.875	4.8544	10.37	7.82	0.42
15pro_9	15	3.6779	5.5411	4.1979	4.1953	14.14	-24.24	0.06
20pro_10	20	4.8325	5.668	6.0899	5.6475	26.02	7.44	7.83
20pro_11	20	3.3205	4.1696	4.0183	4.0896	21.01	-3.63	-1.75
20pro_16	20	3.9251	4.1188	4.0776	4.0621	3.88	-1.0	0.38
20pro_19	20	4.6958	4.6967	5.0213	4.9104	6.93	6.91	2.26
20pro_22	20	3.8567	4.6192	5.2722	4.6942	36.7	14.14	12.31
20pro_23	20	3.7025	3.7012	4.2606	3.6982	15.07	15.11	15.21
20pro_26	20	4.3189	4.348	4.7648	4.5228	10.33	9.59	5.35
20pro_3	20	1.8107	1.8584	1.859	1.8431	2.67	0.03	0.86
20pro_30	20	4.0281	4.077	4.4783	4.3241	11.18	9.85	3.57
20pro_9	20	4.036	4.8564	4.7013	4.0479	16.48	-3.19	16.14
25pro_12	25	3.3013	3.3247	3.7376	3.7483	13.22	12.42	-0.28
25pro_13	25	3.5098	3.916	4.211	3.5055	19.98	7.53	20.13
25pro_15	25	5.2054	5.1726	5.248	5.2078	0.82	1.46	0.77
25pro_18	25	2.9477	2.9523	3.1454	3.1279	6.71	6.54	0.56
25pro_19	25	1.9223	1.905	3.0673	2.2979	59.56	61.01	33.48
25pro_21	25	3.7641	5.0129	4.3636	4.1883	15.93	-12.95	4.18
25pro_26	25	4.7596	4.8357	5.3288	4.7833	11.96	10.2	11.4
25pro_27	25	3.1662	3.1786	3.433	3.1657	8.43	8.01	8.44
25pro_4	25	2.1937	3.3352	3.3616	2.2707	53.24	0.79	48.04
25pro_5	25	3.761	5.3965	4.9069	3.9107	30.47	-9.07	25.47